

The Stochastic Smoluchowski Kramers Approximation with a Mean Field Nonlinearity

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Abstract

We consider the Smoluchowski Kramers Approximation for the stochastic damped mean field wave equation (SDMFW) $\varepsilon^2 \partial_t^2 u_\varepsilon + \partial_t u_\varepsilon + (1 - \Delta)u_\varepsilon + \mathbb{E}[:u_\varepsilon^2:]u_\varepsilon = \sqrt{2}\xi$ to the stochastic mean field heat equation (SMFH) $\partial_t u_0 + (1 - \Delta)u_0 + \mathbb{E}[:u_0^2:]u_0 = \sqrt{2}\xi$, with a white-noise forcing term ξ , posed on the two-dimensional torus $\mathbb{T}^2$. The main contributions of this thesis are: (i) We apply the *I*-Method to derive a global bound on u_ε that is uniform in ε , allowing for the strong global convergence of u_ε to u_0 as $\varepsilon \rightarrow 0$. (ii) We establish an argument, utilizing self-adjoint operator theory, to prove that $\rho_\varepsilon := \mathcal{N}\left(0, \begin{pmatrix} (1 - \Delta)^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}\right) \in \mathcal{P}(H^{-\sigma}(\mathbb{T}^2) \times H^{-1-\sigma}(\mathbb{T}^2))$ is the unique invariant measure of the SDMFW in the class $\{\mu \in \mathcal{P}(H^{-\sigma}(\mathbb{T}^2) \times H^{-1-\sigma}(\mathbb{T}^2)) : \mathbb{E}_\mu[:u^2:] \in H^{-\sigma}(\mathbb{T}^2)\}$, where $:u^2:$ denotes a Wick renormalization. (iii) We prove the convergence of ρ_ε to ρ_0 as $\varepsilon \rightarrow 0$ using continuous bounded test functions that vanish at infinity, where $\rho_0 := \mathcal{N}(0, (1 - \Delta)^{-1}) \in \mathcal{P}(H^{-\sigma}(\mathbb{T}^2))$ is the invariant measure of the SMFH, and discuss why this convergence fails in a weak sense and is trivial in a vague sense.

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1 Introduction

In this thesis, we investigate the existence of solutions, invariant measures, and the convergence behavior of a family of stochastic damped wave equations, referred to as the stochastic damped mean field wave equation (SDMFW) on $\mathbb{T}^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2$. Namely, let ξ be a stochastic forcing term and $\varepsilon \in (0, 1]$ then the SDMFW is given by

$$\begin{cases} \varepsilon^2 \partial_t^2 u_\varepsilon + \partial_t u_\varepsilon + (1 - \Delta)u_\varepsilon + \mathbb{E}[u_\varepsilon^2]u_\varepsilon = \xi, \\ (u_\varepsilon, \partial_t u_\varepsilon)|_{t=0} = (\phi_0, \phi_1). \end{cases}$$

We prove that the SDMFW is globally well-posed for all $\varepsilon \in (0, 1]$ and converges to the stochastic mean field heat equation (SMFH):

$$\begin{cases} \partial_t u + (1 - \Delta)u + \mathbb{E}[u^2]u = \xi, \\ u|_{t=0} = \phi_0 \end{cases}$$

as $\varepsilon \rightarrow 0$. The convergence of a damped wave equation to a heat equation is known as the Smoluchowski-Kramers Approximation (SKA), consequently, in this thesis, we refer to convergence of the SDMFW to the SMFH as the mean field Smoluchowski-Kramers Approximation (mean field SKA).

Furthermore, we show that the SDMFW admits a unique invariant measure that converges to the invariant measure of the SMFH for continuous bounded test functions that vanish at infinity.

Chapter 1 motivates the mean field SKA based on its historical origin and its current physical context in the stochastic quantization program. Chapter 2 introduces the notation and definitions necessary for this thesis. In particular, we formalize the definition of the mean field SKA and identify the invariant measure of the SDMFW. Furthermore, we motivate our ideas via formal computations which we extend to rigorous proofs in Chapters 4 through 6. In Chapter 3, we present some preliminary functional analytical results and estimates to understand the behavior of the propagators involved in the SKA. Additionally, we analyze the stochastic convolution of white noise with the damped wave equation propagator, proving its well-definedness as well as its convergence to the stochastic convolution involving the heat equation propagator. Subsequently, in Chapters 4 and 5, we prove the local and global well-posedness of the mean field SKA, respectively. Finally, in Chapter 6 we prove the existence and conditional uniqueness of an invariant measure for the SDMFW and its convergence to the invariant measure of the SMFH.

1.1 Origin of the Smoluchowski-Kramers Approximation

Smoluchowski [1] observed that physical processes modeled by a Brownian motion can only be interpreted in a statistical sense. Rather than being able to predict the exact trajectory of a particle subject to random forces we can only predict the probability of the particle being at a certain location in space at a given time. Following these considerations, he derived an equation describing the evolution of the density function of particle motion in the overdamped regime: When inertia (due to small mass or strong damping) is negligible, the motion of a particle exposed to a certain random external force can be described by

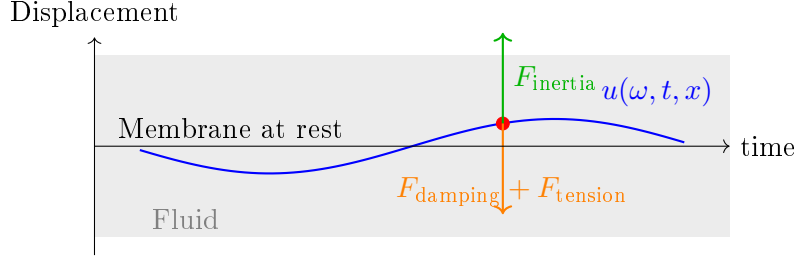


Figure 1: A sketch of how the forces affect the displacement of the membrane giving it its characteristic wave shape. For simplicity, we do not display the effect of the random force.

the Brownian motion. Kramers [2] extended these ideas to incorporate inertia into the description of particle movement in phase space. In particular, he considered the movement of a particle in an external field subject to the random forces of a surrounding medium in temperature equilibrium.

In contrast to Smoluchowski's and Kramers' work, which described the evolution of the density function (the Fokker-Planck equation), we follow the standard mathematical literature on this topic and study a pathwise description where the probabilistic nature of the problem is introduced through a stochastic forcing term ξ . To illustrate this, consider a membrane under tension where $u(\omega, t, x)$ describes the transverse displacement of the membrane at a spatial point x and time t (see Figure 1). Due to the tension a restoring force proportional to $F_{tension}(\omega, t, x) = \Delta u(\omega, t, x)$ acts on the membrane. Furthermore, assume that the membrane lies in some fluid with viscosity γ , hence, by Stokes' law a damping force $F_{damping}(\omega, t, x) = -\gamma \partial_t u(\omega, t, x)$ acts on the membrane. By Newton's second law, the membrane is also subject to an inertial force proportional to the mass m , given by $F_{inertia} = m \partial_t^2 u(\omega, t, x)$. If the molecules constituting the fluid affect the membrane directly (rather than only through the damping force), we would need to model a very high-dimensional system of differential equations. As is standard in thermodynamics, we consider an ensemble of such systems, or in mathematical terms, a sample space Ω . We model the force exerted on the membrane by the molecules as a random force $\xi(\omega, t, x)$ which varies for each system $\omega \in \Omega$. Consequently, the transverse displacement of the membrane $u(\omega, t, x)$ also depends on the realization ω . Balancing these forces via Newton's second law, the transverse displacement of the membrane $u(\omega, t, x)$ can be described by the stochastic damped wave equation:

$$m \partial_t^2 u + \gamma \partial_t u - \Delta u = \xi.$$

Kramers observed that in the case of overdamping (i.e., a large γ factor), the stochastic damped wave equation reduces to a diffusion equation, the equation studied by Smoluchowski years earlier. Thus, for large viscosity γ or equivalently small mass m the transverse displacement of the membrane can be approximated by the heat equation:

$$\gamma \partial_t u - \Delta u = \xi.$$

This limit is the origin of the term SKA. For a more in-depth background on the derivation see the original papers by Smoluchowski [1] and Kramers [2] as well as detailed works on

the larger picture [3]. In Section 2.1 we motivate the validity of the SKA, mathematically.

1.2 The Mean Field Smoluchowski Kramers Approximation

In this thesis, we investigate the SKA with a mean field nonlinearity. This nonlinearity has been studied in the context of the stochastic quantization of the N -component Φ_2^4 model also known as the $O(N)$ linear sigma model. More specifically, [4] and [5] showed that in the limit as $N \rightarrow \infty$, the N -component cubic stochastic damped nonlinear wave equation (SDNLW) and cubic stochastic nonlinear heat equation (SNLH) converge to an SDNLW and an SNLH with a mean field nonlinearity, respectively. As the name "stochastic quantization" suggests, the specific formulation of this problem is motivated by Quantum Field Theory (QFT). Therefore, to provide context, we give a brief introduction to QFT and its relation to the study of stochastic partial differential equations.

QFT extends Schrödinger's quantum mechanical framework to incorporate Einstein's special relativity theory (see [6]). It postulates that forces between particles arise from field equations, and a particle is defined as a state of a field associated with an irreducible representation of the Poincaré group which is the symmetry group of special relativity. For an introduction to QFT see [7]. In order to develop a rigorous mathematical model that incorporates these physical assumptions Wightman [8] published a set of axioms characterizing physically relevant quantum field theories. However, it remains an open problem to show that a mathematical model satisfies all Wightman Axioms for interacting theories in four dimensions (see, for example, the Millennium Problem regarding the Yang-Mills existence and mass gap).

One promising mathematical approach of a probabilistic nature is that of Euclidean Quantum Field Theory (EQFT). EQFT describes the dynamics of quantum fields in terms of probability measures that satisfy the Osterwalder-Schrader (OS) Axioms. This set of Axioms is equivalent to the Wightman Axioms (see Chapter 19 in [9]). The name EQFT originates from the definition of quantum fields via their analytical continuation to imaginary time. This continuation changes the Minkowski spacetime metric (which is invariant under relativistic transformations, namely Lorentz transformations) into the Euclidean metric. Such fields are characterized by a probability measure μ on the space of real distributions $\mathcal{D}'(\mathbb{R}^d)$, with d being the dimension of space-time. For a thorough introduction to EQFT see [9].

The modern program of stochastic quantization was introduced by Parisi and Wu ([10]), originally as a method to evaluate path integrals in gauge theories without the need for gauge-fixing. This approach aims to resolve the problem of sampling from the infinite-dimensional measure μ via the construction of a stochastic differential equation (SDE) that satisfies: (i) μ is the unique invariant measure of this SDE, and (ii) for an initial measure the Law of a solution X_t to this SDE converges to μ . Then, numerical techniques like the Hamiltonian Monte Carlo procedure can be applied to sample from μ , reducing the problem of sampling from an infinite-dimensional space.

In this thesis, we examine the mean field limit of the stochastic quantization of the Hyperbolic Φ_2^4 model, also called the Hyperbolic $O(N)$ linear sigma model (HLSM_N) [5]. The dynamics of this model's state variable, denoted by $u_{N,j}$, are described by an N -system of damped wave equations with a cubic nonlinearity driven by a cylindrical Wiener Process

W_t on $L^2(\mathbb{T}^2)$:

$$\begin{cases} \partial_t^2 u_{N,j} + \partial_t u_{N,j} + (1 - \Delta)u_{N,j} + \frac{1}{N} \sum_{k=1}^N : (u_{N,k})^2 u_{N,j} : = \sqrt{2} dW_t, \\ (u_{N,j}, \partial_t u_{N,j})|_{t=0} = (\phi_{0,N,j}, \phi_{1,N,j}) \end{cases} \quad (1.1)$$

with $1 \leq j \leq N$. Note that $: (u_{N,k})^2 u_{N,j} :$ corresponds to the Wick-renormalization, which we discuss later.

Generalizing the result by [11] (see Remark 1.7 in [5]) this system admits the unique invariant measure

$$d\rho_N(\mathbf{u}_N, \partial_t \mathbf{u}_N) := Z_N^{-1} \exp \left[-\frac{1}{4N} \int_{\mathbb{T}^2} : \left(\sum_{j=1}^N u_{N,j}^2 \right)^2 : dx \right] d\rho^{\otimes N}(\mathbf{u}_N, \partial_t \mathbf{u}_N)$$

with

$$d\rho(u, \partial_t u) := Z^{-1} e^{-\frac{1}{2} \|u\|_{H_x^1}^2} e^{-\frac{1}{2} \|\partial_t u\|_{L_x^2}^2} du d(\partial_t u) \quad (1.2)$$

where Z_N^{-1} and Z^{-1} are appropriate normalization constants. Thus, in the context of stochastic quantization, $d\rho_N$ is the measure that potentially describes an interacting quantum field theory, and (1.1) is the SDE which we want to use to sample from $d\rho_N$. Clearly, the first step is to show that the SDE admits a global solution. Indeed, [5] showed that there exists a unique global solution to (1.1). Furthermore, they showed that this solution converges to the SDMFV:

$$\begin{cases} \partial_t^2 u + \partial_t u + (1 - \Delta)u + \mathbb{E}[:u^2:]u = \sqrt{2} dW_t, \\ (u, \partial_t u)|_{t=0} = (\phi_0, \phi_1), \end{cases} \quad (1.3)$$

with which we are concerned. [4] showed the same type of convergence for the Parabolic $O(N)$ linear sigma model (PLSM_N) to the SMFH:

$$\begin{cases} \partial_t u + (1 - \Delta)u + \mathbb{E}[:u^2:]u = \sqrt{2} dW_t, \\ u|_{t=0} = \phi_0. \end{cases} \quad (1.4)$$

From here, we can make the connection to the mean field SKA. Recent work by [12] proved that the SKA occurs for the SDNLW, specifically, that the SDNLW with an ε -factor coupled to the second-order time derivative converges to the stochastic heat equation as this ε factor converges to zero. This result can be generalized to the HLSM_N. Following this idea, we show the mean field SKA, namely that a solution to the SDNLW,

$$\begin{cases} \varepsilon^2 \partial_t^2 u_\varepsilon + \partial_t u_\varepsilon + (1 - \Delta)u_\varepsilon + \mathbb{E}[:u_\varepsilon^2:]u_\varepsilon = \sqrt{2} dW_t, \\ (u_\varepsilon, \partial_t u_\varepsilon)|_{t=0} = (\phi_0, \phi_1), \end{cases} \quad (1.5)$$

converges to (1.4) as $\varepsilon \rightarrow 0$.

We summarize the relationships between the HLSM_N and the PLSM_N, their mean field limits, and the respective SKAs in Figure 2.

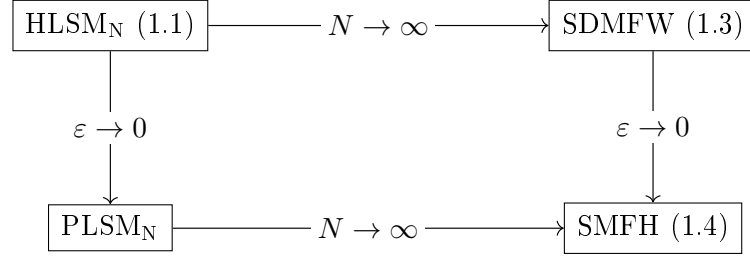


Figure 2: Diagram between SDNLW and SNLH.

1.3 Invariant Dynamics of the Mean Field SKA

Motivated by the stochastic quantization program, the next step is to identify an invariant measure and show its uniqueness. Since we are concerned with the mean field SKA, we are particularly interested in the invariant measure of the SDMFw (1.5) and whether it converges to the invariant measure of the SMFH (1.4) as the mass term converges to zero.

In recent years, there has been considerable progress in identifying invariant measures of SPDEs (for example, [13], [14] [15]) based on the Gibbs measure, which is induced by the conservation of the Hamiltonian dynamics that describe a physical system.

Consider a Hamiltonian system of ODEs on $\mathbb{R}^{2n}$:

$$\begin{aligned}\partial_t q_i &= \frac{\partial H}{\partial p_i} \\ \partial_t p_i &= -\frac{\partial H}{\partial q_i}\end{aligned}$$

for $i \in 1, \dots, n$.

This system gives rise to a divergence-free vector field $X = (\frac{\partial H}{\partial p_1}, \dots, \frac{\partial H}{\partial p_n}, -\frac{\partial H}{\partial q_1}, \dots, -\frac{\partial H}{\partial q_n})$. Then, Liouville's Theorem yields that the Lebesgue measure on $\mathbb{R}^{2n}$ is invariant under the flow generated by X . Furthermore, the Hamiltonian of the system is conserved in time. Hence, we can see that the so-called Gibbs measure

$$d\mu = Z^{-1} e^{-H(p,q)} dp dq$$

for some normalizing constant $Z > 0$, turns out to be invariant under the flow generated by the vector field X . When passing to infinite-dimensional systems, such as the stochastic nonlinear damped wave, or heat equations, we have to change the reference measure, as an infinite-dimensional Lebesgue measure is known to be ill-defined. Typically, we replace it with a Gaussian-like measure.

Previous works [4] and [5] have already established that the invariant measures for the N -component SNLH and the N -component SDNLW ((1.1)) converge to the invariant measures of their respective mean field limits. They show that the mean field limits, that is, the SMFH (1.4) and the SDMFw (1.3), are invariant under the Gibbs measures $(\pi_1)_* \rho$ and ρ , respectively, where π_j is the canonical projection onto the j -th component, for $j \in \{1, 2\}$, and ρ is as defined in (1.2).

In this work, we extend the result in [5] showing that

$$d\rho_\varepsilon(u, \partial_t u) := Z_\varepsilon^{-1} e^{-\frac{1}{2}\|u\|_{H_x^1}^2} e^{-\frac{1}{2}\varepsilon^2\|\partial_t u\|_{L_x^2}^2} du d(\partial_t u) \quad (1.6)$$

is the (conditional) unique invariant measure for (1.5).

Intuitively, if (1.5) converges to (1.4) the solution no longer depends on the second initial value, hence, we expect $(\pi_2)_*\rho_\varepsilon$ to converge to some kind of uniform distribution on the second initial datum. However, a uniform probability measure does not exist on an infinite-dimensional space. Thus, we must treat the $\varepsilon \rightarrow 0$ -behavior of ρ_ε more carefully. In particular, we show that it does not converge weakly to a probability measure, but rather that $(\pi_2)_*\rho_\varepsilon$ converges for continuous bounded test functions that vanish at infinity to the zero measure. This aligns with our intuition that the limiting distribution of ρ_ε no longer depends on the second value as ε tends to 0. Since this result only concerns the second component of ρ_ε it can be directly generalized to the convergence of the invariant measure of the HLSM_N to the invariant measure of the PLSM_N. Thus, summarizing these results, Diagram 2 also applies to the invariant measures of the respective systems.

2 Mild Solution of the Mean Field SKA

This section motivates and defines the concept of (i) a solution to the mean field SKA on the two-dimensional torus $\mathbb{T}^2 := \mathbb{R}^2/(2\pi\mathbb{Z})^2$, as well as (ii) the invariant dynamics of the mean field SKA. As we will see, solutions to the SDNLW are distributions. Consequently, a solution to the mean field SKA must be understood in a distributional sense. In particular, it does not satisfy (1.5) in a classical sense. In addition to motivating the main definitions, we introduce most of the notation used throughout this thesis. Notation specific to a particular problem will be introduced as needed.

2.1 Mathematical Motivation of the SKA

To start mathematically formalizing the SKA motivated in Section 1.1, let $0 < \varepsilon \leq 1$ and consider the following non-homogeneous damped wave equation for some function F :

$$\begin{cases} \varepsilon^2 \partial_t^2 u_\varepsilon + \partial_t u_\varepsilon + (1 - \Delta)u_\varepsilon = F, \\ (u_\varepsilon, \partial_t u_\varepsilon)|_{t=0} = (\phi_0, \phi_1). \end{cases} \quad (2.1)$$

It has been proven in various contexts (for example, see [16], [17], [12]) that the solution u_ε to (2.1) converges to the solution u of the non-homogeneous heat equation:

$$\begin{cases} \partial_t u + (1 - \Delta)u = F, \\ u|_{t=0} = \phi_0, \end{cases} \quad (2.2)$$

as $\varepsilon \rightarrow 0$.

Following the arguments provided in [12], we give a formal argument for why the SKA occurs in this setting. Solving the characteristic polynomial of the homogeneous linear

damped wave equation

$$\begin{cases} \varepsilon^2 \partial_t^2 u_\varepsilon + \partial_t u_\varepsilon + (1 - \Delta) u_\varepsilon = 0, \\ (u_\varepsilon, \partial_t u_\varepsilon)|_{t=0} = (\phi_0, \phi_1) \end{cases}$$

for each frequency $n \in \mathbb{Z}^2$ we obtain the (complex) roots

$$\Lambda_\varepsilon^\pm(n) := \frac{-1 \pm \sqrt{1 - 4\langle n \rangle^2 \varepsilon^2}}{2\varepsilon^2} \quad (2.3)$$

where we denote the Japanese bracket $\langle n \rangle := \sqrt{1 + |n|^2}$. Note that because of the ε -factor in the equation, we must separate two frequency regimes. Recall the Fourier Transform $\mathcal{F}(f)$, also denoted as $\widehat{\mathcal{F}}(f) = \widehat{f}$, and its inverse $\mathcal{F}^{-1}(g)$ on $\mathbb{T}^2$:

$$\begin{aligned} \mathcal{F}(f)(n) &:= \frac{1}{(2\pi)^2} \int_{\mathbb{T}^2} e^{-in \cdot x} f(x) dx \\ \mathcal{F}^{-1}(g)(x) &:= \sum_{n \in \mathbb{Z}^2} e^{in \cdot x} g(n). \end{aligned}$$

Define the sharp projections:

$$\mathbf{P}_\varepsilon^{\text{low}} f := \mathcal{F}^{-1}(\mathbf{1}_{\langle n \rangle \leq (2\varepsilon)^{-1}} \widehat{f}(n)) \quad \text{and} \quad \mathbf{P}_\varepsilon^{\text{high}} f := \mathcal{F}^{-1}(\mathbf{1}_{\langle n \rangle > (2\varepsilon)^{-1}} \widehat{f}(n)).$$

Then a solution of the homogeneous damped linear wave equation (DLW) is given by

$$u_\varepsilon(t) = P_\varepsilon(t)(\phi_0, \phi_1) := (\varepsilon^{-2} + \partial_t) \mathcal{D}_\varepsilon(t) \phi_0 + \mathcal{D}_\varepsilon(t) \phi_1 \quad (2.4)$$

where we denote the damped wave propagator

$$\mathcal{D}_\varepsilon(t) := e^{-\frac{t}{2\varepsilon^2}} \left(\frac{\sinh(\lambda_\varepsilon(\nabla)t)}{\lambda_\varepsilon(\nabla)} \mathbf{P}_\varepsilon^{\text{low}} + \frac{\sin(\zeta_\varepsilon(\nabla)t)}{\zeta_\varepsilon(\nabla)} \mathbf{P}_\varepsilon^{\text{high}} \right) \quad (2.5)$$

with coefficients

$$\lambda_\varepsilon(n) := \frac{\sqrt{1 - 4\langle n \rangle^2 \varepsilon^2}}{2\varepsilon^2} \quad \text{and} \quad \zeta_\varepsilon(n) := \frac{\sqrt{4\langle n \rangle^2 \varepsilon^2 - 1}}{2\varepsilon^2}. \quad (2.6)$$

Define the damped wave Duhamel integral operator:

$$\mathcal{I}_\varepsilon(t)(F) := \int_0^t \varepsilon^{-2} \mathcal{D}_\varepsilon(t - t') F(t') dt'. \quad (2.7)$$

Then, by the Duhamel principle, the solution u_ε of (2.1) is given by

$$u_\varepsilon(t) = P_\varepsilon(t)(\phi_0, \phi_1) + \mathcal{I}_\varepsilon(t)(F). \quad (2.8)$$

Note that by the same line of reasoning, a formal solution of the non-homogeneous heat equation (2.2) is given by

$$u(t) = P_0(t)(\phi_0) + \mathcal{I}_0(t)(F),$$

where we define the heat propagator and the heat Duhamel integral operator:

$$P_0(t)(f) := e^{(\Delta-1)t}(f). \quad (2.9)$$

$$\mathcal{I}_0(t)(F) := \int_0^t P_0(t-t')F(t') dt'. \quad (2.10)$$

To see that u_ε formally converges to u as $\varepsilon \rightarrow 0$, define, for $N \in \mathbb{N}$, the frequency projection:

$$\mathbf{P}_N f := \mathcal{F}^{-1}(\mathbf{1}_{\langle n \rangle \leq N} \widehat{f}(n)). \quad (2.11)$$

Rearranging the characteristic roots in (2.3) and examining their Taylor expansion, we observe that

$$\begin{aligned} \Lambda_\varepsilon^+(n) &= \frac{-2\langle n \rangle^2}{1 + \sqrt{1 - 4\langle n \rangle^2 \varepsilon^2}} = -\langle n \rangle^2 + \mathcal{O}(\langle n \rangle^4 \varepsilon^2) \rightarrow -\langle n \rangle^2, \\ \Lambda_\varepsilon^-(n) &= \frac{-2\langle n \rangle^2}{1 - \sqrt{1 - 4\langle n \rangle^2 \varepsilon^2}} \rightarrow -\infty, \end{aligned}$$

in the regime $\langle n \rangle = o(\varepsilon^{-\frac{1}{2}})$ as $\varepsilon \rightarrow 0$. Then, for $0 < \theta \ll 1$, at a formal level, as $\varepsilon \rightarrow 0$, we expect

$$\varepsilon^{-2} \mathcal{D}_\varepsilon(t) \mathbf{P}_{\varepsilon^{-\frac{1}{2}+\theta}} = \frac{e^{\Lambda_\varepsilon^+(\nabla)t} - e^{\Lambda_\varepsilon^-(\nabla)t}}{\sqrt{1 - 4\langle \nabla \rangle^2 \varepsilon^2}} \mathbf{P}_{\leq \varepsilon^{-\frac{1}{2}+\theta}} \rightarrow P_0(t),$$

and,

$$\begin{aligned} \partial_t \mathcal{D}_\varepsilon(t) \mathbf{P}_{\varepsilon^{-\frac{1}{2}+\theta}} &= \left(\left(1 - \frac{1}{\sqrt{1 - 4\langle \nabla \rangle^2 \varepsilon^2}} \right) \frac{e^{\Lambda_\varepsilon^+(\nabla)t}}{2} + \left(1 + \frac{1}{\sqrt{1 - 4\langle \nabla \rangle^2 \varepsilon^2}} \right) \frac{e^{\Lambda_\varepsilon^-(\nabla)t}}{2} \right) \mathbf{P}_{\varepsilon^{-\frac{1}{2}+\theta}} \\ &\rightarrow 0. \end{aligned}$$

Hence, a solution of the non-homogeneous linear damped wave equation (2.1) converges formally to a solution of the non-homogeneous heat equation (2.2) as $\varepsilon \rightarrow 0$.

2.2 The Mild Solution of the Mean Field SKA

The next step in understanding the mean field SKA is to analyze the stochastic forcing term dW_t . In this thesis, we work with a fixed probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with a normal filtration $\{F_t\}_{t \in \mathbb{R}^+}$.

We consider a specific space-time white noise given by the distributional derivative with respect to time of the cylindrical Wiener process. Define

$$\mathcal{J} := (\mathbb{Z}_+ \times \{0\}) \cup (\mathbb{Z} \times \mathbb{Z}_+) \cup \{(0, 0)\}. \quad (2.12)$$

We define the cylindrical Wiener process with respect to the $L^2(\mathbb{T}^2)$ Fourier basis $\{e^{in \cdot x}\}_{n \in \mathbb{Z}^2}$ as follows:

Definition 2.1 (Cylindrical Wiener process on $L^2(\mathbb{T}^2)$). *Let $\{\beta_n\}_{n \in \mathcal{J}}$ be a family of mutually independent Brownian motions on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where β_0 is real-valued*

and β_n are complex-valued for $n \neq (0,0)$. Set $\beta_{-n} := \overline{\beta_n}$ for $n \in \mathcal{J}$. On $\mathbb{T}^2$ the cylindrical Wiener process is given by

$$W(t) := \beta_0(t) + \frac{1}{\sqrt{2}} \sum_{n \in \mathbb{Z}^2 \setminus \{0\}} \beta_n(t) e^{in \cdot x}. \quad (2.13)$$

Note that $\text{Var}(\beta_n(t)) = 2t$ for $n \in \mathbb{Z}^2 \setminus \{0\}$, while $\text{Var}(\beta_0(t)) = t$.

In Section 2.1 we motivated what a mild solution of (1.5) is, namely, if it had been well-defined, we would call u_ε a mild solution of (1.5) if

$$u_\varepsilon(t) = P_\varepsilon(t)(\phi_0, \phi_1) - \mathcal{I}_\varepsilon(t) (\mathbb{E}[u_\varepsilon^2] u_\varepsilon) + \Psi_\varepsilon(t) \quad (2.14)$$

where we define the stochastic convolution

$$\Psi_\varepsilon(t) := \sqrt{2} \int_0^t \varepsilon^{-2} \mathcal{D}_\varepsilon(t-t') dW(t'). \quad (2.15)$$

However, the nonlinear term in (2.14) is ill-defined because the variance of the stochastic convolution diverges. The computations in Equation (3.15) in Section 3 imply

$$\sigma_N(t) := \mathbb{E}[(\mathbf{P}_N \Psi_\varepsilon(x, t))^2] = 2 \sum_{\langle n \rangle \leq N} \int_0^t (\varepsilon^{-2} D_\varepsilon(t-t'))^2 dt' \sim \log N \quad (2.16)$$

uniformly in t , ε and x . We therefore expect $\mathbb{E}[\Psi_\varepsilon^2]$ to be ill-defined, implying that $\mathbb{E}[u_\varepsilon^2]$ is likewise ill-defined. To avoid this problem, we consider a renormalized version of this problem. Following the standard approach in the literature (see [18]), we consider the Da Prato-Debussche decomposition

$$u_\varepsilon = v_\varepsilon + \Psi_\varepsilon \quad (2.17)$$

where Ψ_ε is defined as in (2.15) and v_ε is the residual term $v_\varepsilon = u_\varepsilon - \Psi_\varepsilon$. Formally, v_ε solves the deterministic nonlinear wave equation:

$$\varepsilon^2 \partial_t^2 v_\varepsilon + \partial_t v_\varepsilon + (1 - \Delta) v_\varepsilon = -(\mathbb{E}[v_\varepsilon^2] + \mathbb{E}[\Psi_\varepsilon^2] + 2\mathbb{E}[v_\varepsilon \Psi_\varepsilon])(v_\varepsilon + \Psi_\varepsilon). \quad (2.18)$$

To avoid the singular variance of Ψ_ε we replace Ψ_ε^2 by its Wick-ordered counterpart

$$:\Psi_\varepsilon^2: := \lim_{N \rightarrow \infty} H_2(\mathbf{P}_N \Psi_\varepsilon, \sigma_N) = \lim_{N \rightarrow \infty} (\mathbf{P}_N \Psi_\varepsilon)^2 - \mathbb{E}[(\mathbf{P}_N \Psi_\varepsilon)^2]. \quad (2.19)$$

Later, in Proposition 3.7, we will see that (2.19) is well-defined, which implies that $\mathbb{E}[:\Psi_\varepsilon^2:] = 0$.

Now, we can expand (2.14) in terms of v_ε and Ψ_ε , replacing $\mathbb{E}[\Psi_\varepsilon^2]$ with $\mathbb{E}[:\Psi_\varepsilon^2:]$ to state a well-defined problem.

Definition 2.2 (Mild Solution of the SDMFV). *Let $\delta \in \mathbb{R}$, $\varepsilon \in (0, 1]$ and define the product Sobolev space*

$$\mathcal{H}^\delta(\mathbb{T}^2) := H^\delta(\mathbb{T}^2) \times H^{\delta-1}(\mathbb{T}^2).$$

Given initial data $(\phi_0, \phi_1) \in \mathcal{H}^\delta(\mathbb{T}^2)$, a mild solution of the SDMFW is defined by $u_\varepsilon = v_\varepsilon + \Psi_\varepsilon$ where Ψ_ε is defined as in (2.15) and v_ε satisfies the Residual SDMFW:

$$\begin{cases} v_\varepsilon(t) = P_\varepsilon(t)(\phi_0, \phi_1) - \mathcal{I}_\varepsilon(t) \left((\mathbb{E}[v_\varepsilon^2] + 2\mathbb{E}[v_\varepsilon \Psi_\varepsilon])(v_\varepsilon + \Psi_\varepsilon) \right) \\ \partial_t v_\varepsilon(t) = \partial_t P_\varepsilon(t)(\phi_0, \phi_1) - \partial_t \mathcal{I}_\varepsilon(t) \left((\mathbb{E}[v_\varepsilon^2] + 2\mathbb{E}[v_\varepsilon \Psi_\varepsilon])(v_\varepsilon + \Psi_\varepsilon) \right) \\ (v_\varepsilon, \partial_t v_\varepsilon)|_{t=0} = (\phi_0, \phi_1) \end{cases} \quad (2.20)$$

in an appropriate space.

Analogously, we define a solution of the SMFH as follows:

Definition 2.3 (Mild Solution of the SMFH). *Let $\delta \in \mathbb{R}$. Given an initial datum $\phi_0 \in H^\delta(\mathbb{T}^2)$ a mild solution of the SMFH is defined by $u_0 = v_0 + \Psi_0$ where we define*

$$\Psi_0(t) := \sqrt{2} \int_0^t P_0(t-t') dW(t') \quad (2.21)$$

and v_0 satisfies

$$\begin{cases} v_0(t) = P_0(t)(\phi_0) - \mathcal{I}_0(t) \left((\mathbb{E}[v_0^2] + 2\mathbb{E}[v_0 \Psi_0])(v_0 + \Psi_0) \right) \\ (v_0)|_{t=0} = \phi_0. \end{cases} \quad (2.22)$$

Finally, we give a general definition of the mean field SKA in our setting:

Definition 2.4 (The Mean Field SKA). *Let $\delta \in \mathbb{R}$. Given initial data $(\phi_0, \phi_1) \in \mathcal{H}^\delta(\mathbb{T}^2)$. Let u_0 satisfy Definition 2.3 with respect to the initial datum ϕ_0 . We say that a family $\{u_\varepsilon\}_{\varepsilon \in (0,1]}$ is a mean field SKA of u_0 , if for all $\varepsilon \in (0,1]$ u_ε satisfies Definition 2.2 with respect to the initial data (ϕ_0, ϕ_1) and $u_\varepsilon \rightarrow u_0$ as $\varepsilon \rightarrow 0$ in an appropriate sense.*

2.3 Invariant Dynamics of the Mean Field SKA

In this section, we discuss the invariant dynamics for the mean-field SKA. Suppose the initial data for (1.5) is sampled from an invariant distribution. The nonlinear dynamics then reduce to linear dynamics:

$$0 = \varepsilon^2 \partial_t^2 u_\varepsilon + \partial_t u_\varepsilon + (1 - \Delta) u_\varepsilon + \mathbb{E}[u_\varepsilon^2] u_\varepsilon - \sqrt{2} dW_t, \quad (2.23)$$

$$= \varepsilon^2 \partial_t^2 u_\varepsilon + \partial_t u_\varepsilon + (1 - \Delta + \mathbb{E}[u_\varepsilon(0)^2]) u_\varepsilon - \sqrt{2} dW_t. \quad (2.24)$$

To formalize the invariant dynamics for the mean field SKA, we make the following observations:

1. Suppose $\mathbf{u}_\varepsilon := (u_\varepsilon, \partial_t u_\varepsilon)$ is a solution to (2.23) with invariant measure μ . Then the pushforward of $\mathbb{P}$ under the flow $\mathbf{u}_\varepsilon$, denoted as $(\mathbf{u}_\varepsilon)_* \mathbb{P}$, and μ must be supported on the same spaces. [19]¹ showed that while the damped wave propagator conserves the spaces $\mathcal{H}^s(\mathbb{T}^2)$ for $s \in \mathbb{R}$, it loses regularity in spaces $W^{p,s}(\mathbb{T}^2) \times W^{p,s-1}(\mathbb{T}^2)$ where $p \neq 2$. Consequently, the invariant measure must be supported on the space $\mathcal{H}^s(\mathbb{T}^2)$ for some $s \in \mathbb{R}$.

¹[19] demonstrated this result specifically for the wave equation. However, because both equations exhibit identical spatial behavior, this result generalizes directly to the damped wave equation.

2. The cylindrical Wiener process, defined in (2.13), has strict negative regularity. Since the damped wave propagator conserves, but does not improve, regularity, any invariant measure must be supported on a space of negative regularity. Combining this observation with the first, we deduce that any invariant measure will be supported on $\mathcal{H}^{-\sigma}(\mathbb{T}^2)$ for $\sigma > 0$.
3. From the second observation, it immediately follows that we must introduce a renormalization $:u_\varepsilon^2:$ to ensure well-definedness. Under invariant dynamics μ , we expect $\mathbb{E}_\mu[:u_\varepsilon(t)^2:] = \mathbb{E}_\mu[:u_\varepsilon(0)^2:]$. Applying the Wick-renormalization with respect to Ψ_ε as we did in Section 2.2, we observe

$$\begin{aligned}\mathbb{E}_\mu[:u_\varepsilon(t)^2:] &= \mathbb{E}_\mu \left[\lim_{N \rightarrow \infty} (P_N u_\varepsilon(t))^2 - \mathbb{E}[(P_N \Psi(t))^2] \right] \\ &= \mathbb{E}_\mu \left[\lim_{N \rightarrow \infty} (P_N u_\varepsilon(0))^2 - \mathbb{E}[(P_N \Psi(t))^2] \right] \\ &\neq \mathbb{E}_\mu \left[\lim_{N \rightarrow \infty} (P_N u_\varepsilon(0))^2 - \mathbb{E}[(P_N \Psi(0))^2] \right] = \mathbb{E}_\mu[:u_\varepsilon(0)^2:]\end{aligned}$$

Note that the term on the right-hand side is likely not even well-defined, as $\Psi(0) = 0$. However, for any $t > 0$, we have $\mathbb{E}[(P_N \Psi(t))^2] \sim \sum_{\langle n \rangle \leq N} \langle n \rangle^{-2}$. In particular, to make this result comparable to both the previous section and existing literature (see, for example, [11]), we consider the renormalization with respect to $\sum_{\langle n \rangle \leq N} \langle n \rangle^{-2}$.

Based on these observations, we define the Wick renormalization as

$$:u^2: := \lim_{N \rightarrow \infty} (P_N u)^2 - \sum_{\langle n \rangle \leq N} \langle n \rangle^{-2} \quad (2.25)$$

whenever this limit exists in $H^{-\sigma}(\mathbb{T}^2)$, and we can thus define the invariant dynamics for the SDMFw:

Definition 2.5 (Invariant Dynamics of the SDMFw). *Let $\sigma > 0$, and denote $\Lambda := 1 - \Delta$, $M_\mu := \mathbb{E}_\mu[:u^2:]$, and $L := \Lambda + M_\mu$. Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, we say that $\mu \in \mathcal{P}(\mathcal{H}^{-\sigma}(\mathbb{T}^2))$ is an invariant measure of the SDMFw if there exists a mild solution to*

$$\begin{cases} dX(t) := d \begin{pmatrix} u_\varepsilon(t) \\ \partial_t u_\varepsilon(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\varepsilon^{-2}L & -\varepsilon^{-2} \end{pmatrix} X(t)dt + \sqrt{2} \begin{pmatrix} 0 \\ \varepsilon^{-2} \end{pmatrix} dW_t \\ (u_\varepsilon, \partial_t u_\varepsilon)|_{t=0} \sim \mu, \end{cases} \quad (2.26)$$

with $\text{Law}(X(t)) = \mu$ for all $t \geq 0$.

Assuming a well-defined solution flow to (2.26) exists, note that any invariant measure to (2.26) must be a Gaussian measure on $\mathcal{H}^{-\sigma}(\mathbb{T}^2)$. This is a standard consequence of the linearity of the equation, as established in [20, Theorem 11.7]. Following the intuition gained from finite-dimensional systems, an invariant measure is the Gaussian measure $\mathcal{N}\left(0, \begin{pmatrix} L^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}\right)$. This will be rigorously established in Chapter 6. Consequently, we

can directly identify one possible invariant measure $\rho_\varepsilon := \mathcal{N}\left(0, \begin{pmatrix} \Lambda^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}\right)$. A direct computation shows $\mathbb{E}_{\rho_\varepsilon}[:u^2:] = 0$. Thus, (2.26) reduces to the linear equation of the SDMFW (2.26) with $M_\mu = 0$, for which ρ_ε is the unique invariant measure. This specific setting was discussed in [5] for $\varepsilon = 1$. We extend this result, showing that ρ_ε is the unique invariant measure for (2.26) under the assumption that $\mathbb{E}_\mu[:u^2:] \in H^{-\sigma}(\mathbb{T}^2)$ for $\sigma \in (0, \frac{1}{2})$. The same line of reasoning shows that $\rho_0 := \mathcal{N}(0, \Lambda^{-1})$ is an invariant measure for the SMFH (discussed in [4]). As for the pathwise convergence in Definition 2.4, we expect the invariant measure of the SDMFW to converge to the invariant measure of the SMFH. However, examining the density function of the invariant measure

$$d\rho_\varepsilon = Z_\varepsilon^{-1} e^{-\frac{1}{2}\|u\|_{H_x^1}^2 - \frac{1}{2}\varepsilon^2\|\partial_t u\|_{L_x^2}^2} dud(\partial_t u),$$

it is evident that this probability measure will not converge to any probability measure as $\varepsilon \rightarrow 0$. To see this, it suffices to look at the truncated finite-dimensional probability measure. Let

$$\begin{aligned} \pi_2 : \mathcal{H}^{-\sigma}(\mathbb{T}^2) &\rightarrow H^{-1-\sigma}(\mathbb{T}^2) \\ (u_1, u_2) &\rightarrow u_2 \end{aligned}$$

be the canonical projection onto the second component. Then $(\mathbf{P}_N \pi_2)_* \rho_\varepsilon$ corresponds to a finite-dimensional centered complex-valued Gaussian random variable with a diagonal covariance matrix where each diagonal entry equals ε^{-2} . In particular, the family of probability measures $(\mathbf{P}_N \pi_2)_* \rho_\varepsilon$ is not tight in $H^{-1-\sigma}(\mathbb{T}^2)$ (indeed, it is not even tight in its respective finite-dimensional subspace of $H^{-1-\sigma}(\mathbb{T}^2)$). Hence, by Prokhorov's theorem, this family does not converge weakly to any probability measure on $H^{-1-\sigma}(\mathbb{T}^2)$ as $\varepsilon \rightarrow 0$. However, this behavior aligns with our intuition: in the limit $\varepsilon \rightarrow 0$, the probability measure ρ_ε loses its dependence on the second component. While this cannot occur in the standard sense of weak convergence of probability measures (where mass is preserved), it can be described in a weaker sense. Let $(X, \|\cdot\|)$ be a normed space. We define the set of continuous bounded functions that vanish at infinity:

$$C_0(X) := \{f \in C_b(X) \mid \lim_{\|u\| \rightarrow \infty} f(u) = 0 \text{ for all } u \in X\}.$$

Thus, we resort to a notion of convergence that allows mass to escape, which better suits this setting.

Definition 2.6 (Mean field SKA of Invariant Measures). *Let $\sigma > 0$ and $\rho_0 := \mathcal{N}(0, \Lambda^{-1})$. We say that $\rho_\varepsilon \rightarrow \rho_0$ as $\varepsilon \rightarrow 0$ if for all $f \in C_0(H^{-1-\sigma}(\mathbb{T}^2))$*

$$\int_{H^{-1-\sigma}(\mathbb{T}^2)} f(u) d((\pi_2)_* \rho_\varepsilon)(u) \rightarrow 0$$

as $\varepsilon \rightarrow 0$.

Remark 2.7. The more classical notion of vague convergence, defined via integration against continuous, compactly supported functions, is trivial in this setting. Because $H^{-1-\sigma}(\mathbb{T}^2)$ is

an infinite-dimensional Banach space, the space of continuous, compactly supported functions on it, $C_c(H^{-1-\sigma}(\mathbb{T}^2))$, contains only the zero function. Assume, toward contradiction, that there exists a non-zero $f \in C_c(H^{-1-\sigma}(\mathbb{T}^2))$. By continuity, there exists a closed ball $\overline{B_r(x)}$ of radius $r > 0$ on which f does not vanish. Consequently, this closed ball is contained within the support of f . Since the support is compact and the ball is a closed subset, $\overline{B_r(x)}$ must itself be compact. However, Riesz's lemma implies that closed balls in a Banach space are compact if and only if the space is finite-dimensional. This yields a contradiction, confirming that $f = 0$.

In contrast, the set $C_0(H^{-1-\sigma}(\mathbb{T}^2))$ is not empty, since, for example, $e^{-\|u\|_{H_x^{-1-1\sigma}}} \in C_0(H^{-1-\sigma}(\mathbb{T}^2))$.

3 Preliminary Results and Well-Definedness of the Stochastic Convolution

In this section, we recall and extend some known results that are important to show that the problem in Definition 2.2 is globally well-posed for every $\varepsilon \in (0, 1]$ in a suitable function space. Furthermore, a goal of this section is to show that the stochastic convolutions defined in (2.15) and (2.21) are well-defined.

3.1 Functional analytical estimates

We start by recording well established functional analytical estimates on Sobolev spaces. These will be important when we analyze nonlinear terms like $(\mathbb{E}[v_\varepsilon^2] + 2\mathbb{E}[v_\varepsilon \Psi_\varepsilon])(v_\varepsilon + \Psi_\varepsilon)$. For a proof and more general results, see [21], Proposition 1 and 2.

Lemma 3.1. *Let $\sigma > 0$.*

1. *Let $1 < p_1, p_2, q_1, q_2 \leq \infty$ and $\frac{1}{2} < r < \infty$ be such that $\frac{1}{r} = \frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{p_2} + \frac{1}{q_2}$ and $\sigma > \frac{2}{r} - 2$. Then we have*

$$\|fg\|_{W^{\sigma,r}} \leq \|f\|_{W^{\sigma,p_1}} \|g\|_{L^{q_1}} + \|f\|_{L^{p_2}} \|g\|_{W^{\sigma,q_2}}.$$

2. *Let $1 < p \leq \infty$ and $1 < q, r < \infty$ be such that $\frac{\sigma}{2} \geq \frac{1}{p} + \frac{1}{q} - \frac{1}{r}$ and $q, r' \geq p'$ where p' and r' denote the Hölder conjugates. Then we have*

$$\|fg\|_{W^{-\sigma,r}} \lesssim \|f\|_{W^{-\sigma,p}} \|g\|_{W^{\sigma,q}}.$$

The next estimate is a standard result for discrete convolutions.

Lemma 3.2. *Let $m \in \mathbb{Z}^2$ and $\alpha \in (1, 2)$. Then we have the bound*

$$S(m) := \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-\alpha} \langle n+m \rangle^{-\alpha} \lesssim \langle m \rangle^{2-2\alpha}.$$

If $\alpha = 2$, then we have the bound

$$S_2(m) := \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2} \langle n+m \rangle^{-2} \lesssim \langle m \rangle^{-2} \ln(1 + |m|^2).$$

Proof. We partition the domain of summation into three regions: $R_1 := \{n \in \mathbb{Z}^2 : |n| \leq \frac{|m|}{2}\}$, $R_2 := \{n \in \mathbb{Z}^2 : |n + m| \leq \frac{|m|}{2}\}$, and $R_3 := \{n \in \mathbb{Z}^2 : |n| > \frac{|m|}{2} \text{ and } |n + m| > \frac{|m|}{2}\}$. Since $\mathbb{Z}^2 \subset R_1 \cup R_2 \cup R_3$, establishing the bound on each of these three subregions is sufficient to prove the statement for the entire sum.

The bound for R_1 follows by applying the reverse triangle inequality. For any $n \in R_1$, we have $|n| \leq \frac{|m|}{2}$, which implies:

$$|n + m| \geq |m| - |n| \geq |m| - \frac{|m|}{2} = \frac{|m|}{2}.$$

Because $\alpha > 0$, this implies $\langle n + m \rangle^{-\alpha} \lesssim \langle m \rangle^{-\alpha}$. Approximating the remaining sum by a polar integral and utilizing the fact that $2 - \alpha > 0$ for $\alpha \in (1, 2)$, we obtain

$$\begin{aligned} S_{R_1}(m) &= \sum_{n \in R_1} \langle n \rangle^{-\alpha} \langle n + m \rangle^{-\alpha} \lesssim \langle m \rangle^{-\alpha} \sum_{|n| \leq \frac{|m|}{2}} \langle n \rangle^{-\alpha} \\ &\lesssim \langle m \rangle^{-\alpha} \int_0^{\frac{|m|}{2}} (1 + r^2)^{-\frac{\alpha}{2}} r \, dr = \langle m \rangle^{-\alpha} \frac{1}{2 - \alpha} \left[(1 + r^2)^{1 - \frac{\alpha}{2}} \right]_0^{\frac{|m|}{2}} \\ &\leq \langle m \rangle^{-\alpha} \langle m \rangle^{2 - 2\alpha} = \langle m \rangle^{2 - 2\alpha}. \end{aligned} \quad (3.1)$$

The bound in region R_2 follows by shifting the summation index to $n' = n + m$.

To show the bound in the final region R_3 , we apply Young's product inequality

$$S_{R_3}(m) \leq \frac{1}{2} \sum_{n \in R_3} \langle n \rangle^{-2\alpha} + \frac{1}{2} \sum_{n \in R_3} \langle n + m \rangle^{-2\alpha}.$$

By symmetry, it suffices to bound the first term. Comparing the sum to a polar integral and noting that $1 - \alpha < 0$, we obtain

$$\sum_{n \in R_3} \langle n \rangle^{-2\alpha} \leq \sum_{|n| > \frac{|m|}{2}} \langle n \rangle^{-2\alpha} \lesssim \int_{\frac{|m|}{2}}^{\infty} (1 + r^2)^{-\alpha} r \, dr = \frac{1}{2 - 2\alpha} \left[(1 + r^2)^{1 - \alpha} \right]_{\frac{|m|}{2}}^{\infty} \lesssim \langle m \rangle^{2 - 2\alpha}.$$

The proof for the bound of S_2 follows analogously, but the estimate in (3.1) changes to

$$\begin{aligned} S_{2R_1}(m) &= \sum_{n \in R_1} \langle n \rangle^{-2} \langle n + m \rangle^{-2} \lesssim \langle m \rangle^{-2} \int_0^{\frac{|m|}{2}} (1 + r^2)^{-1} r \, dr = \frac{1}{2} \langle m \rangle^{-2} \left[\ln(1 + r^2) \right]_0^{\frac{|m|}{2}} \\ &\leq \langle m \rangle^{-2} \ln(1 + |m|^2). \end{aligned}$$

This yields the desired estimate. $\square$

3.2 Propagator estimates

Following the work in [12], we record important estimates for the propagator $\mathcal{D}_\varepsilon$ and the operators P_0 , $\mathcal{I}_0$, P_ε , and $\mathcal{I}_\varepsilon$ defined in (2.5), (2.9), (2.10), (2.4), and (2.7), respectively.

Lemma 3.3 ([12] Lemma 3.4). *Fix $0 < \theta \ll 1$. We have the following bounds*

$$\varepsilon^{-2}|\widehat{\mathcal{D}}_\varepsilon(n, t)| \lesssim_\theta \begin{cases} e^{-\theta t \langle n \rangle^2} & \text{for } \langle n \rangle \leq (1 + \theta)(2\varepsilon)^{-1} \\ e^{-\frac{t}{2\varepsilon^2}} \varepsilon^{-1} \langle n \rangle^{-1} & \text{otherwise} . \end{cases} \quad (3.2)$$

$$|\partial_t \widehat{\mathcal{D}}_\varepsilon(n, t)| \lesssim_\theta \begin{cases} e^{-\theta t \langle n \rangle^2} & \text{for } \langle n \rangle \leq (1 + \theta)(2\varepsilon)^{-1} \\ e^{-\frac{t}{2\varepsilon^2}} & \text{otherwise} . \end{cases} \quad (3.3)$$

Lemma 3.4 ([12] Lemma 3.6). *Fix $0 < \theta \ll 1$. The following estimate holds*

$$|\varepsilon^{-2} \widehat{\mathcal{D}}_\varepsilon(n, t) - e^{-t \langle n \rangle^2}| \mathbf{1}_{\langle n \rangle \lesssim \varepsilon^{-1+\theta}} \lesssim_\theta e^{-\frac{t}{2\varepsilon^2}} + \varepsilon^{2\theta} e^{-\frac{t}{2} \langle n \rangle^2}, \quad (3.4)$$

for any $t \geq 0$.

The following Lemma is essential in bounding the propagators. Since we are concerned with the SKA, it is important to track the ε -dependence carefully. However, estimates (3.7) and (3.8) we will only need for fixed $\varepsilon \in (0, 1]$. Consequently, for once we suppress the exact ε -dependence in the estimate.

Lemma 3.5. *Let f be any function. The following inequalities hold.*

1. *Let $\sigma \in \mathbb{R}$, $0 \leq \gamma \leq 1$, and $t > 0$. Then*

$$\|\mathcal{D}_\varepsilon(t)f\|_{H_x^\sigma} \lesssim \varepsilon^{2-\gamma} \|f\|_{H_x^{\sigma-\gamma}} \quad (3.5)$$

with an implicit constant independent of $\varepsilon > 0$ and $t > 0$.

2. *Let $\sigma \in \mathbb{R}$. Then*

$$\sup_{\varepsilon, t > 0} \|\partial_t \mathcal{D}_\varepsilon(t)f\|_{H_x^\sigma} \lesssim \|f\|_{H_x^\sigma}. \quad (3.6)$$

3. *Let $\sigma \in \mathbb{R}$. Then*

$$\sup_{t > 0} \|\partial_t^2 \mathcal{D}_\varepsilon(t)f\|_{H_x^{\sigma-1}} \lesssim_\varepsilon \|f\|_{H_x^\sigma}. \quad (3.7)$$

4. *Let $\sigma \in \mathbb{R}$. Then*

$$\sup_{t > 0} \|\partial_t^3 \mathcal{D}_\varepsilon(t)f\|_{H_x^{\sigma-2}} \lesssim_\varepsilon \|f\|_{H_x^\sigma}. \quad (3.8)$$

Proof. See [12] Lemma 3.7 (iii) and (iv) for the proof of (3.5) and (3.6). To prove (3.7) we focus on the low frequency regime $\langle n \rangle \leq (2\varepsilon)^{-1}$ as the high frequency regime follows analogously. Fix $\varepsilon \in (0, 1]$, let $n \in \mathbb{Z}^2$, and $t \in \mathbb{R}^+$. Then we directly compute (for $\langle n \rangle \leq (2\varepsilon)^{-1}$)

$$\partial_t^2 \widehat{D}_\varepsilon(t, n) = -(2\varepsilon)^{-2} \partial_t \widehat{D}_\varepsilon(t, n) - (2\varepsilon)^{-2} e^{-\frac{t}{2\varepsilon^2}} \cosh(\lambda_\varepsilon(n)t) + \lambda_\varepsilon^2(n) \widehat{D}_\varepsilon(t, n).$$

Note that in this regime $\lambda_\varepsilon(n) \leq (2\varepsilon^2)^{-1}$. Expanding the hyperbolic cosine reveals that $e^{-\frac{t}{2\varepsilon^2}} \cosh(\lambda_\varepsilon(n)t) \leq 1$ for all $t \geq 0$, meaning it acts as a Fourier multiplier bounded

uniformly by 1 (just like $\cos(\cdot)$ and $\sin(\cdot)$). From this, it follows that

$$\|(2\varepsilon)^{-2}e^{-\frac{t}{2\varepsilon^2}}\cosh(\lambda_\varepsilon(\nabla)t)f\|_{H_x^{\sigma-1}} \lesssim_\varepsilon \|f\|_{H_x^{\sigma-1}} \lesssim_\varepsilon \|f\|_{H_x^\sigma} \quad (3.9)$$

As ε is fixed, $|\lambda_\varepsilon(n)| \sim \langle n \rangle$, which implies $\|\lambda_\varepsilon(\nabla)f\|_{H_x^{\sigma-1}} \lesssim_\varepsilon \|f\|_{H_x^\sigma}$. Combined with (3.9), (3.5) and (3.6), this bound yields

$$\sup_{t>0} \|\partial_t^2 D_\varepsilon(t)f\|_{H_x^{\sigma-1}} \lesssim_\varepsilon \|f\|_{H_x^\sigma}.$$

Note that we lose one regularity due to the $\lambda_\varepsilon^2(n)\widehat{D}_\varepsilon(t,n)$ term.

(3.8) follows from the same arguments, (3.7), and the fact that (for the low frequency regime $\langle n \rangle \leq (2\varepsilon)^{-1}$)

$$\begin{aligned} \partial_t^3 \widehat{D}_\varepsilon(t,n) &= - (2\varepsilon)^{-2} \partial_t^2 \widehat{D}_\varepsilon(t,n) - (2\varepsilon)^{-2} \lambda_\varepsilon^2(n) \widehat{D}_\varepsilon(t,n) + (2\varepsilon)^{-4} e^{-\frac{t}{2\varepsilon^2}} \cosh(\lambda_\varepsilon(n)t) \\ &\quad + \lambda_\varepsilon^2(n) \partial_t \widehat{D}_\varepsilon(t,n). \end{aligned}$$

Note that we lose two orders of regularity due to the $\lambda_\varepsilon^2(n)\partial_t \widehat{D}_\varepsilon(t,n)$ term. $\square$

Proposition 3.6 ([12] Proposition 3.1). *Let $\sigma \in \mathbb{R}$ and $T > 0$. For any functions $\phi_0, \phi_1 : \mathbb{T}^2 \rightarrow \mathbb{R}$ and $F : [0, 1] \times \mathbb{T}^2 \rightarrow \mathbb{R}$ we have the following bounds*

$$\|P_\varepsilon(t)(\phi_0, \phi_1)\|_{C_{\varepsilon,T}H_x^\sigma} \lesssim \|(\phi_0, \phi_1)\|_{\mathcal{H}_x^\sigma}, \quad (3.10)$$

$$\|\mathcal{I}_\varepsilon(t)(F(\varepsilon, \cdot))\|_{C_{\varepsilon,T}H_x^\sigma} \lesssim T^{\frac{1}{2}} \|F\|_{C_{\varepsilon,T}H_x^{\sigma-1}}, \quad (3.11)$$

Moreover, for any $0 < \theta \ll 1$, we have

$$\|(P_\varepsilon - P_0)(\phi_0, \phi_1)\|_{C_T H_x^\sigma} \lesssim \varepsilon^{\frac{\theta}{2}} \|(\phi_0, \phi_1)\|_{\mathcal{H}_x^{\sigma+\theta}}, \quad (3.12)$$

$$\|(\mathcal{I}_\varepsilon - \mathcal{I}_0)(F)\|_{C_T H_x^\sigma} \lesssim T^{\frac{1}{2}} \varepsilon^{\frac{\theta}{2}} \|F\|_{L_T^\infty H_x^{\sigma-1+\theta}}. \quad (3.13)$$

3.3 Well-Definedness of the Stochastic Convolution

Here, we show the well-definedness of the stochastic convolutions Ψ_ε and Ψ_0 defined in (2.15) and (2.21), respectively. The main difficulty, compared to the works of [18] and [12], is that we require a uniform bound in time and in our approximation parameter ε under the $L^p(\Omega)$ -norm. We follow the arguments presented in Proposition 4.1 of [12] but provide additional details. Furthermore, for independent copies of stochastic convolutions Ψ and Ψ' , we establish a uniform bound for the product $\Psi\Psi'$. The strategy is to find a sufficient bound on the variation in time and in ε , which can be extended to a uniform bound in time and in ε under the $L^p(\Omega)$ -norm using an extension of the Garsia-Rodemich-Rumsey Lemma (see Appendix 7.1).

Proposition 3.7. *Fix $p \geq 1$, $T > 0$ and $\sigma > 0$. Let Ψ_ε be as in (2.15) and denote $\Psi_{\varepsilon,N} := \mathbf{P}_N \Psi_\varepsilon$. Then,*

1. the sequence $\{(\varepsilon, t) \mapsto: \Psi_{\varepsilon, N}^j : \}_{N \in \mathbb{N}}$ converges to the process $(\varepsilon, t) \mapsto: \Psi_\varepsilon :$ in $L^p(\Omega; C([0, 1] \times [0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$ where we denote for $j = 1$ the stochastic convolution as defined in (2.15) and for $j = 2$ the Wick-ordered one defined in (2.19). Furthermore, Ψ_0 satisfies (2.21) in $L^p(\Omega; C([0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$. In particular, $\Psi_\varepsilon \rightarrow \Psi_0$ in $L^p(\Omega; C([0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$ as $\varepsilon \rightarrow 0$.
2. Let Ψ'_ε be an independent copy of Ψ_ε . Then $\Psi_\varepsilon \Psi'_\varepsilon$ is well-defined in $L^p(\Omega; C([0, 1] \times [0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$.

Proof. For a proof that $\{(\varepsilon, t) \mapsto \Psi_{\varepsilon, N}\}_{N \in \mathbb{N}}$ converges to the process $(\varepsilon, t) \mapsto \Psi_\varepsilon$ almost surely in $C([0, 1] \times [0, T]; W^{-\sigma, \infty}(\mathbb{T}^2))$ see [12] Proposition 4.1. We extend the proof showing that $\Psi_\varepsilon \in L^p(\Omega; C([0, 1] \times [0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$.

First, we show that the second part can be shown analogously to the first part which is due to the following computation. In the following, we suppress the ε and t dependencies as all estimates are uniform in these parameters. Recall the definitions of the stochastic convolutions (2.21) and (2.15) which are given by

$$\Psi(t, x) = \sqrt{2} \sum_{n \in \mathbb{Z}^2} e^{in \cdot x} \int_0^t F(t - t', n) d\beta_n(t')$$

where we set $F(t, n) := \varepsilon^{-2} \widehat{\mathcal{D}}_\varepsilon(t, n)$ if $\varepsilon > 0$ and $F(t, n) := e^{-t\langle n \rangle^2}$ if $\varepsilon = 0$. Let Ψ' be an independent copy of Ψ :

$$\Psi'(t, x) = \sqrt{2} \sum_{n \in \mathbb{Z}^2} e^{in \cdot x} \int_0^t F(t - t', n) d\beta'_n(t')$$

where β'_n is an independent copy of β_n .

Note, that $\widehat{\Psi}(n)$ and $\widehat{\Psi}'(n)$ are centered independent Gaussian variables (for fixed t). Furthermore, by Itô's Isometry and the cylindrical independence of β_n for different $n \in \mathbb{Z}^2$, we have

$$\mathbb{E}[\widehat{\Psi}(n) \widehat{\Psi}(m)] = \begin{cases} 2 \int_0^t F(t - t', n)^2 dt' & \text{if } n = m, \\ 0 & \text{else.} \end{cases}$$

From this, it follows that for $n, m \in \mathbb{Z}^2$:

$$\begin{aligned} \mathbb{E}[\widehat{\Psi} \cdot \widehat{\Psi}'(n) \cdot \widehat{\Psi} \cdot \widehat{\Psi}'(m)] &= \sum_{k, l \in \mathbb{Z}^2} \mathbb{E}[\widehat{\Psi}(n - k) \widehat{\Psi}'(k) \cdot \widehat{\Psi}(m - l) \widehat{\Psi}'(l)] \\ &= \sum_{k, l \in \mathbb{Z}^2} \mathbb{E}[\widehat{\Psi}(n - k) \widehat{\Psi}(m - l)] \mathbb{E}[\widehat{\Psi}'(k) \widehat{\Psi}'(l)] \\ &= \begin{cases} 4 \sum_{k \in \mathbb{Z}^2} \int_0^t F(t - t', n - k)^2 dt' \int_0^t F(t - t', k)^2 dt' & \text{if } n = m, \\ 0 & \text{else.} \end{cases} \end{aligned}$$

Using this, and (3.2), from which it follows that $\int_0^t F(t - t', n)^2 dt' \lesssim_\theta \langle n \rangle^2$, Lemma 3.2, and

the fact that the logarithm grows slower than any monomial, we compute

$$\begin{aligned}
\mathbb{E}[|\langle \nabla \rangle^{-\sigma}(\Psi\Psi')|^2] &= \mathbb{E}\left[\left|\sum_{n \in \mathbb{Z}^2} e^{in \cdot x} \langle n \rangle^{-\sigma} \widehat{\Psi \cdot \Psi'}(n)\right|^2\right] \\
&= \sum_{n, m \in \mathbb{Z}^2} e^{in \cdot x} e^{-im \cdot x} \langle n \rangle^{-2\sigma} \mathbb{E}\left[\widehat{\Psi \cdot \Psi'}(n) \cdot \overline{\widehat{\Psi \cdot \Psi'}(m)}\right] \\
&= 4 \sum_{n, k \in \mathbb{Z}^2} \langle n \rangle^{-2\sigma} \int_0^t F(t-t', n-k)^2 dt' \cdot \int_0^t F(t-t', k)^2 dt' \\
&\lesssim \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2\sigma} \sum_{k \in \mathbb{Z}^2} \langle n-k \rangle^{-2} \langle k \rangle^{-2} \\
&\lesssim \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2\sigma} \langle n \rangle^{-2} \ln(1+|n|^2) \\
&\lesssim_{\sigma} \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2\sigma} \langle n \rangle^{-2} \langle n \rangle^{\sigma} \\
&\lesssim_{\sigma} 1.
\end{aligned} \tag{3.14}$$

Analogous computations show that

$$\mathbb{E}[|\langle \nabla \rangle^{-\sigma} \Psi|^2] = 2 \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2\sigma} \int_0^t F(t-t', n)^2 dt' \lesssim \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2-2\sigma} \lesssim 1. \tag{3.15}$$

As we see in the following, computing $\|\cdot\|_{L_{\omega}^2 H_x^{-\sigma}}$ suffices to derive the desired bounds. In particular, the former computations allow us to fall back onto the less computationally tedious case of Ψ instead of $\Psi\Psi'$. Therefore, we continue to show that $\Psi \in L^p(\Omega; C([0, 1] \times [0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$.

Since $L^{p'}(\Omega) \subset L^p(\Omega)$ for $1 \leq p' < p$, it suffices to show the statement for sufficiently large $p \geq 1$. By Sobolev's embedding theorem, Fubini's theorem, and noting that for fixed t , $\mathbf{P}_N \Psi$ is just a finite sum of Gaussian variables and therefore Gaussian, the Gaussian hypercontractivity property (see Lemma 2.4 [18]) applies, and we have for $\sigma p > 4$ and any $N > 0$:

$$\|\mathbf{P}_N \Psi\|_{L_{\omega}^p W_x^{-\sigma, \infty}} \lesssim \|\mathbf{P}_N \Psi\|_{L_{\omega}^p W_x^{-\frac{\sigma}{2}, p}} = \left\| \langle \nabla \rangle^{-\frac{\sigma}{2}} \mathbf{P}_N \Psi \right\|_{L_x^p L_{\omega}^p} \lesssim \sqrt{p} \left\| \langle \nabla \rangle^{-\frac{\sigma}{2}} \mathbf{P}_N \Psi \right\|_{L_x^p L_{\omega}^2}. \tag{3.16}$$

From (3.15) it follows that $\|\mathbf{P}_N \Psi\|_{L_{\omega}^p W_x^{-\sigma, \infty}} \lesssim 1$.

A direct computation shows that $\{\Psi_N\}_{N \in \mathbb{N}}$ is a Cauchy sequence in $L^p(\Omega, W^{-\sigma, \infty}(\mathbb{T}^2))$ (see for example (4.5) in [12]), and we define Ψ as its limit in this space.

To show that Ψ is also continuous in ε and t we have to bound the variations of Ψ in ε and t . Following the approach in [12] we define the operator

$$\delta_{\mathbf{h}} X(\mathbf{x}) := X(x_1 + h_1, x_2 + h_2) - X(x_1, x_2),$$

for $\mathbf{h} = (h_1, h_2) \in (0, 1]$. Then Equation (4.7) in Proposition 4.1 in [12] and (3.16) yield

$$\mathbb{E}[\|\delta_{\mathbf{h}}\Psi\|]_{W_x^{-\infty,\sigma}}^p \lesssim p^{\frac{p}{2}}|\mathbf{h}|^{p\gamma}, \quad (3.17)$$

for all $\varepsilon \in [0, 1]$, $t \in [0, T]$, and $\mathbf{h} = (h_1, h_2)$ such that $\varepsilon + h_1 \in [0, 1]$ and $t + h_2 \geq 0$ and some $0 < \gamma \ll 1$. Hence, for $p > 0$ such that $p\gamma > 2$, we can apply a bi-parameter Kolmogorov continuity criterion (see [12] Lemma 2.3) to see that Ψ is almost surely continuous as a function $(\varepsilon, t) \rightarrow \Psi_\varepsilon(t) \in W^{-\sigma,\infty}(\mathbb{T}^2)$.

Next, we follow Lemma 2.3 in [15] to show that using this continuity we can infer that $\Psi \in L^p(\Omega; C([0, 1] \times [0, T]; W^{-\sigma,\infty}(\mathbb{T}^2)))$. Note, for all $\mathbf{x} := (\varepsilon_1, t_1)$, $\mathbf{x}' := (\varepsilon_2, t_2) \in [0, 1] \times [0, T]$ such that $|\mathbf{x} - \mathbf{x}'| < 1$, (3.17) implies that

$$\mathbb{E}[\|\Psi(\mathbf{x}) - \Psi(\mathbf{x}')\|]_{W_x^{-\infty,\sigma}}^p \lesssim p^{\frac{p}{2}}|\mathbf{h}|^{p\gamma}.$$

Let ζ be defined as in Lemma 7.2. Applying Lemma 4.5 in [22] and choosing $\delta > 0$ sufficiently small (note, the δ in Lemma 4.5 [22] is only bounded from above but may be chosen arbitrarily small), we obtain

$$\mathbb{E} \left[\exp \left(\delta \zeta(1) \left(\frac{\|\Psi(\mathbf{x}) - \Psi(\mathbf{x}')\|_{W_x^{-\sigma,\infty}}}{|\mathbf{x} - \mathbf{x}'|^\gamma} \right)^2 \right) \right] \leq 1,$$

for $|\mathbf{x} - \mathbf{x}'| < 1$. Integrating in $\mathbf{x}$, and $\mathbf{x}'$ and swapping the order of integration with Fubini's theorem, we see that the assumption (7.3) of Lemma 7.2 is satisfied up to a set of zero probability where Ψ might not be continuous. Thus, applying Lemma 7.2 (if necessary using a simple change of variables to obtain the correct domains) yields for $|\mathbf{x} - \mathbf{x}'| < 1$:

$$\mathbb{E} \left[\exp \left(\frac{\delta \zeta(1)}{(8\gamma)^2} \left(\sup_{\mathbf{x} \leq \mathbf{y} < \mathbf{y}' \leq \mathbf{x}'} \frac{\|\Psi(\mathbf{y}) - \Psi(\mathbf{y}')\|_{W_x^{-\sigma,\infty}}}{\zeta(|\mathbf{y} - \mathbf{y}'|)} \right)^2 \right) \right] \leq 1, \quad (3.18)$$

where we denote with $\mathbf{x} \leq \mathbf{y} < \mathbf{y}' \leq \mathbf{x}'$ the component wise comparison and without loss of generality we assume $\mathbf{x} < \mathbf{x}'$. Now, let $\lambda > 0$. From $\zeta(|\mathbf{y} - \mathbf{y}'|) \leq \zeta(1) < \infty$, $0 < \gamma \ll 1$ such that $\frac{1}{(8\gamma)^2} > 1$, Chebychev's inequality, and (3.18), it follows that

$$\begin{aligned} & \mathbb{P} \left(\sup_{\mathbf{x} \leq \mathbf{y} < \mathbf{y}' \leq \mathbf{x}'} \|\Psi(\mathbf{y}) - \Psi(\mathbf{y}')\|_{W_x^{-\sigma,\infty}} > \frac{\lambda}{2} \right) \\ &= \mathbb{P} \left(\exp \left(\delta \left(\sup_{\mathbf{x} \leq \mathbf{y} < \mathbf{y}' \leq \mathbf{x}'} \|\Psi(\mathbf{y}) - \Psi(\mathbf{y}')\|_{W_x^{-\sigma,\infty}} \right)^2 \right) > e^{\delta(\frac{\lambda}{2})^2} \right) \\ &\leq e^{-\delta(\frac{\lambda}{2})^2} \mathbb{E} \left[\exp \left(\delta \left(\sup_{\mathbf{x} \leq \mathbf{y} < \mathbf{y}' \leq \mathbf{x}'} \|\Psi(\mathbf{y}) - \Psi(\mathbf{y}')\|_{W_x^{-\sigma,\infty}} \right)^2 \right) \right] \\ &\leq e^{-\delta(\frac{\lambda}{2})^2} \mathbb{E} \left[\exp \left(\frac{\delta \zeta(1)}{(8\gamma)^2} \left(\sup_{\mathbf{x} \leq \mathbf{y} < \mathbf{y}' \leq \mathbf{x}'} \frac{\|\Psi(\mathbf{y}) - \Psi(\mathbf{y}')\|_{W_x^{-\sigma,\infty}}}{\zeta(|\mathbf{y} - \mathbf{y}'|)} \right)^2 \right) \right] \\ &\leq e^{-\delta(\frac{\lambda}{2})^2}. \end{aligned} \quad (3.19)$$

Next, assume $T \geq 1$, otherwise the following step may be skipped. Consider the partition $[0, T] = [0, 1] \cup [0, 2] \cup \dots \cup [j, j+1] \cup \dots \cup [n, T]$. Then

$$\begin{aligned} \mathbb{P}(\|\Psi\|_{C_{\varepsilon, T} W_x^{-\sigma, \infty}} > \lambda) &\leq \sum_{j=0}^n \mathbb{P}(\|\Psi\|_{C([0, 1] \times [j, j+1]) W_x^{-\sigma, \infty}} > \lambda) \\ &\leq T \max_{0 \leq j \leq n} \mathbb{P}(\|\Psi\|_{C([0, 1] \times [j, j+1]) W_x^{-\sigma, \infty}} > \lambda) \end{aligned}$$

where we denote $n+1 := T$.

We can bound the last term using Chebychev's inequality and the fact that for fixed t $\Psi(t) \in L^p(\Omega, W^{-\sigma, \infty}(\mathbb{T}^2))$ for all $p \geq 1$, and (3.19),

$$\begin{aligned} &\mathbb{P}(\|\Psi\|_{C([0, 1] \times [j, j+1]) W_x^{-\sigma, \infty}} > \lambda) \\ &\leq \mathbb{P}(\|\Psi(j)\|_{W_x^{-\sigma, \infty}} > \frac{\lambda}{2}) + \mathbb{P}\left(\sup_{\mathbf{x} \leq \mathbf{y} < \mathbf{y}' \leq \mathbf{x}'} \|\Psi(\mathbf{y}) - \Psi(\mathbf{y}')\|_{W_x^{-\sigma, \infty}} > \frac{\lambda}{2}\right) \\ &\leq e^{-c\lambda^2}, \end{aligned}$$

where $\mathbf{x} := (0, j)$, $\mathbf{x}' := (1, j+1)$, and $c > 0$ depends only on δ from (3.19).

Finally, we use the layer-cake representation to get the desired bound

$$\|\Psi\|_{L_{\omega}^p C_{\varepsilon, T} W_x^{-\sigma, \infty}}^p = p \int_0^\infty \lambda^{p-1} \mathbb{P}(\|\Psi\|_{C_{\varepsilon, T} W_x^{-\sigma, \infty}} > \lambda) d\lambda \leq pT \int_0^\infty \lambda^{p-1} e^{-c\lambda^2} d\lambda < \infty.$$

□

The next result, needed to show the convergence $v_\varepsilon \rightarrow v_0$ as $\varepsilon \rightarrow 0$, is a direct corollary of the calculations for independent Ψ and Ψ' and the computations in Proposition 4.1 in [12].

Corollary 3.8. *Let $T > 0$, $\sigma > 0$, $\varepsilon \in (0, 1]$, let Ψ_ε and Ψ_0 be defined as in (2.15) and (2.21), and let Ψ'_0 be an independent copy of Ψ_0 . Then there exists $\gamma > 0$ such that*

$$\|\Psi'_0(\Psi_\varepsilon - \Psi_0)\|_{L_{\omega}^2 C_T W_x^{-\sigma, \infty}} \lesssim \max(T, 1) \varepsilon^\gamma.$$

Proof. The proof follows directly from combining Equation (4.29) in Proposition 4.1 with the calculations in (3.14), (3.16), and that $\|\Psi'_0\|_{L_{\omega}^2 C_T W_x^{-\sigma, \infty}} \lesssim 1$ by Proposition 3.7. (Note that uniform $L^2(\Omega)$ integrability over time and the approximation parameter ε is justified by the uniform exponential tail bounds we derived to show $\|\Psi\|_{L_{\omega}^2 C_{\varepsilon, T} W_x^{-\sigma, \infty}} < \infty$, rather than solely by Kolmogorov's continuity criterion as stated in [12].) □

4 Local Well-Posedness

As outlined in Section 2, our goal is to prove the existence of a solution $u_\varepsilon = v_\varepsilon + \Psi_\varepsilon$ satisfying Definition 2.2 for all $\varepsilon \in (0, 1]$ and that $u_\varepsilon \rightarrow u_0$ as $\varepsilon \rightarrow 0$, where $u_0 = v_0 + \Psi_0$ satisfies Definition 2.3. In Proposition 3.7, we established that Ψ_ε is well-defined in $L^2(\Omega; C([0, 1] \times [0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$ and, in particular, that Ψ_ε converges to Ψ_0 in this space. We are left to show that the residual term v_ε is well-defined and that it converges

to v_0 . Existing pathwise local well-posedness results of SDNLW typically require the initial data to lie in a Sobolev space of positive regularity (see [18] and [15]). This assumption on the initial data yields a solution to (2.20) with positive regularity. Note that this positive regularity will also be necessary to prove global well-posedness later.

4.1 Local Well-Posedness of the Residual SDMFw

In the following, we adopt the approach of [5] who established the local well-posedness of (2.20) for $\varepsilon = 1$.

Proposition 4.1. *Let $0 < \delta \leq \frac{1}{2}$. Given initial data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$, there exist $T = T(\|(\phi_0, \phi_1)\|_{L_\omega^2 \mathcal{H}_x^{1-\delta}}) \in (0, 1]$ and a unique solution $v \in L^2(\Omega; C([0, 1] \times [0, T]; H^{1-\delta}(\mathbb{T}^2)))$ to (2.20). Furthermore, $\partial_t v_\varepsilon \in L^2(\Omega; C([0, T]; H^{-\delta}(\mathbb{T}^2)))$ for any $\varepsilon \in (0, 1]$.*

Proof. We are going to prove the existence using the Banach fixedpoint theorem. Consider the map

$$\begin{aligned} \Gamma : \overline{B_R} &\subset L^2(\Omega; C([0, 1] \times [0, T], H^{1-\delta}(\mathbb{T}^2))) \mapsto \overline{B_R} \\ \Gamma(v) &= (\varepsilon^{-2} + \partial_t) \mathcal{D}_\varepsilon(t) \phi_0 + \mathcal{D}_\varepsilon(t) \phi_1 - \int_0^t \varepsilon^{-2} \mathcal{D}_\varepsilon(t-t') (\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi) dt', \end{aligned} \quad (4.1)$$

where we determine R, T , and $\delta > 0$ later.

From Proposition 3.6 it follows that

$$\|(\varepsilon^{-2} + \partial_t) \mathcal{D}_\varepsilon(t) \phi_0 + \mathcal{D}_\varepsilon(t) \phi_1\|_{L_\omega^2 C_{\varepsilon, T} H_x^{1-\delta}} \leq \|(\phi_0, \phi_1)\|_{L_\omega^2 H_x^{1-\delta}}, \quad (4.2)$$

and

$$\begin{aligned} &\left\| \int_0^t \varepsilon^{-2} \mathcal{D}_\varepsilon(t-t') (\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi) dt' \right\|_{L_\omega^2 C_{\varepsilon, T} H_x^{1-\delta}} \\ &\leq \sqrt{T} (\|\mathbb{E}[v^2]v\|_{L_\omega^2 C_{\varepsilon, T} H_x^{-\delta}} + \|\mathbb{E}[v^2]\Psi\|_{L_\omega^2 C_{\varepsilon, T} H_x^{-\delta}} \\ &\quad + 2\|\mathbb{E}[v\Psi]v\|_{L_\omega^2 C_{\varepsilon, T} H_x^{-\delta}} + 2\|\mathbb{E}[v\Psi]\Psi\|_{L_\omega^2 C_{\varepsilon, T} H_x^{-\delta}}). \end{aligned} \quad (4.3)$$

Next, we bound the four summands in (4.3). First, we obtain the following by applying Sobolev's inequality in the first line, Hölder's inequality in the second, Minkowski's integral inequality in the third, Cauchy-Schwarz in ω in the fourth, and Sobolev's inequality

combined with the fact that $0 < \delta \leq \frac{1}{2}$ in the last line:

$$\begin{aligned}
\|\mathbb{E}[v^2]v\|_{L_\omega^2 C_{\varepsilon,T} H_x^{-\delta}} &\lesssim \|\mathbb{E}[v^2]v\|_{L_\omega^2 C_{\varepsilon,T} L_x^{\frac{2}{\delta+1}}} \\
&\leq \left\| \|\mathbb{E}[v^2]\|_{L_x^{\frac{6}{2+\delta}}} \|v\|_{L_x^{\frac{6}{1+\delta}}} \right\|_{L_\omega^2 C_{\varepsilon,T}} \\
&\leq \left\| \mathbb{E} \left[\|v\|_{L_x^{\frac{6}{1+\delta}}}^2 \right] \|v\|_{L_x^{\frac{6}{1+\delta}}} \right\|_{L_\omega^2 C_{\varepsilon,T}} \\
&= \|v\|_{L_\omega^2 C_{\varepsilon,T} L_x^{\frac{6}{1+\delta}}}^3 \\
&\lesssim \|v\|_{L_\omega^2 C_{\varepsilon,T} H_x^{1-\delta}}^3.
\end{aligned} \tag{4.4}$$

Applying Sobolev's inequality in the first, Lemma 3.1 2 in the second, Jensen's inequality and Lemma 3.1 1, Sobolev's inequality in the third and choosing $\varepsilon > 0$ such that $3\varepsilon \leq 1 - \delta$ and the assumption $0 < \delta < \frac{1}{2}$, and Proposition 3.7 in the last line, we obtain

$$\begin{aligned}
\|\mathbb{E}[v^2]\Psi\|_{L_\omega^2 C_{\varepsilon,T} H_x^{-\delta}} &\lesssim \|\mathbb{E}[v^2]\Psi\|_{L_\omega^2 C_{\varepsilon,T} W_x^{-\varepsilon, \frac{2}{1+\delta-\varepsilon}}} \\
&\leq \left\| \|\mathbb{E}[v^2]\|_{W_x^{\varepsilon, \frac{4}{1+\delta-\varepsilon}}} \|\Psi\|_{W_x^{-\varepsilon, \infty}} \right\|_{L_\omega^2 C_{\varepsilon,T}} \\
&\lesssim \|v\|_{L_\omega^2 C_{\varepsilon,T} H_x^{\frac{1-\delta+3\varepsilon}{2}}}^2 \|\Psi\|_{L_\omega^2 C_{\varepsilon,T} W_x^{-\varepsilon, \infty}} \\
&\lesssim \|v\|_{L_\omega^2 C_{\varepsilon,T} H_x^{1-\delta}}^2.
\end{aligned} \tag{4.5}$$

The same computations (though in a different order) yield

$$\begin{aligned}
\|\mathbb{E}[v\Psi]v\|_{L_\omega^2 C_{\varepsilon,T} H_x^{-\delta}} &\lesssim \|v\|_{L_\omega^2 C_{\varepsilon,T} H_x^{1-\delta}}^2 \|\Psi\|_{L_\omega^2 C_{\varepsilon,T} W_x^{-\varepsilon, \infty}} \\
&\lesssim \|v\|_{L_\omega^2 C_{\varepsilon,T} H_x^{1-\delta}}^2.
\end{aligned} \tag{4.6}$$

To bound the last term we proceed as in [4, Lemma 3.1]. Let v', Ψ' be independent copies of v, Ψ . Then we use conditional Jensen's inequality in the second, Lemma 3.1 2 in the third, conditional Hölder's inequality in the fourth, the fact that v' is independent of Ψ , and Proposition 3.7 in the last line to obtain

$$\begin{aligned}
\|\mathbb{E}[v\Psi]\Psi\|_{L_\omega^2 C_{\varepsilon,T} H_x^{-\delta}} &= \|\mathbb{E}[v'\Psi'\Psi|\Psi]\|_{L_\omega^2 C_{\varepsilon,T} H_x^{-\delta}} \\
&\leq \left\| \mathbb{E} \left[\|v'\Psi'\Psi\|_{C_{\varepsilon,T} H_x^{-\delta}} |\Psi| \right] \right\|_{L_\omega^2} \\
&\leq \left\| \mathbb{E} \left[\|v'\|_{C_{\varepsilon,T} H_x^\delta} \|\Psi'\Psi\|_{C_{\varepsilon,T} W_x^{-\delta, \infty}} |\Psi| \right] \right\|_{L_\omega^2} \\
&\leq \left\| \mathbb{E} \left[\|v'\|_{C_{\varepsilon,T} H_x^\delta}^2 |\Psi| \right]^{\frac{1}{2}} \mathbb{E} \left[\|\Psi'\Psi\|_{C_{\varepsilon,T} W_x^{-\delta, \infty}}^2 |\Psi| \right]^{\frac{1}{2}} \right\|_{L_\omega^2} \\
&= \|v\|_{L_\omega^2 C_{\varepsilon,T} H_x^\delta} \|\Psi'\Psi\|_{L_\omega^2 C_{\varepsilon,T} W_x^{-\delta, \infty}} \\
&\lesssim \|v\|_{L_\omega^2 C_{\varepsilon,T} H_x^{1-\delta}}.
\end{aligned} \tag{4.7}$$

Finally, choosing $0 < \delta \leq \frac{1}{2}$ we combine the estimates (4.2), (4.4), (4.5), (4.6), and (4.7) to obtain the bound

$$\|\Gamma v\|_{L^2_{\omega} C_{\varepsilon, T} H_x^{1-\delta}} \lesssim \|(\phi_0, \phi_1)\|_{L^2_{\omega} \mathcal{H}_x^{1-\delta}} + \sqrt{T}(1 + \|v\|_{L^2_{\omega} C_{\varepsilon, T} H_x^{1-\delta}})^3.$$

Therefore, there exists $T(\|(\phi_0, \phi_1)\|_{L^2_{\omega} \mathcal{H}_x^{1-\delta}}) > 0$ and $R(T, \|(\phi_0, \phi_1)\|_{L^2_{\omega} \mathcal{H}_x^{1-\delta}}) > 0$ such that $\|\Gamma v\|_{L^2_{\omega} C_{\varepsilon, T} H_x^{1-\delta}} \leq R$ for $v \in \overline{B_R}$, guaranteeing that the map $\Gamma : \overline{B_R} \mapsto \overline{B_R}$ is well-defined.

Similarly, we see that for $v, w \in \overline{B_R}$:

$$\|\Gamma v - \Gamma w\|_{L^2_{\omega} C_{\varepsilon, T} H_x^{1-\delta}} \lesssim \sqrt{T}\|v - w\|_{L^2_{\omega} C_{\varepsilon, T} H_x^{1-\delta}}(1 + R^2).$$

Consequently, we can choose T sufficiently small such that the map Γ is a contraction. Hence, by Banach's fixed point theorem there exists a unique $v \in \overline{B_R}$.

Next, we extend the uniqueness to the entire space $L^2(\Omega; C([0, 1] \times [0, \tilde{T}]; H^{1-\delta}(\mathbb{T}^2)))$ for some $\tilde{T} > 0$, possibly smaller than T up to a constant factor, using a standard continuity argument: Let $\frac{R}{2} = \|(\phi_0, \phi_1)\|_{L^2_{\omega} \mathcal{H}_x^{1-\delta}}$. Our goal is to show that if v is a solution to (2.20), then $\|v\|_{L^2_{\omega} C_{\tilde{T}, \varepsilon} H_x^{1-\delta}} \leq R$. Consequently, by the uniqueness of v within the closed ball $\overline{B_R}$, we obtain uniqueness in the entire space $L^2(\Omega; C([0, 1] \times [0, \tilde{T}]; H^{1-\delta}(\mathbb{T}^2)))$. Let $v \in L^2(\Omega; C([0, 1] \times [0, \tilde{T}]; H^{1-\delta}(\mathbb{T}^2)))$ satisfy (2.20). Since $\|v(0)\|_{L^2_{\omega} C_{\varepsilon} H_x^{1-\delta}} = \|\phi_0\|_{L^2_{\omega} \mathcal{H}_x^{1-\delta}} \leq \frac{R}{2}$, the uniform continuity of v in t implies that there exists $0 < \tau \leq \tilde{T}$ such that

$$\|v\|_{L^2_{\omega} C([0, 1] \times [0, \tau]) H_x^{1-\delta}} \leq 2R.$$

By using (2.20), we get

$$\|v\|_{L^2_{\omega} C([0, 1] \times [0, \tau]) H_x^{1-\delta}} = \|\Gamma v\|_{L^2_{\omega} C([0, 1] \times [0, \tau]) H_x^{1-\delta}} \leq \frac{R}{2} + \tilde{T}C(1 + 2R)^3 \leq R$$

for some $C > 0$ as long as we have $0 < \tilde{T} \leq \frac{R}{2C(1+2R)^3}$.

Due to the uniform continuity, we can iterate this procedure to obtain $\|v\|_{L^2_{\omega} C([0, 1] \times [0, \tilde{T}]) H_x^{1-\delta}} \leq R$, which finishes the proof of uniqueness.

Lastly, we fix $\varepsilon \in (0, 1]$ to show that $\partial_t v_{\varepsilon} \in L^2(\Omega; C([0, T]; H^{-\delta}(\mathbb{T}^2)))$.

$$\|\partial_t v_{\varepsilon}\|_{L^2_{\omega} C_T H_x^{-\delta}} \leq \|\partial_t P_{\varepsilon}(t)(\phi_0, \phi_1)\|_{L^2_{\omega} C_T H_x^{-\delta}} + \|\partial_t I_{\varepsilon}(t)(\mathbb{E}[v_{\varepsilon}^2 + 2v_{\varepsilon}\Psi_{\varepsilon}](v_{\varepsilon} + \Psi_{\varepsilon}))\|_{L^2_{\omega} C_T H_x^{-\delta}}.$$

Using Lemma 3.5 (3.5), (3.6) and (3.7), we obtain a bound on the first summand on the right

$$\begin{aligned} \|\partial_t P_{\varepsilon}(t)(\phi_0, \phi_1)\|_{L^2_{\omega} C_T H_x^{-\delta}} &\lesssim \|D_{\varepsilon}(t)\phi_0\|_{L^2_{\omega} C_T H_x^{-\delta}} + \|\partial_t D_{\varepsilon}(t)\phi_1\|_{L^2_{\omega} C_T H_x^{-\delta}} + \|\partial_t^2 D_{\varepsilon}(t)\phi_0\|_{L^2_{\omega} C_T H_x^{-\delta}} \\ &\lesssim \|\phi_0\|_{H_x^{-\delta}} + \|\phi_1\|_{H_x^{-\delta}} + \|\phi_0\|_{H_x^{1-\delta}} \lesssim \|(\phi_0, \phi_1)\|_{\mathcal{H}_x^{1-\delta}}. \end{aligned}$$

Lemma 3.5 (3.6) and the same computations used to bound the nonlinearity in the Duhamel

formulation of v yield

$$\begin{aligned} \|\partial_t I_\varepsilon(t)((\mathbb{E}[v_\varepsilon^2] + 2\mathbb{E}[v_\varepsilon \Psi_\varepsilon])(v_\varepsilon + \Psi_\varepsilon))\|_{L_\omega^2 C_T H_x^{1-\delta}} &\lesssim_\varepsilon T \|\mathbb{E}[v_\varepsilon^2] + 2\mathbb{E}[v_\varepsilon \Psi_\varepsilon]\|_{L_\omega^2 C_T H_x^{1-\delta}} \|v_\varepsilon + \Psi_\varepsilon\|_{L_\omega^2 C_T H_x^{1-\delta}} \\ &\lesssim_\varepsilon T(1 + \|v\|_{L_\omega^2 C_T H_x^{1-\delta}})^3 \\ &< \infty, \end{aligned}$$

since $\|v\|_{L_\omega^2 C_T H_x^{1-\delta}} < \infty$. $\square$

Remark 4.2. As noted in [5], it is possible to lower the regularity threshold by using Strichartz estimates. Since we aim to show Global Well-Posedness, for which we need to increase the regularity threshold anyway, this regularity result suffices in this work.

4.2 Local Well-Posedness of the Mean Field SKA

Proposition 4.1 establishes that the map $\varepsilon \rightarrow v_\varepsilon$ is continuous including $\varepsilon = 0$. However, we must explicitly verify that the limit v_0 satisfies (2.22). The continuity of v_ε with respect to ε is crucial, as the following proposition relies on it. We follow the approach of Lemma 4.6 in [12] to show that $v_\varepsilon \rightarrow v_0$ as $\varepsilon \rightarrow 0$.

Proposition 4.3. *Let $0 < \theta < \delta < \frac{1}{2}$ such that $0 < 1 - \delta + \theta \leq \frac{1}{2}$. Consider initial data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta+\theta}(\mathbb{T}^2))$. Let $T > 0$ and assume that $v_\varepsilon \in L^2(\Omega; C([0, 1] \times [0, T]; H^{1-\delta+\theta}(\mathbb{T}^2)))$ is a solution to (2.20) with initial data (ϕ_0, ϕ_1) . Furthermore, let $v_0 \in L^2(\Omega; C([0, T]; H^{1-\delta+\theta}(\mathbb{T}^2)))$ be the solution to (2.22) with initial data ϕ_0 .*

Then there exist constants $K(T, \|v_0\|_{L_\omega^2 C_T H_x^{1-\delta}}, \|v_\varepsilon\|_{L_\omega^2 C_{\varepsilon, T} H_x^{1-\delta+\theta}}) > 0$ and $\gamma > 0$ such that

$$\|v_\varepsilon - v_0\|_{L_\omega^2 C_T H_x^{1-\delta}} \leq K \max(\varepsilon^\gamma, \|\Psi_\varepsilon - \Psi_0\|_{L_\omega^2 C_T W_x^{-\delta, \infty}}, \varepsilon^{\frac{\theta}{2}} \|(\phi_0, \phi_1)\|_{L_\omega^2 \mathcal{H}_x^{1-\delta+\theta}}).$$

Proof. Recall the Duhamel representation

$$\begin{aligned} v_\varepsilon(t) &= P_\varepsilon(t)(\phi_0, \phi_1) - \mathcal{I}_\varepsilon(t)(G_\varepsilon) \\ v_0 &= P_0(t)(\phi_0) - \mathcal{I}_0(t)(G_0), \end{aligned}$$

where we set $G(\varepsilon) := (\mathbb{E}[v_\varepsilon^2] + 2\mathbb{E}[v_\varepsilon \Psi_\varepsilon])(v_\varepsilon + \Psi_\varepsilon)$ and P_ε , $\mathcal{I}_\varepsilon$, P_0 , and $\mathcal{I}_0$ are defined as in (2.4), (2.7), (2.9), and (2.10) respectively.

Then

$$\begin{aligned} \|v_\varepsilon - v_0\|_{L_\omega^2 C_T H_x^{1-\delta}} &\leq \|P_\varepsilon(\phi_0, \phi_1) - P_0(\phi_0)\|_{L_\omega^2 C_T H_x^{1-\delta}} + \|(\mathcal{I}_\varepsilon - \mathcal{I}_0)(G_\varepsilon)\|_{L_\omega^2 C_T H_x^{1-\delta}} \\ &\quad + \|\mathcal{I}_0(G_\varepsilon - G_0)\|_{L_\omega^2 C_T H_x^{1-\delta}}. \end{aligned} \tag{4.8}$$

The first two terms, we can bound similar as in Proposition 4.1. However, for the third term, we introduce the norm

$$\|\cdot\|_{\lambda, \delta} := \|e^{-\lambda t} \cdot\|_{L_\omega^2 C_T H_x^\delta},$$

which is equivalent to the $\|\cdot\|_{L^2_{\omega}C_TH_x^\delta}$ -norm, since,

$$\|\cdot\|_{\lambda,\delta} \leq \|\cdot\|_{L^2_{\omega}C_TH_x^\delta} \leq e^{\lambda T} \|\cdot\|_{\lambda,\delta}.$$

Consequently, it suffices to find a bound in $\|\cdot\|_{\lambda,1-\delta}$.

Using Proposition 3.6 and the computations (4.3) - (4.7) in Proposition 4.1, we obtain

$$\begin{aligned} \|P_\varepsilon(\phi_0, \phi_1) - P_0(\phi_0)\|_{\lambda,\delta} &\leq \varepsilon^{\theta/2} \|(\phi_0, \phi_1)\|_{\mathcal{H}_x^{1-\delta+\theta}}. \\ \|(\mathcal{I}_\varepsilon - \mathcal{I}_0)(G_\varepsilon)\|_{\lambda,\delta} &\lesssim \sqrt{T} \varepsilon^{\theta/2} \|G_\varepsilon\|_{L^2_{\omega}C_TH_x^{-\delta+\theta}} \lesssim \|v_\varepsilon\|_{L^2_{\omega}C([0,1]\times[0,T])H_x^{1-\delta+\theta}} \sqrt{T} \varepsilon^{\theta/2}. \end{aligned} \quad (4.9)$$

From Lemma 2.5 in [12] it follows that

$$\begin{aligned} \|\mathcal{I}_0(G_\varepsilon - G_0)\|_{\lambda,1-\delta} &= \|e^{-\lambda t} \int_0^t e^{(\Delta-1)(t-t')}(G_\varepsilon - G_0)(t') dt'\|_{L^2_{\omega}C_TH_x^{1-\delta}} \\ &= \left\| \int_0^t e^{-\lambda(t-t')} e^{(\Delta-1)(t-t')}(e^{-\lambda t'}(G_\varepsilon - G_0)(t')) dt' \right\|_{L^2_{\omega}C_TH_x^{1-\delta}} \\ &\leq \left\| \int_0^t e^{-\lambda(t-t')} \|e^{(\Delta-1)(t-t')} e^{-\lambda t'}(G_\varepsilon - G_0)(t')\|_{H_x^{1-\delta}} dt' \right\|_{L^2_{\omega}C_T} \\ &\leq \left\| e^{-\lambda t}(G_\varepsilon - G_0)(t) \right\|_{L^2_{\omega}C_TH_x^{-\delta}} \left\| \int_0^t e^{-\lambda(t-t')}(t-t')^{-\frac{1}{2}} dt' \right\|_{C_T} \\ &\leq \lambda^{-\frac{1}{2}} \sqrt{\pi} \|G_\varepsilon - G_0\|_{\lambda,-\delta}. \end{aligned}$$

Let Ψ'_0 be an independent copy of Ψ_0 . From computations analogous to (4.3) - (4.7) in Proposition 4.1, by factoring the cubic and quadratic monomial differences and bounding the remainder by the uniform norms of v_ε and v_0 , and from Corollary 3.8 it follows that

$$\begin{aligned} \|\mathcal{I}_0(G_\varepsilon - G_0)\|_{\lambda,1-\delta} &\leq \lambda^{-\frac{1}{2}} \sqrt{\pi} \|(G_\varepsilon - G_0)\|_{\lambda,-\delta} \\ &\leq K \lambda^{-\frac{1}{2}} (\|v_\varepsilon - v_0\|_{\lambda,1-\delta} + \|\Psi_\varepsilon - \Psi_0\|_{L^2_{\omega}C_TW_x^{-\delta,\infty}} + \|\Psi'_0(\Psi_\varepsilon - \Psi_0)\|_{L^2_{\omega}C_TW_x^{-\delta,\infty}}) \\ &\leq K \lambda^{-\frac{1}{2}} (\|v_\varepsilon - v_0\|_{\lambda,1-\delta} + \|\Psi_\varepsilon - \Psi_0\|_{L^2_{\omega}C_TW_x^{-\delta,\infty}} + \max(T, 1) \varepsilon^\gamma) \end{aligned} \quad (4.10)$$

for some sufficiently small $\gamma > 0$ and K depending on $\|v_0\|_{L^2_{\omega}C_TH_x^{1-\delta}}$ and $\|v_\varepsilon\|_{L^2_{\omega}C([0,1]\times[0,T])H_x^{1-\delta}}$.

Finally, we choose $\lambda > 0$ such that $K \lambda^{-\frac{1}{2}} \leq \frac{1}{2}$. Then, by absorbing the $\frac{1}{2} \|v_\varepsilon - v_0\|_{\lambda,1-\delta}$ term onto the left-hand side of the combined inequality of (4.8), (4.9), and (4.10), we obtain the desired result. $\square$

Remark 4.4. Note that the constant K does not depend on ε as we assume that $v_\varepsilon \in L^2(\Omega; C([0,1] \times [0,T]; H^{1-\delta+\theta}(\mathbb{T}^2)))$. In particular, Proposition 4.3 implies that $\|v_\varepsilon - v_0\|_{L^2_{\omega}C_TH_x^{1-\delta}} \rightarrow 0$ as $\varepsilon \rightarrow 0$. We would obtain the same result if we only required $v_\varepsilon \in L^2(\Omega; [0,T]; H^{1-\delta+\theta}(\mathbb{T}^2))$ with $\|v_\varepsilon\|_{L^2_{\omega}C_TH_x^{1-\delta+\theta}}$ bounded uniformly in ε .

Remark 4.5. Note that we lose some regularity in Proposition 4.3. Consequently, while the local solution v_ε to (2.20) exists in $L^p(\Omega; C([0, 1] \times [0, T]; H^{1-\delta}(\mathbb{T}^2)))$ for $0 < \delta \leq \frac{1}{2}$, the convergence $v_\varepsilon \rightarrow v_0$ as $\varepsilon \rightarrow 0$ (where v_0 solves (2.22)) holds only for $0 < \delta < \frac{1}{2}$.

Finally, we can prove that the mean field SKA is well-posed, that is, there exists a family $\{u_\varepsilon\}_{\varepsilon \in (0,1]}$ satisfying Definition 2.4.

Theorem 4.6. *Let $0 < \delta < \frac{1}{2}$ and $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$. Then there exists $T > 0$ such that for each $\varepsilon \in (0, 1]$, there exists a solution u_ε in the sense of Definition 2.2 with initial data (ϕ_0, ϕ_1) and a solution u_0 in the sense of Definition 2.3 with initial data ϕ_0 , both belonging to $L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ for any $\sigma > 0$. Moreover, $\{u_\varepsilon\}_{\varepsilon \in (0,1]}$ converges to the solution u_0 in $L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ as $\varepsilon \rightarrow 0$.*

Proof. First, we show that for $\sigma > 0$ and $\varepsilon \in (0, 1]$, there exists $T > 0$ such that $u_\varepsilon = v_\varepsilon + \Psi_\varepsilon \in L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ satisfies Definition 2.2. By Proposition 4.1 there exists $T > 0$ such that $v \in L^2(\Omega; C([0, 1] \times [0, T]; H^{1-\delta}(\mathbb{T}^2)))$ which satisfies (2.20). Proposition 3.7 yields that $\Psi_\varepsilon \in L^2(\Omega; C([0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$. Define $v_\varepsilon = v(\varepsilon, \cdot)$. Then $u_\varepsilon = v_\varepsilon + \Psi_\varepsilon$ is the desired solution.

[4] proved in Theorem 3.6 that there exists a solution $u_0 = v_0 + \Psi_0 \in L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ to (2.3). (Actually, they prove a stronger result, namely $u_0 \in L^2(\Omega; C([0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$, which is due to the stronger smoothing property of the heat propagator.)

Finally, from Proposition 3.7 and Proposition 4.3, it follows that $u_\varepsilon \rightarrow u_0$ in $L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ as $\varepsilon \rightarrow 0$. Note that the loss of regularity remarked in 4.5 does not affect the result, as the convergence is only needed in $L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ and $-\sigma < 0 < 1 - \delta - \theta$ for any θ such that $0 < \theta < \delta$ and $0 < \delta < \frac{1}{2}$. $\square$

4.3 Independent Convergence of the Residual SDMFV

Before we start proving the global SKA, we present an alternative convergence result for $v_\varepsilon(t) \rightarrow v_0(t)$ compared to Proposition 4.3. The following Proposition establishes convergence independent of the solution norms of either v_ε or v_0 . In contrast, the convergence result in Proposition 4.3 depends on both $\|v_0\|_{L_\omega^2 C_T H_x^{1-\delta}}$ and $\|v_\varepsilon\|_{L_\omega^2 C_T H_x^{1-\delta+\theta}}$. Consequently, applying the previous estimate requires a priori knowledge of the growth of both solutions. Specifically, the convergence depends on ε through the $\|v_\varepsilon\|_{L_\omega^2 C_T H_x^{1-\delta+\theta}}$ norm. The advantage of Proposition 4.3 is that it yields uniform-in-time $L^2(\Omega)$ convergence for $v_\varepsilon \rightarrow v_0$, a stronger result that is not attainable using the approach in the next proposition. In Chapter 5, we show that we can bound $\|v_\varepsilon\|_{L_\omega^2 C_T H_x^{1-\delta+\theta}}$ uniformly in ε , allowing the stronger convergence result via Proposition 4.3. Nevertheless, the alternative approach, proposed in the next Proposition, is necessary to establish the global SKA of the HL SM_N (1.1) to the PL SM_N , as discussed in [23], because finding an a priori bound for v_ε that is uniform in ε was not possible.

Proposition 4.7. *Let $0 < \theta < \delta \leq \frac{1}{2}$ and $\sigma > 0$. Consider some data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta+\theta}(\mathbb{T}^2))$.*

Let $T > 0$ and assume that $v_\varepsilon \in L^2(\Omega; C([0, T]; H^{1-\delta+\theta}(\mathbb{T}^2)))$ is a solution to (2.20) with initial data (ϕ_0, ϕ_1) . Furthermore, let $v_0 \in L^2(\Omega; C([0, T]; H^{1-\delta+\theta}(\mathbb{T}^2)))$ be a solution to (2.22) with initial data ϕ_0 .

Then, there exist $\varepsilon_0 \in (0, 1]$ and $\gamma > 0$ such that for all $\varepsilon \in (0, \varepsilon_0]$ and $t \in [0, T]$

$$\|v_\varepsilon(t) - v_0(t)\|_{L_\omega^2 H_x^{1-\delta}} \leq C\varepsilon^\gamma (1 + \|(\phi_0, \phi_1)\|_{L_\omega^2 H_x^{1-\delta+\theta}}) \exp \left[(1 + \|v_0\|_{L_\omega^2 C_T H_x^{1-\delta+\theta}})^3 \sqrt{1+t} \right] \quad (4.11)$$

$$\|v_\varepsilon(t) - v_0(t)\|_{L_\omega^2 H_x^{1-\delta}} \leq C\varepsilon^\gamma (1 + \|(\phi_0, \phi_1)\|_{L_\omega^2 H_x^{1-\delta+\theta}}) \exp \left[(1 + \|v_\varepsilon\|_{L_\omega^2 C_T H_x^{1-\delta+\theta}})^3 \sqrt{1+t} \right] \quad (4.12)$$

with $C > 0$ being some constant independent of the parameters T and ε .

Proof. The proof relies on a continuity argument: we show that v_ε and v_0 , starting from the same initial data, remain close on a short time interval. This allows us to absorb the nonlinear terms appearing in the Duhamel formulation of both v_ε and v_0 and apply a Gronwall inequality.

First, we show that there exists $\varepsilon_0 \in (0, 1]$ such that

$$\|v_\varepsilon(t) - v_0(t)\|_{L_\omega^2 H_x^{1-\delta}} < 1$$

for all $\varepsilon \in (0, \varepsilon_0]$ and $t \in [0, T]$. While proving this smallness condition, we already obtain the desired bound for all $\varepsilon \in (0, \varepsilon_0]$.

Suppose toward contradiction that there exists $t_0 \in [0, T]$ such that for all $\varepsilon \in (0, 1]$

$$\|v_\varepsilon(t_0) - v_0(t_0)\|_{L_\omega^2 H_x^{1-\delta}} \geq 1. \quad (4.13)$$

Without loss of generality, we can assume there exists $\varepsilon_0 \in (0, 1]$ such that for all $t \in [0, t_0]$ it holds that

$$\|v_{\varepsilon_0}(t) - v_0(t)\|_{L_\omega^2 H_x^{1-\delta}} \leq 1, \quad (4.14)$$

because v_ε and v_0 are continuous in $L^2(\Omega)$. Since

$$\|v_\varepsilon(0) - v_0(0)\|_{L_\omega^2 H_x^{1-\delta}} = \|\phi_0 - \phi_0\|_{L_\omega^2 H_x^{1-\delta}} = 0,$$

continuity implies that $\|v_{\varepsilon_0}(t) - v_0(t)\|_{L_\omega^2 H_x^{1-\delta}}$ stays small for small t and some arbitrary ε_0 .

Next, we use the Duhamel representations to derive a contradiction. We only show (4.11) as (4.12) follows similar. Recall the Duhamel representation

$$\begin{aligned} v_\varepsilon(t) &= P_\varepsilon(t)(\phi_0, \phi_1) - \mathcal{I}_\varepsilon(t)(G_\varepsilon) \\ v_0(t) &= P_0(t)(\phi_0) - \mathcal{I}_0(t)(G_0) \end{aligned}$$

where $G(\varepsilon) = (\mathbb{E}[v_\varepsilon^2] + 2\mathbb{E}[v_\varepsilon \Psi_\varepsilon])(v_\varepsilon + \Psi_\varepsilon)$ and P_ε , $\mathcal{I}_\varepsilon$, P_0 , and $\mathcal{I}_0$ are defined as in (2.4),

(2.7), (2.9), and (2.10) respectively. Then,

$$\begin{aligned}
\|v_{\varepsilon_0}(t_0) - v_0(t_0)\|_{L_{\omega}^2 H_x^{1-\delta}} &\leq \|P_{\varepsilon}(t_0)(\phi_0, \phi_1) - P_0(t_0)(\phi_0)\|_{L_{\omega}^2 H_x^{1-\delta}} + \|(\mathcal{I}_{\varepsilon} - \mathcal{I}_0)(t_0)(G_0)\|_{L_{\omega}^2 H_x^{1-\delta}} \\
&\quad + \|\mathcal{I}_{\varepsilon}(t_0)(G_{\varepsilon} - G_0)\|_{L_{\omega}^2 H_x^{1-\delta}} \\
&\leq \varepsilon_0^{\frac{\theta}{2}} \|(\phi_0, \phi_1)\|_{L_{\omega}^2 H_x^{1-\delta+\theta}} + \varepsilon_0^{\frac{\theta}{2}} \sqrt{t_0} (1 + \|v_0\|_{L_{\omega}^2 C_T H_x^{1-\delta+\theta}})^3 \\
&\quad + \int_0^{t_0} \|\varepsilon^{-2} \mathcal{D}_{\varepsilon}(t_0 - t')(G_{\varepsilon}(t') - G_0(t'))\|_{L_{\omega}^2 H_x^{1-\delta}} dt' \\
&\leq \varepsilon_0^{\frac{\theta}{2}} \|(\phi_0, \phi_1)\|_{L_{\omega}^2 H_x^{1-\delta+\theta}} + \varepsilon_0^{\frac{\theta}{2}} \sqrt{t_0} (1 + \|v_0\|_{L_{\omega}^2 C_T H_x^{1-\delta+\theta}})^3 \\
&\quad + \int_0^{t_0} (t_0 - t')^{-\frac{1}{2}} \|G_{\varepsilon}(t') - G_0(t')\|_{L_{\omega}^2 H_x^{-\delta}} dt'
\end{aligned} \tag{4.15}$$

where we used (3.12), (3.13) combined with the computations (4.4)-(4.6) in Proposition 4.1 in the second, and (3.39) in [12] in the last line.

To bound $\|G_{\varepsilon}(t') - G_0(t')\|_{L_{\omega}^2 H_x^{-\delta}}$, we use that $\|\Psi_{\varepsilon_0}\|_{L_{\omega}^2 W_x^{-\delta, \infty}}, \|\Psi_0\|_{L_{\omega}^2 W_x^{-\delta, \infty}} \lesssim 1$ and $\|(\Psi_{\varepsilon_0} - \Psi_0)\Psi_0'\|_{L_{\omega}^2 W_x^{-\delta, \infty}}, \|\Psi_{\varepsilon_0} - \Psi_0\|_{L_{\omega}^2 W_x^{-\delta, \infty}} \lesssim \varepsilon_0^{\gamma}$ where Ψ_0' is an independent copy of Ψ_0 (see Propositions 3.7 and 3.8) for some sufficiently small $\gamma > 0$. By completing the square, and utilizing the computations in Proposition 4.1, along with the smallness assumption (4.14) we can find a bound for the last term depending only on v_0 . Because all terms can be computed similarly, we do the computations only for cubic v_{ε} term in $\|G_{\varepsilon}(t') - G_0(t')\|_{L_{\omega}^2 H_x^{-\delta}}$.

Let $t \in (0, t_0]$. Then,

$$\begin{aligned}
&\|(\mathbb{E}[v_{\varepsilon_0}^2]v_{\varepsilon_0} - \mathbb{E}[v_0^2]v_0)(t)\|_{L_{\omega}^2 H_x^{-\delta}} \\
&\leq \|(\mathbb{E}[(v_{\varepsilon_0} - v_0)^2](v_{\varepsilon_0} - v_0))(t)\|_{L_{\omega}^2 H_x^{-\delta}} + \|(\mathbb{E}[(v_{\varepsilon_0} - v_0)^2]v_0)(t)\|_{L_{\omega}^2 H_x^{-\delta}} \\
&\quad + 2\|(\mathbb{E}[v_0(v_{\varepsilon_0} - v_0)](v_{\varepsilon_0} - v_0))(t)\|_{L_{\omega}^2 H_x^{-\delta}} + 2\|(\mathbb{E}[v_0(v_{\varepsilon_0} - v_0)]v_0)(t)\|_{L_{\omega}^2 H_x^{-\delta}} \\
&\quad + \|(\mathbb{E}[v_0^2](v_{\varepsilon_0} - v_0))(t)\|_{L_{\omega}^2 H_x^{-\delta}} \\
&\leq \|(v_{\varepsilon_0} - v_0)(t)\|_{L_{\omega}^2 H_x^{1-\delta}}^3 + 3\|(v_{\varepsilon_0} - v_0)(t)\|_{L_{\omega}^2 H_x^{1-\delta}}^2 \|v_0(t)\|_{L_{\omega}^2 H_x^{1-\delta}} \\
&\quad + 3\|(v_{\varepsilon_0} - v_0)(t)\|_{L_{\omega}^2 H_x^{1-\delta}} \|v_0(t)\|_{L_{\omega}^2 H_x^{1-\delta}}^2 \\
&\lesssim \|(v_{\varepsilon_0} - v_0)(t)\|_{L_{\omega}^2 H_x^{1-\delta}} \left(1 + \|v_0(t)\|_{L_{\omega}^2 H_x^{1-\delta}}\right)^2.
\end{aligned}$$

Computing the other terms, we obtain

$$\begin{aligned}
&\int_0^{t_0} (t_0 - t')^{-\frac{1}{2}} \|G_{\varepsilon}(t') - G_0(t')\|_{L_{\omega}^2 H_x^{-\delta}} dt' \\
&\lesssim \left(1 + \|v_0\|_{L_{\omega}^2 C_T H_x^{1-\delta}}\right)^2 \int_0^{t_0} (t_0 - t')^{-\frac{1}{2}} (\|(v_{\varepsilon_0} - v_0)(t')\|_{L_{\omega}^2 H_x^{1-\delta}} \\
&\quad + \|(\Psi_{\varepsilon_0} - \Psi_0)(t')\|_{L_{\omega}^2 W_x^{-\sigma, \infty}}) dt' \\
&\lesssim \left(1 + \|v_0\|_{L_{\omega}^2 C_T H_x^{1-\delta}}\right)^2 (\sqrt{t_0} \varepsilon_0^{\gamma} + \int_0^{t_0} (t_0 - t')^{-\frac{1}{2}} \|(v_{\varepsilon_0} - v_0)(t')\|_{L_{\omega}^2 H_x^{1-\delta}} dt').
\end{aligned} \tag{4.16}$$

Plugging (4.16) into (4.15) we see that we can apply Gronwall's inequality. Additionally, we use that $x \leq e^x$ to obtain

$$\begin{aligned} & \|v_{\varepsilon_0}(t_0) - v_0(t_0)\|_{L_{\omega}^2 H_x^{1-\delta}} \\ & \leq C\varepsilon_0^\gamma (1 + \|(\phi_0, \phi_1)\|_{L_{\omega}^2 H_x^{1-\delta+\theta}}) \exp \left[(1 + \|v_0\|_{L_{\omega}^2 C_T H_x^{1-\delta+\theta}})^3 \sqrt{1+t_0} \right]. \end{aligned}$$

Since the ε_0 was arbitrary, we can choose it sufficiently small such that $\|v_{\varepsilon_0}(t_0) - v_0(t_0)\|_{L_{\omega}^2 H_x^{1-\delta}} < 1$ in contradiction to the assumption (4.13). In particular, all the computations from above are valid for arbitrary $t \in [0, T]$, thus, we immediately obtain the desired bound (4.11) for ε_0 . Clearly, the same argument works for any $\varepsilon \in (0, \varepsilon_0]$. $\square$

Remark 4.8. The bounds (4.11) and (4.12) are likely suboptimal with respect to their dependence on t . In our proof, we did not exploit the damping effect inherent to the damped wave equation. Consequently, it is highly probable that one could establish a bound that depends on t only through the $\|v_0\|_{L_{\omega}^2 C_T H_x^{1-\delta+\theta}}$ or $\|v_{\varepsilon}\|_{L_{\omega}^2 C_T H_x^{1-\delta+\theta}}$ norms.

5 Global Well-Posedness

In this section, we show that for any $T > 0$ and all $\varepsilon \in (0, 1]$, $\sigma > 0$ there exists $u_{\varepsilon} = v_{\varepsilon} + \Psi_{\varepsilon} \in L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ satisfying Definition 2.2, and that the convergence $u_{\varepsilon} \rightarrow u_0$ holds in $L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$. Since we already know that Ψ_{ε} is globally well-defined and converges to Ψ_0 , we only need to show that the residual term v_{ε} is globally defined. The convergence then follows from Proposition 4.3. The strategy is to establish an a priori bound on $\|(v_{\varepsilon}(t), \partial_t v_{\varepsilon}(t))\|_{L_{\omega}^2 \mathcal{H}_x^{1-\delta}}$ for any $t > 0$, which allows us to iterate the local well-posedness result of Proposition 4.1. We follow the approach in [5], who showed global well-posedness for the SDMFWE, adapted from [15], who showed global well-posedness for the cubic stochastic nonlinear wave equation (SNLW). The main contribution in this thesis is to derive an a priori bound that is uniform in ε such that we can apply Proposition 4.3 to obtain the global mean-field SKA as described in Definition 2.4.

In the following, let $\varepsilon > 0$ be fixed. We omit the ε -subscript in the function notations for brevity. Consider the nonlinear mean field wave equation:

$$\begin{cases} \varepsilon^2 \partial_t^2 u + (1 - \Delta)u + \mathbb{E}[u^2]u = 0 \\ (u, \partial_t u)|_{t=0} = (\phi_0, \phi_1). \end{cases} \quad (5.1)$$

Formally, a solution u to this equation conserves the energy functional defined by

$$E(\mathbf{u}(t)) := \frac{1}{2} \mathbb{E} \left[\int_{\mathbb{T}^2} (\varepsilon \partial_t u)^2 + u^2 + |\nabla u|^2 dx \right] + \frac{1}{4} \int_{\mathbb{T}^2} (\mathbb{E}[u^2])^2 dx, \quad (5.2)$$

where we denote $\mathbf{u} = (u, \partial_t u)$.

If $(\phi_0, \phi_1) \in \mathcal{H}^1(\mathbb{T}^2)$ we would immediately obtain an a priori bound, since

$$\|\mathbf{u}(t)\|_{L_{\omega}^2 \mathcal{H}_x^1}^2 \leq 2\varepsilon^{-2} E(\mathbf{u}(t)) = 2\varepsilon^{-2} E(\mathbf{u}(0)) \lesssim 2\varepsilon^{-2} \|(\phi_0, \phi_1)\|_{L_{\omega}^2 \mathcal{H}_x^1}^2 < \infty.$$

However, we encounter three challenges:

1. First, Proposition 4.1 only yields $v(t) \in L^2(\Omega; H^{1-\delta}(\mathbb{T}^2))$ for $0 < \delta \leq \frac{1}{2}$, hence, the energy functional (5.2) is ill-defined. We follow the approach in [15] employing the I-method to find a global a priori bound for the energy functional in the context of the cubic SNLW.
2. Second, v satisfies (5.1) with perturbations. In particular, v does not conserve (5.2). Again following [15], we show that $E(\mathbf{v})$ can be bounded sufficiently well using a Gronwall-type argument.
3. Finally, our main concern, as the former two are being addressed in [15], the bound is not uniform in ε . However, upon closer inspection, only the term $\|\partial_t u\|_{L_\omega^2 L_x^2}$ cannot be bounded uniformly in ε , whereas $\|u(t)\|_{L_\omega^2 H_x^1}^2 \leq 2E(\mathbf{u}(t))$ for any $\varepsilon > 0$. In particular, if we can bound $E(\mathbf{v}(t))$ uniformly in ε , Proposition 4.3 implies that $v_\varepsilon(t) \rightarrow v_0(t)$ in $L^2(\Omega; H^{1-\delta}(\mathbb{T}^2))$ for $0 < \delta < \frac{1}{2}$ and all $t \in [0, T]$.

5.1 The I-Method

The I-operator applied to the solution $\mathbf{v} := (v, \partial_t v)$ to (2.20) improves the regularity such that $E(I\mathbf{v})$ is well-defined.

Definition 5.1 (I-operator). *Let $0 < \delta < 1$. Let $N \in \mathbb{N}$ and $m_N \in C^\infty(\mathbb{R}^2; [0, 1])$ be a radially symmetric, non-increasing function satisfying*

$$m_N(\xi) = \begin{cases} 1, & \text{if } |\xi| \leq N, \\ \left(\frac{N}{|\xi|}\right)^\delta, & \text{if } |\xi| \geq 2N. \end{cases}$$

We define the I-operator $I := I_N$ to be the Fourier multiplier operator with multiplier m_N :

$$\widehat{I_N f}(n) := m_N(n) \widehat{f}(n). \quad (5.3)$$

Let $\delta \in (0, \frac{1}{2})$. Given initial data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$, we consider the I-SDMFW:

$$\begin{cases} \varepsilon^2 \partial_t^2 I v + \partial_t I v + (1 - \Delta) I v + I(\mathbb{E}[v^2]v) + 2I(\mathbb{E}[v\Psi]v) + 2I(\mathbb{E}[v\Psi]\Psi) + I(\mathbb{E}[v^2]\Psi) = 0 \\ (v, \partial_t v)|_{t=0} = (\phi_0, \phi_1). \end{cases} \quad (5.4)$$

Since I acts simply as a linear Fourier multiplier on the equation, any $\mathbf{v}$ solving the SDMFW is also a solution to the I-SDMFW. In particular, from Proposition 4.1 it follows that there exists $T > 0$ and a solution $v \in L^2(\Omega; C([0, 1] \times [0, T]; H^{1-\delta}(\mathbb{T}^2)))$ to (5.4) in the sense of (2.20). Denote

$$E(t) := E(I\mathbf{v}) = \frac{1}{2} \mathbb{E} \left[\int_{\mathbb{T}^2} (\varepsilon \partial_t I v)^2 + (I v)^2 + |\nabla I v|^2 dx \right] + \frac{1}{4} \int_{\mathbb{T}^2} (\mathbb{E}[(I v)^2])^2 dx. \quad (5.5)$$

From Definition (5.3), a direct computation shows that $I f \in H^1(\mathbb{T}^2)$ if and only if $f \in H^{1-\delta}(\mathbb{T}^2)$ with the bounds

$$\|f\|_{H_x^{1-\delta}} \lesssim \|I f\|_{H_x^1} \lesssim N^\delta \|f\|_{H_x^{1-\delta}}. \quad (5.6)$$

By Littlewood-Paley theory, for all $s_0 \in \mathbb{R}$, $0 \leq s_1 \leq \delta$, and $1 < p < \infty$, we have

$$\|If\|_{W^{s_0+s_1,p}} \lesssim N^{s_1} \|f\|_{W^{s_0,p}}, \quad (5.7)$$

$$\|f\|_{W^{s_0,p}} \lesssim \|If\|_{W^{s_0+\delta,p}}. \quad (5.8)$$

Thus, finding a bound on $E(t)$ immediately yields a bound on $\|\mathbf{v}(t)\|_{L_\omega^2 \mathcal{H}_x^{1-\delta}}$:

$$\|\mathbf{v}(t)\|_{L_\omega^2 \mathcal{H}_x^{1-\delta}}^2 \lesssim \|I\mathbf{v}(t)\|_{L_\omega^2 \mathcal{H}_x^1}^2 \leq 2\varepsilon^{-2} E(t).$$

To find such a bound, we use the equation satisfied by the I -SDMFW (5.4). Assume that $\frac{d}{dt} E(t)$ is integrable and continuous. Then, it follows from Lebesgue's differentiation theorem that

$$E(t) - E(t_0) = \int_{t_0}^t \frac{d}{dt'} E(t') dt'.$$

Adding $\mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right]$ and assuming that v satisfies the I -SDMFW (5.4) in a classical sense, we obtain

$$\begin{aligned} E(t) - E(t_0) + \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] \\ = \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} \partial_t I v \cdot (\mathbb{E}[(I v)^2] I v - I(\mathbb{E}[v^2] v)) dx dt' \right] \\ - \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} \partial_t I v \cdot (I(\mathbb{E}[v^2] \Psi) + 2I(\mathbb{E}[\Psi v] v)) dx dt' \right] \\ - 2\mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} \partial_t I v \cdot I(\mathbb{E}[\Psi v] \Psi) dx dt' \right] \\ := B_1 - B_2 - B_3. \end{aligned} \quad (5.9)$$

This equality is a priori unjustified, as we do not yet know if the derivatives are well-defined. Furthermore, to apply Lebesgue's differentiation theorem, we need the integrand on the right-hand side to be continuous. In the next section, we rigorously show that this equality holds.

At this point, we remark that the right-hand side depends on ε only via v and Ψ , there is no explicit factor of ε . Hence, to bound B_1 , B_2 , and B_3 , we can follow the steps in [5] closely. An issue arises if we attempt to bound $\partial_t I v$ using $E(I\mathbf{v})$, as this would introduce an ε factor. Fortunately, by utilizing Young's product inequality and absorbing terms into the damping term $+\mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right]$ on the left-hand side, we obtain bounds independent of ε .

5.2 The Energy estimate

To establish that equation (5.9) holds true, we follow a standard regularization approach. We show that we can approximate our irregular solution to (5.4) by a smooth solution. Then, we show that equation (5.9) holds for this smooth solution.

Before proving this, we collect several estimates required both to justify equation (5.9) and

to derive the necessary global a priori bound on $E(t)$.

Define

$$B_4 := \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} \partial_t I v \left(\mathbb{E}[(Iv)^2] I \Psi + 2\mathbb{E}[I \Psi I v] I v \right) dx dt' \right]. \quad (5.10)$$

First, we need estimates on the commutative terms. Let us record some useful estimates for the I -operator. The following lemma was proved in [15, Lemma 3.1].

Lemma 5.2. *Let $0 < \delta \leq \frac{1}{3}$. Then, we have*

$$\|(If)^2 Ig - I(f^2 g)\|_{L^2} \leq C(\delta) N^{-1+3\delta} \|If\|_{H^1}^2 \|Ig\|_{H^1}$$

for some constant $C(\delta) > 0$ depending only on δ .

We also have the following lemma, whose proof follows from [15, Lemma 3.2 and Lemma 3.3].

Lemma 5.3. *Let $0 < \delta \leq \frac{1}{3}$. Then, given a small $\nu > 0$, there exist a small $\sigma_0 = \sigma_0(\nu) > 0$ and a large $p_0 = p_0(\nu) > 1$ such that*

$$\|I(fh) - If Ih\|_{L^2} \leq C(\delta, \nu) N^{-\frac{1}{2} + \delta + \nu} \|If\|_{H^1} \|h\|_{W^{-\sigma_0, p_0}} \quad (5.11)$$

and

$$\|I(fgh) - If Ig Ih\|_{L^2} \leq C(\delta, \nu) N^{-\frac{1}{2} + \delta + \nu} \|If\|_{H^1} \|Ig\|_{H^1} \|h\|_{W^{-\sigma_0, p_0}} \quad (5.12)$$

for any sufficiently large $N \gg 1$, and some constant $C(\delta, \nu) > 0$ depending only on δ and ν .

In the following series of lemmas, we estimate the terms B_1 , B_2 , B_3 , and B_4 .

Lemma 5.4. *Let $0 < \delta \leq \frac{1}{3}$, $N \in \mathbb{N}$, and $0 \leq t_0 \leq t \leq T$. Then, for B_1 defined in (5.9), we have*

$$|B_1| \leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + C(\delta) N^{-2+6\delta} \int_{t_0}^t E(t')^3 dt'$$

for some constant $C(\delta) > 0$.

Proof. By Minkowski's integral inequality and Lemma 5.2, we have

$$\begin{aligned} & \mathbb{E} \left[\left\| \mathbb{E}[(Iv)^2] I v - I(\mathbb{E}[v^2] v) \right\|_{L_x^2}^2 \right] \\ & \leq \int_{\Omega} \left(\int_{\Omega} \left\| (Iv(\omega'))^2 I v(\omega) - I(v(\omega')^2 v(\omega)) \right\|_{L_x^2} d\omega' \right)^2 d\omega \\ & \leq C(\delta) \int_{\Omega} \left(\int_{\Omega} N^{-1+3\delta} \|Iv(\omega')\|_{H_x^1}^2 \|Iv(\omega)\|_{H_x^1} d\omega' \right)^2 d\omega \\ & = C(\delta) N^{-2+6\delta} \|Iv\|_{L_{\omega}^2 H_x^1}^6. \end{aligned} \quad (5.13)$$

for some constant $C(\delta) > 0$.

Thus, by Young's inequality, (5.13), and the definition of the energy functional $E(t)$ in (5.5), we obtain

$$\begin{aligned} |B_1| &\leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + \int_{t_0}^t \mathbb{E} \left[\left\| \mathbb{E}[(I v)^2] I v - I(\mathbb{E}[v^2] v) \right\|_{L_x^2}^2 \right] dt' \\ &\leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + C(\delta) N^{-2+6\delta} \int_{t_0}^t E(t')^3 dt'. \end{aligned}$$

□

Lemma 5.5. *Let $0 < \nu \leq \delta \leq \frac{1}{3}$, and $0 \leq t_0 \leq t \leq T$, such that $|t - t_0| \leq 1$. Let $\sigma_0 = \sigma_0(\nu) > 0$, $p_0 = p_0(\nu) > 1$, and $N \gg 1$ be given as in Lemma 5.3. Then, we have*

$$|B_2 - B_4| \leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + C(\delta, \nu) N^{-1+2\delta+2\nu} \int_{t_0}^t E(t')^2 dt'$$

for some constant $C(\nu, \delta, T) > 0$.

Proof. By Minkowski's integral inequality, (5.12) in Lemma 5.3, and the Cauchy-Schwarz inequality in ω' , we have

$$\begin{aligned} &\mathbb{E} \left[\left\| I(\mathbb{E}[\Psi v] v) - \mathbb{E}[I \Psi I v] I v \right\|_{L_x^2}^2 \right] \\ &\leq \int_{\Omega} \left(\int_{\Omega} \left\| I(\Psi(\omega') v(\omega') v(\omega)) - I \Psi(\omega') I v(\omega') I v(\omega) \right\|_{L_x^2} d\omega' \right)^2 d\omega \\ &\leq C_1(\delta, \nu) N^{-1+2\delta+2\nu} \int_{\Omega} \left(\int_{\Omega} \left\| \Psi(\omega') \right\|_{W_x^{-\sigma_0, p_0}} \left\| I v(\omega') \right\|_{H_x^1} \left\| I v(\omega) \right\|_{H_x^1} d\omega' \right)^2 d\omega \\ &\leq C_1(\delta, \nu) N^{-1+2\delta+2\nu} \|I v\|_{L_{\omega}^2 H_x^1}^4 \|\Psi\|_{L_{\omega}^2 W_x^{-\sigma_0, p_0}}^2. \end{aligned} \tag{5.14}$$

for some constant $C_1(\delta, \nu) > 0$. Using similar steps, we obtain

$$\mathbb{E} \left[\left\| I(\mathbb{E}[v^2] \Psi) - \mathbb{E}[(I v)^2] I \Psi \right\|_{L_x^2}^2 \right] \leq C_1(\delta, \nu) N^{-1+2\delta+2\nu} \|I v\|_{L_{\omega}^2 H_x^1}^4 \|\Psi\|_{L_{\omega}^2 W_x^{-\sigma_0, p_0}}^2. \tag{5.15}$$

Thus, by Youngs' product inequality, (5.15), (5.14), (5.5), and Proposition 3.7, we get

$$\begin{aligned} &|B_2 - B_4| \\ &\leq \frac{1}{8} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + 2 \int_{t_0}^t \mathbb{E} \left[\left\| I(\mathbb{E}[v^2] \Psi) - \mathbb{E}[(I v)^2] I \Psi \right\|_{L_x^2}^2 \right] dt' \\ &\quad + \frac{1}{8} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + 8 \int_{t_0}^t \mathbb{E} \left[\left\| I(\mathbb{E}[\Psi v] v) - \mathbb{E}[I \Psi I v] I v \right\|_{L_x^2}^2 \right] dt' \\ &\leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + C_1(\delta, \nu) N^{-1+2\delta+2\nu} \|\Psi\|_{L_{\omega}^2 C_{[t_0, t]} W_x^{-\sigma_0, p_0}}^2 \int_{t_0}^t E(t')^2 dt' \\ &\leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + C_2(\delta, \nu) N^{-1+2\delta+2\nu} \int_{t_0}^t E(t')^2 dt' \end{aligned}$$

for some constant $C_2(\delta, \nu) > 0$.

□

Lemma 5.6. *Let $0 < \delta < 1$. It holds that for $N \geq 2$*

$$|B_4| \leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + C(\delta) \log N \int_{t_0}^t E(t')^{1+2\eta} dt'$$

uniformly in $0 < \eta \leq \frac{1}{2}$, $t \geq t_0 \geq 0$, and some constant $C(\delta) > 0$.

Proof. By Hölder's inequality in x , (5.5), Minkowski's integral inequality, and Sobolev's inequality, we have

$$\begin{aligned} \|\mathbb{E}[(Iv)^2]\|_{L_x^{\frac{2}{1-\eta}}} &= \|\mathbb{E}[(Iv)^2]^{1-2\eta} \mathbb{E}[(Iv)^2]^{2\eta}\|_{L_x^{\frac{2}{1-\eta}}} \\ &\leq \|\mathbb{E}[(Iv)^2]^{1-2\eta}\|_{L_x^{\frac{2}{1-2\eta}}} \|\mathbb{E}[(Iv)^2]^{2\eta}\|_{L_x^{\frac{2}{\eta}}} \\ &= \|\mathbb{E}[(Iv)^2]\|_{L_x^2}^{1-2\eta} \|Iv\|_{L_x^8 L_\omega^2}^{4\eta} \\ &\leq C E^{\frac{1-2\eta}{2}} \|Iv\|_{L_\omega^2 H_x^1}^{4\eta} \\ &\leq C E^{\frac{1+2\eta}{2}}. \end{aligned} \tag{5.16}$$

for η such that $0 < \eta \leq \frac{1}{2}$ and some constant $C > 0$ independent of η . Then, applying the Cauchy-Schwarz inequality in ω , Hölder's inequality in x , Youngs' product inequality, and (5.16), we obtain

$$\begin{aligned} |B_4| &\leq 3 \int_{t_0}^t \int_{\mathbb{T}^2} \|\partial_t I v\|_{L_\omega^2} \|Iv\|_{L_\omega^2}^2 \|I\Psi\|_{L_\omega^2} dx dt' \\ &\leq 3 \int_{t_0}^t \|\partial_t I v\|_{L_x^2 L_\omega^2} \|\mathbb{E}[(Iv)^2]\|_{L_x^{\frac{2}{1-\eta}}} \|I\Psi\|_{L_x^{\frac{2}{\eta}} L_\omega^2} dt' \\ &\leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + 9 \int_{t_0}^t \|\mathbb{E}[(Iv)^2]\|_{L_x^{\frac{2}{1-\eta}}}^2 \|I\Psi\|_{L_x^{\frac{2}{\eta}} L_\omega^2}^2 dt' \\ &\leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + 9C \int_{t_0}^t E(t')^{1+2\eta} \|I\Psi\|_{L_x^{\frac{2}{\eta}} L_\omega^2}^2 dt'. \end{aligned}$$

To estimate $\|I\Psi(t)\|_{L_x^{\frac{2}{\eta}} L_\omega^2}^2$, write $\Psi = \mathbf{P}_n \Psi + \mathbf{P}_n^\perp \Psi$. In the low frequency regime (2.16) yields

$$\mathbb{E}[(I\mathbf{P}_N \Psi(x, t))^2] = \mathbb{E}[(\mathbf{P}_N \Psi(x, t))^2] \sim \log N$$

uniformly in $x \in \mathbb{T}^2$ and $t \in [0, \infty)$. In the high frequency regime, it follows from the Itô isometry and (3.2) that

$$\mathbb{E}[(I\mathbf{P}_N^\perp \Psi(x, t))^2] = \int_0^t \sum_{|n|>N} \mathbb{E}[|\widehat{\Psi}(n, t')|^2] m_N^2(n) dt' \lesssim \sum_{|n|>N} \frac{N^{2\delta}}{(2|n|)^{2+2\delta}} \lesssim C(\delta)$$

uniformly in $x \in \mathbb{T}^2$ and $t \in [0, \infty)$, for some constant $C(\delta) > 0$. Combining both estimates

yields

$$\|I\Psi(t)\|_{L_x^{\frac{2}{\gamma}} L_\omega^2}^2 \lesssim C(\delta) \log N.$$

This proves Lemma 5.6. $\square$

Lemma 5.7. *Let $0 < \gamma \leq \delta \leq \frac{1}{3}$, $N \in \mathbb{N}$, and $0 \leq t_0 \leq t \leq T$, such that $|t - t_0| \leq 1$. Then, we have*

$$|B_3| \leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + C(\gamma) N^{2\gamma} \int_{t_0}^t E(t')^{\frac{1}{2} + \frac{\delta}{2} + \frac{\gamma}{2}} dt'$$

for some constant $C(\gamma, T) > 0$.

Proof. We proceed as in Proposition 4.1 (or rather as in [4, Lemma 3.1]) by writing

$$\mathbb{E}[\Psi v] \Psi = \mathbb{E}[v' \Psi' \Psi | \Psi],$$

where (Ψ', v') is an independent copy of (Ψ, v) . Thus, by (5.7), Jensen's inequality, the product estimate in Lemma 3.1 (ii), and conditional Hölder's inequality, we have

$$\begin{aligned} \|I(\mathbb{E}[\Psi v] \Psi)\|_{L_\omega^2 L_x^2} &= \|I(\mathbb{E}[v' \Psi' \Psi | \Psi])\|_{L_\omega^2 L_x^2} \\ &\lesssim N^\gamma \|\mathbb{E}[\|v' \Psi' \Psi\|_{H_x^{-\gamma}} | \Psi]\|_{L_\omega^2} \\ &\lesssim_\gamma N^\gamma \|\mathbb{E}[\|v'\|_{H_x^\gamma} \|\Psi' \Psi\|_{W_x^{-\gamma, \infty}} | \Psi]\|_{L_\omega^2} \\ &\leq N^\gamma \|v\|_{L_\omega^2 H_x^\gamma} \|\Psi' \Psi\|_{L_\omega^2 W_x^{-\gamma, \infty}}. \end{aligned} \tag{5.17}$$

We bound $\|v\|_{L_\omega^2 H_x^\gamma}$ by using (5.6), Gagliardo-Nirenberg interpolation, the continuous embedding $L^4(\mathbb{T}^2) \rightarrow L^2(\mathbb{T}^2)$, and (5.5) to obtain

$$\begin{aligned} \|v\|_{L_\omega^2 H_x^\gamma} &\lesssim \|Iv\|_{L_\omega^2 H_x^{\gamma+\delta}} \\ &\lesssim \|Iv\|_{L_\omega^2 H_x^1}^{\gamma+\delta} \|Iv\|_{L_\omega^2 L_x^2}^{1-\gamma-\delta} \\ &\leq \|Iv\|_{L_\omega^2 H_x^1}^{\gamma+\delta} \|Iv\|_{L_x^4 L_\omega^2}^{1-\gamma-\delta} \\ &\lesssim E^{\frac{1}{4} + \frac{\gamma}{4} + \frac{\delta}{4}}. \end{aligned} \tag{5.18}$$

Thus, by Youngs' product inequality, (5.17), (5.18), and Proposition 3.7, we get

$$\begin{aligned} |B_3| &\leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + 4 \int_{t_0}^t \|I(\mathbb{E}[\Psi v] \Psi)\|_{L_\omega^2 L_x^2}^2 dt' \\ &\leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + C_1(\gamma) N^{2\gamma} \|\Psi' \Psi\|_{L_\omega^2 C_{[t_0, t]} W_x^{-\gamma, \infty}}^2 \int_{t_0}^t E(t')^{\frac{1}{2} + \frac{\gamma}{2} + \frac{\delta}{2}} dt' \\ &\leq \frac{1}{4} \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] + C_2(\gamma) N^{2\gamma} \int_{t_0}^t E(t')^{\frac{1}{2} + \frac{\gamma}{2} + \frac{\delta}{2}} dt' \end{aligned}$$

for some constants $C_1(\gamma), C_2(\gamma) > 0$, as desired. $\square$

Now, we proceed to verify equation (5.9). First, we demonstrate that if we begin with smooth initial data, the solution to the residual SDMFV remains smooth. To achieve this, we must work with an extended set of initial data, specifically, we replace the stochastic convolution Ψ_ε with a smooth approximation. Since we verify equation (5.9) for a fixed ε , we treat ε as a constant in the following discussion and omit it from the notation of the smooth approximations.

Definition 5.8 (Mild Solution of the Residual SDMFV with respect to an extended set of initial data). *Let $p \geq 1$, $s \in \mathbb{R}$. Let $P(t)$ and $\mathcal{I}(t)$ be defined as in (2.4) and (2.7), respectively. Given an extended set of initial data $(\phi_0, \phi_1) \in \mathcal{H}^s(\mathbb{T}^2)$ and a function $\Psi \in L^2(\Omega; C([0, T]; W^{s,p}(\mathbb{T}^2)))$, we say that $(v, \partial_t v)$ is a mild solution of the Residual SDMFV with respect to an extended set of initial data if it satisfies*

$$\begin{cases} v(t) = P(t)(\phi_0, \phi_1) - \mathcal{I}(t) ((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi)) . \\ \partial_t v(t) = \partial_t P(t)(\phi_0, \phi_1) - \partial_t \mathcal{I}(t) ((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi)) \\ (v, \partial_t v)|_{t=0} = (\phi_0, \phi_1) \end{cases} \quad (5.19)$$

in an appropriate space.

Proposition 5.9 (Persistence of regularity). *Let $s \geq 1$, $p \geq 2$, $T > 0$, $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^s(\mathbb{T}^2))$, and $\Psi \in L^2(\Omega; C([0, T]; W^{s,p}(\mathbb{T}^2)))$ be an extended set of initial data. If $(v, \partial_t v) \in L^2(\Omega; C([0, T]; \mathcal{H}^{\frac{3}{4}}(\mathbb{T}^2)))$ is a solution in the sense of Definition 5.8 with respect to the extended set of initial data, then $(v, \partial_t v) \in L^\infty([0, T]; L^2(\Omega; \mathcal{H}^s(\mathbb{T}^2)))$.*

Proof. Fix $t \in [0, T]$. Then Proposition 3.6 (3.10) and Lemma 3.5 (3.5) yield

$$\begin{aligned} \|v(t)\|_{L_\omega^2 H_x^s} &\leq \|P(t)(\phi_0, \phi_1)\|_{L_\omega^2 H_x^s} + \|\mathcal{I}(t)((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi))\|_{L_\omega^2 H_x^s} \\ &\lesssim \|(\phi_0, \phi_1)\|_{\mathcal{H}_x^s} + \int_0^t \|(\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi)(t')\|_{L_\omega^2 H_x^{s-1}} dt'. \end{aligned}$$

To estimate the term within the time integral, observe that from Sobolev's embedding theorem and Lemma 3.1 1 it follows that for $f, g \in H^s(\mathbb{T}^2)$

$$\|fg\|_{H_x^{s-1}} \lesssim \|fg\|_{W^{s-\frac{1}{2}, \frac{4}{3}}} \leq \|f\|_{H^{s-\frac{1}{2}}} \|g\|_{L^4} + \|g\|_{H^{s-\frac{1}{2}}} \|f\|_{L^4}.$$

Repeating this argument with a third term $h \in H^s(\mathbb{T}^2)$, we obtain

$$\|fgh\|_{H_x^{s-1}} \lesssim_{\|f\|_{L_x^4}, \|g\|_{L_x^8}, \|h\|_{L_x^8}} \|f\|_{H_x^s} + \|g\|_{H_x^s} + \|h\|_{H_x^s}. \quad (5.20)$$

Since $\Psi \in L^2(\Omega; C([0, T]; W^{s,p}(\mathbb{T}^2)))$, for $p \geq 2$, and $v \in L^2(\Omega; C([0, T]; H^{\frac{3}{4}}(\mathbb{T}^2)))$ Sobolev's embedding theorem yields $\|v\|_{L_\omega^2 C_T L_x^8} \lesssim \|v\|_{L_\omega^2 C_T H_x^{\frac{3}{4}}}$ and $\|\Psi\|_{L_\omega^2 C_T L_x^8} \lesssim \|\Psi\|_{L_\omega^2 C_T W_x^{s,p}}$. Thus, applying (5.20) repeatedly along with Minkowski's integral inequality, we obtain

$$\|v(t)\|_{L_\omega^2 H_x^s} \lesssim \|(\phi_0, \phi_1)\|_{\mathcal{H}_x^s} + \int_0^t \|v(t')\|_{L_\omega^2 H_x^s} dt'.$$

Therefore, Gronwall's inequality yields $v \in L^\infty([0, T]; L^2(\Omega; H^s(\mathbb{T}^2)))$.

Next, we show the statement for $\partial_t v = \partial_t P(t)(\phi_0, \phi_1) + \partial_t \mathcal{I}(t)((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi))$ where we treat the two summands separately. From Lemma 3.5 (3.5), (3.6) and (3.7) it follows that

$$\begin{aligned} \|\partial_t P(t)(\phi_0, \phi_1)\|_{L_\omega^2 H_x^{s-1}} &\lesssim \|\mathcal{D}(t)\phi_0\|_{L_\omega^2 H_x^{s-1}} + \|\partial_t \mathcal{D}(t)\phi_1\|_{L_\omega^2 H_x^{s-1}} + \|\partial_t^2 \mathcal{D}(t)\phi_0\|_{L_\omega^2 H_x^{s-1}} \\ &\lesssim \|\phi_0\|_{H_x^{s-1}} + \|\phi_1\|_{H_x^{s-1}} + \|\phi_0\|_{H_x^s} \lesssim \|(\phi_0, \phi_1)\|_{\mathcal{H}_x^s}. \end{aligned}$$

Applying Lemma 3.5 (3.6) we use (5.20) again along with Minkowski's integral inequality to obtain

$$\|\partial_t \mathcal{I}(t)((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi))\|_{L_\omega^2 H_x^{s-1}} \lesssim_t \|v\|_{L_T^\infty L_\omega^2 H_x^s}.$$

Thus, $\partial_t v \in L^\infty([0, T]; L^2(\Omega; H^{s-1}(\mathbb{T}^2)))$. $\square$

Next, we prove that an irregular solution can be approximated by a smooth one.

Proposition 5.10 (Continuity with respect to initial data). *Let $p \geq 2$, $0 < \delta < \frac{1}{4}$, $T > 0$, $\sigma > 0$, and $(v, \partial_t v) \in L^2(\Omega; C([0, T]; \mathcal{H}^{1-\delta}(\mathbb{T}^2)))$ be a solution to (2.20) with respect to the initial data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$.*

Furthermore, let $(v_n, \partial_t v_n) \in L^2(\Omega; C([0, T]; \mathcal{H}^{1-\delta}(\mathbb{T}^2)))$ be a solution in the sense of Definition 5.8 with respect to the extended set of initial data $(\phi_{0,n}, \phi_{1,n}) \in L^2(\Omega; \mathcal{H}^\infty(\mathbb{T}^2))$, Ψ be as in (2.15), and $\Psi_n \in L^2(\Omega; C([0, T]; W^{\infty,p}(\mathbb{T}^2)))$ such that

$$\begin{aligned} \|(\phi_0, \phi_1) - (\phi_{0,n}, \phi_{1,n})\|_{L_\omega^2 \mathcal{H}_x^{1-\delta}} &\rightarrow 0, \\ \|\Psi - \Psi_n\|_{L_\omega^2 C_T W_x^{-\sigma,p}} &\rightarrow 0, \end{aligned}$$

as $n \rightarrow \infty$. Then,

$$\|(v, \partial_t v) - (v_n, \partial_t v_n)\|_{L_T^\infty L_\omega^2 \mathcal{H}_x^{1-\delta}} \rightarrow 0$$

as $n \rightarrow \infty$.

Proof. The proof that $\|v - v_n\|_{L_T^\infty L_\omega^2 H_x^{1-\delta}} \rightarrow 0$ as $n \rightarrow \infty$ follows from a Gronwall argument similar to the one presented in Proposition 4.7, though it is simpler here as v_n and v satisfy the same Duhamel formulation. The convergence $\|\partial_t v - \partial_t v_n\|_{L_T^\infty L_\omega^2 H_x^{1-\delta}} \rightarrow 0$ as $n \rightarrow \infty$ follows from computations similar to those for the $\partial_t v$ -term in Proposition 4.1. $\square$

In the next proposition, we show that the smooth solution satisfies equation (5.9).

Proposition 5.11. *Let $T > 0$, $p \geq 2$, $0 < \delta < \frac{1}{4}$, and assume $(v, \partial_t v) \in L^2(\Omega; C([0, T]; \mathcal{H}^{1-\delta}(\mathbb{T}^2)))$ is a solution in the sense of Definition 5.8 with respect to the extended set of initial data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^\infty(\mathbb{T}^2))$ and $\Psi \in L^2(\Omega; C([0, T]; W^{\infty,p}(\mathbb{T}^2)))$. Then $(Iv, I\partial_t v)$ satisfies equation (5.9).*

Proof. First, we show that $(Iv, I\partial_t v)$ satisfies equation (5.4) in an $L^2(\Omega; L^2(\mathbb{T}^2))$ -sense. To do this, we differentiate the Duhamel formulation. In particular, we must compute $\partial_t^2 Iv$ and $\Delta \partial_t Iv$ and verify that the terms are well-defined in $L^\infty([0, T]; L^2(\Omega; L^2(\mathbb{T}^2)))$. Since this argument does not depend on the ε factors, we omit them in the following estimates.

Furthermore, as ϕ_0 and ϕ_1 have the same regularity, we can assume without loss of generality that $\phi_0 = \phi_1$. Let $t \in [0, T]$. To bound $\|\partial_t^2 I v(t)\|_{L_\omega^2 L_x^2}$, we must find bounds for terms of the form

$$\|I \partial_t^2 \mathcal{D}(t) \phi_0\|_{L_\omega^2 L_x^2}, \quad \|I \partial_t^3 \mathcal{D}(t) \phi_0\|_{L_\omega^2 L_x^2}, \text{ and} \quad (5.21)$$

$$\left\| \int_0^t \partial_t^2 \mathcal{D}(t-t') I((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi)) dt' \right\|_{L_\omega^2 L_x^2}. \quad (5.22)$$

Lemma 3.5 (3.7) and (3.8) yield, respectively, with (5.7) that the terms (5.21) are bounded (as we assume that the initial data is smooth, namely, $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^\infty(\mathbb{T}^2))$).

To bound the integral term (5.22), we use Lemma 3.5 (3.7), (5.7) and the same computations as in Proposition 4.1 to compute

$$\begin{aligned} & \left\| \int_0^t \partial_t^2 \mathcal{D}(t-t') I((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi)) dt' \right\|_{L_\omega^2 L_x^2} \\ & \lesssim t \|I((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi))\|_{L_\omega^2 C_T L_x^2} \\ & \lesssim_N t \|(\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi)\|_{L_\omega^2 C_T H_x^{-\delta}} \\ & \lesssim_N T(1 + \|v\|_{L_\omega^2 C_T H_x^{1-\delta}})^3 \\ & < \infty. \end{aligned}$$

Computing $\|\Delta I v(t)\|_{L_\omega^2 L_x^2}$ we see that we have to bound terms of the form

$$\|I \Delta \mathcal{D}(t) \phi_0\|_{L_\omega^2 L_x^2}, \quad \|I \Delta \partial_t \mathcal{D}(t) \phi_0\|_{L_\omega^2 L_x^2}, \text{ and} \quad (5.23)$$

$$\left\| \int_0^t \Delta \mathcal{D}(t-t') I((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi)) dt' \right\|_{L_\omega^2 L_x^2}. \quad (5.24)$$

Lemma 3.5 (3.5) and (3.6) yield, respectively, with (5.7) that the terms in (5.23) are bounded. To bound the integral term (5.24), we use Lemma 3.5 (3.5), (5.7), and the same computations as in Proposition 5.9 to compute

$$\begin{aligned} & \left\| \int_0^t \Delta \mathcal{D}(t-t') I((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi)) dt' \right\|_{L_\omega^2 L_x^2} \\ & \lesssim t \|I((\mathbb{E}[v^2] + 2\mathbb{E}[v\Psi])(v + \Psi))\|_{L_T^\infty L_\omega^2 H_x^1} \\ & \lesssim_N T \|v\|_{L_T^\infty L_\omega^2 H_x^{2-\delta}}. \end{aligned}$$

Using persistence of regularity (Proposition 5.9), the smooth initial data guarantees that $\|v\|_{L_T^\infty L_\omega^2 H_x^{2-\delta}} < \infty$.

Let $0 \leq t_0 < t \leq T$. These computations justify differentiating the Duhamel formulations of $(Iv, \partial_t Iv)$, which implies that $\int_{t_0}^t \frac{d}{dt} E(t') + \|\partial_t Iv(t')\|_{L_\omega^2 L_x^2}^2 dt'$ satisfies equation (5.9) and is finite. Furthermore, Lemmas 5.4, 5.5, 5.6, and 5.7, combined with the assumption $(Iv, \partial_t Iv) \in L^2(\Omega; C([0, T]; \mathcal{H}^1(\mathbb{T}^2)))$ imply that $\frac{d}{dt} E(t)$ is continuous. This justifies using Lebesgue's differentiation theorem and interchanging differentiation and integration in ω

and x to obtain

$$\begin{aligned}
E(t) - E(t_0) + \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] &= \int_{t_0}^t \frac{d}{dt} E(t') dt' + \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] \\
&= \int_{t_0}^t \mathbb{E} \left[\int_{\mathbb{T}^2} \partial_t I v(t') (\varepsilon^2 \partial_t^2 I v(t') + \partial_t I v(t') + (1 - \Delta) I v(t') + \mathbb{E}[(I v(t'))^2] I v(t')) dx \right] dt' \\
&= (5.9).
\end{aligned}$$

□

Finally, we combine the previous results to show that a solution to (2.20) with respect to the low regularity initial data $(\phi_0, \phi_1) \in \mathcal{H}^{1-\delta}(\mathbb{T}^2)$ for $0 < \delta < \frac{1}{4}$ satisfies (5.9).

Proposition 5.12. *Let $0 < \delta < \frac{1}{4}$. Given initial data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$, there exists $T > 0$ and $(v, \partial_t v) \in L^2(\Omega; C([0, T]; \mathcal{H}^{1-\delta}(\mathbb{T}^2)))$ such that equation (5.9) holds true.*

Proof. From Proposition 4.1 it follows that there exists some $T_1 > 0$ and a solution $(v, \partial_t v) \in L^2(\Omega; C([0, T_1]; \mathcal{H}^{1-\delta}(\mathbb{T}^2)))$ to the Residual SDMFV in the sense of (2.20). Let $\sigma > 0$. By density, we can find sequences $(\phi_{0,n}, \phi_{1,n})_{n \in \mathbb{N}} \subset L^2(\Omega; \mathcal{H}^\infty(\mathbb{T}^2))$, and $(\Psi_n)_{n \in \mathbb{N}} \in L^2(\Omega; C([0, T_1]; W^{\infty,p}(\mathbb{T}^2)))$ such that they converge to the initial data (ϕ_0, ϕ_1) in $L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$ and to the stochastic convolution Ψ_ε in $L^2(\Omega; C([0, T_1]; W^{-\sigma,p}(\mathbb{T}^2)))$, respectively. (Recall that the proof of Proposition 3.7 implies $\Psi_\varepsilon \in L^2(\Omega; C([0, T]; W^{-\sigma,p}(\mathbb{T}^2)))$ for $p \geq 2$). Similar to Proposition 4.1, one can show that there exists $T_2 > 0$ and a solution $(v_n, \partial_t v_n) \in L^2(\Omega, C([0, T_2]; \mathcal{H}^{1-\delta}(\mathbb{T}^2)))$ in the sense of Definition 5.8 with respect to the extended initial data $(\phi_0, \phi_1, \Psi)_n$ for all $n \in \mathbb{N}$. Set $T = \min(T_1, T_2)$.

Denote

$$E_n(t) := \frac{1}{2} \mathbb{E} \left[\int_{\mathbb{T}^2} (\varepsilon \partial_t I v_n)^2 + (I v_n)^2 + |\nabla I v_n|^2 dx \right] + \frac{1}{4} \int_{\mathbb{T}^2} (\mathbb{E}[(I v_n)^2])^2 dx.$$

Using (5.6), we note that Proposition 5.10 applies to $(I v, \partial_t I v)$. Thus, approximating $(I v, \partial_t I v)$ with $(I v_n, \partial_t I v_n)$ combined with Proposition 5.11 yields

$$\begin{aligned}
E(t) - E(t_0) + \int_{t_0}^t \mathbb{E} \left[\int_{\mathbb{T}^2} (\partial_t I v)^2 dx \right] dt' &= \lim_{n \rightarrow \infty} \left(E_n(t) - E_n(t_0) + \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v_n)^2 dx dt' \right] \right) \\
&= \lim_{n \rightarrow \infty} \int_{t_0}^t \mathbb{E} \left[\int_{\mathbb{T}^2} \partial_t I v_n \left((\mathbb{E}[(I v_n)^2] I v_n - I(\mathbb{E}[v_n^2] v_n)) - (I(\mathbb{E}[v_n^2] \Psi_n) + 2I(\mathbb{E}[\Psi_n v_n] v_n)) \right. \right. \\
&\quad \left. \left. - 2I(\mathbb{E}[\Psi_n v_n] \Psi_n) \right) dx dt' \right]
\end{aligned} \tag{5.25}$$

for all $0 \leq t_0 \leq T$. Note that we can approximate $\mathbb{E}[\int_{\mathbb{T}^2} (\partial_t I v)^2 dx]$ under the integral because $(I v_n, \partial_t I v_n)$ and $(I v, I \partial_t v)$ are bounded uniformly in time.

Finally, using Lemmas (5.4), (5.5), (5.6), and (5.7) combined with the fact that $(Iv_n, I\partial_t v_n) \rightarrow (Iv, I\partial_t v)$ in $L^\infty([0, T]; L^2(\Omega; H^1(\mathbb{T}^2)))$, we see that dominated convergence applies. Consequently, the final limit (5.25) converges to the desired term. $\square$

Now, we are ready to prove GWP of the SDMFV.

5.3 Global Well-Posedness of the Mean Field SKA

We adopt the approach in [15], who applied an iterative scheme to prove GWP of the cubic SNLW. The following proposition is based on Proposition 3.5 in [15], however, as we are working with a damped wave equation instead of a simple wave equation, the damping effect allows us to modify the proof so that the time step τ no longer depends on the global time T .

Assuming we have a bound on the energy functional as defined in (5.2) for some initial time, this Proposition yields a bound on the energy functional after a time step which, importantly, is independent of the initial bound. Later, this allows us to iterate these time steps to obtain a global bound without risking that the time steps converge to zero.

Proposition 5.13. *Let $0 < \delta < \frac{1}{3}$, $T > 1$. Then there exist constants α and β satisfying $0 < \beta < \alpha < 1 - 3\delta$, and a time step $0 < \tau = \tau(\delta, \beta, \alpha) \leq 1$, such that for $N \gg 1$, if*

$$E(t_0) \leq N^\beta$$

for some $0 \leq t_0 < T$, then

$$E(t) \leq N^\alpha$$

for all t satisfying $t_0 \leq t \leq \min(T, t_0 + \tau)$.

Proof. Assuming without loss of generality that $E(t) \geq 1$ for all $t \geq 0$ (otherwise, replace E with $E + 1$). Since $N^\alpha > N^\beta$, the continuity of $E(t)$ implies there exists some $t_1 \in (t_0, T]$ such that

$$E(t) \leq 2N^\alpha \tag{5.26}$$

for all $t_0 \leq t \leq t_1$. We may assume that $t_1 - t_0 \leq 1$.

Set $\nu := \frac{1}{2}\delta$ and let $\eta \in (0, \frac{1}{2}]$ and $0 < \gamma \leq \delta$ be constants to be specified later. Let $N \gg 1$ be sufficiently large such that, combined with these constants, we can apply Lemmas 5.4,

5.5, 5.6, and 5.7 to (5.9). It follows that for all $t_0 \leq t \leq t_1$

$$\begin{aligned}
E(t) - E(t_0) + \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right] &\leq C(\delta) N^{-2+6\delta} \int_{t_0}^t E(t')^3 dt' \\
&\quad + C(\delta) N^{-1+3\delta} \int_{t_0}^t E(t')^2 dt' \\
&\quad + C(\delta) \log N \int_{t_0}^t E(t')^{1+2\eta} dt' \\
&\quad + C(\delta) N^{2\gamma} \int_{t_0}^t E(t')^{\frac{1}{2} + \frac{\delta}{2} + \frac{\gamma}{2}} dt' \\
&\quad + \mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right].
\end{aligned} \tag{5.27}$$

In particular, we can just subtract the $\mathbb{E} \left[\int_{t_0}^t \int_{\mathbb{T}^2} (\partial_t I v)^2 dx dt' \right]$ term from both sides. In the following, our goal is to establish a differential inequality to obtain a Gronwall-type inequality.

Define

$$F(t) = \max_{t_0 \leq \tau \leq t} E(\tau) - E(t_0) + N^\beta. \tag{5.28}$$

Note that

$$N^\beta \leq F(t) \leq 3N^\alpha, \quad \text{and consequently,} \quad \log F(t) \sim \log N. \tag{5.29}$$

Fix $\gamma < \min \left(\frac{(1-\delta)\beta}{4+\beta}, \delta \right)$, which allows us to apply Lemma 5.7 as $\gamma < \delta$ and yields $-2\gamma\beta(\frac{1}{2} - \frac{\delta}{2} - \frac{\gamma}{2}) \leq 0$. If necessary enlarging $N \gg 1$ we define $\eta := (2\alpha \log N)^{-1} \leq \frac{1}{2}$. Then, from the fact that $\alpha < 1 - 3\delta$ and (5.29) it follows that

$$\begin{aligned}
N^{-2+6\delta} F(t)^3 &\leq N^{-2\alpha} F(t)^3 \leq 9F(t) \\
N^{-1+3\delta} F(t)^2 &\leq N^{-\alpha} F(t)^2 \leq 3F(t) \\
N^{2\gamma} F(t)^{\frac{1}{2} + \frac{\delta}{2} + \frac{\gamma}{2}} &\leq N^{2\gamma} F(t) N^{-\beta(\frac{1}{2} - \frac{\delta}{2} - \frac{\gamma}{2})} \leq F(t). \\
\log(N) F(t)^{1+2\eta} &\leq 3 \log(N) F(t) N^{\frac{1}{\log N}} \leq 3e \log(N) F(t)
\end{aligned} \tag{5.30}$$

Next, we combine (5.28), (5.27), (5.30), and (5.29) to obtain

$$\begin{aligned}
F(t) - F(t_0) &= \max_{t_0 \leq \tau \leq t} (E(\tau) - E(t_0)) \lesssim_{\delta, \beta, \alpha} \int_{t_0}^t F(t') (1 + \log N) dt' \\
&\lesssim_{\delta, \beta, \alpha} \int_{t_0}^t F(t') (1 + \log F(t')) dt'.
\end{aligned} \tag{5.31}$$

Comparing the above inequality to the ODE

$$G(t) = G(0) + C \int_0^t G(t') (1 + \log G(t')) dt'$$

with explicit solution

$$G(t) = \exp(e^{Ct}(1 + \log G(0)) - 1),$$

we infer the Gronwall-type inequality

$$F(t) \leq \exp \left[e^{C_{\delta, \beta, \alpha}(t-t_0)}(1 + \log F(t_0)) - 1 \right] \quad (5.32)$$

where $C_{\delta, \beta, \alpha}$ is the implicit constant from (5.31).

The fact that $\beta < \alpha$ implies $1 + \beta \log N < 1 + \alpha \log N$. Consequently, we can choose t sufficiently close to t_0 such that

$$e^{C_{\delta, \beta, \alpha}(t-t_0)}(1 + \beta \log N) - 1 \leq \alpha \log N.$$

Combining this with (5.32) and recalling that $F(t_0) = N^\beta$, yields

$$E(t) \leq F(t) \leq N^\alpha.$$

In particular, there exists $0 < \tau = \tau(\delta, \beta, \alpha) \leq 1$ such that for all $t_0 \leq t \leq \min(t_1, t_0 + \tau)$

$$E(t) \leq N^\alpha.$$

By a standard continuity argument we can extend this result to hold for all $t_0 \leq t \leq t_0 + \tau$: Assume toward contradiction that the statement does not hold for all $t_0 \leq t \leq t_0 + \tau$. In particular, we may choose $t_1 < t_0 + \tau$ maximal such that for $t \in [t_0, t_1]$, $F(t) \leq N^\alpha$ and for $t \in (t_1, t_0 + \tau]$ it holds that $F(t) > N^\alpha$. Since $F(t_1) \leq N^\alpha$, continuity of F implies there exists $t_2 > t_1$ such that $F(t) \leq 2N^\alpha$ for all $t \in [t_1, t_2]$. By repeating the argument above we see that $F(t) \leq N^\alpha$ for $t \in [t_1, \min(t_2, t_1 + \tau)]$ in contradiction to the prior assumption. $\square$

Now, that we have fixed-size time steps (only depending on the constants δ , α , and β , which we can fix a priori) that allow to enlarge the interval on which $E(t)$ is bounded on, we are ready to implement the iterative scheme to prove GWP of the Residual SDMFV.

Proposition 5.14. *Let $0 < \delta < \frac{1}{5}$, $0 < \varepsilon \leq 1$, and $T > 0$. Given initial data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$ there exists a unique solution $\mathbf{v}_\varepsilon := (v_\varepsilon, \partial_t v_\varepsilon) \in L^2(\Omega; C([0, T]; \mathcal{H}^{1-\delta}(\mathbb{T}^2)))$ to the Residual SDMFV in the sense of (2.20) for all $\varepsilon \in (0, 1]$. Furthermore,*

$$\sup_{\varepsilon \in (0, 1]} \|v_\varepsilon\|_{L_\omega^2 C_T H_x^{1-\delta}} < \infty.$$

Proof. First, we establish an a priori bound on $\|\mathbf{v}_\varepsilon(t)\|_{L_\omega^2 \mathcal{H}_x^{1-\delta}}$ for all $0 < t \leq T$. Then, we iterate a local well-posedness argument similar to Proposition 4.1.

Let $N_0 \gg 1$ and $c > 0$ be specified later. We define an increasing sequence $\{N_k\}_{k \in \mathbb{N}}$:

$$N_k = N_0^{\lambda^k}$$

with $\lambda > \max(\frac{2\alpha}{\beta}, \frac{\alpha}{\beta-2\delta})$, $0 < \beta < \alpha \leq 1 - 3\delta$ such that

$$N_{k+1}^{2\delta} N_k^\alpha + N_k^{2\alpha} \leq c^{-1} N_{k+1}^\beta \quad (5.33)$$

which requires $\beta > 2\delta$. Combined with the condition on α , this restricts $\delta < \frac{1}{5}$.

Let I_N be defined as in (5.3). Suppose that $E(I_{N_k} \mathbf{v}_\varepsilon)(t) \leq N_k^\alpha$. Then inserting the definition (5.5) of $E(t)$ in the first line, applying (5.6), (5.7), Jensen's inequality, (5.7), and Sobolev's inequality in the second line, (5.6) and (5.8) in the third line, Cauchy-Schwarz in ω and the definition of $E(t)$ in the fourth line, and the assumption $E(I_{N_k} \mathbf{v})(t) \leq N_k^\alpha$ in the last line, we obtain

$$\begin{aligned}
E(I_{N_{k+1}} \mathbf{v}_\varepsilon)(t) &= \frac{1}{2} \mathbb{E} \left[\|I_{N_{k+1}} v_\varepsilon\|_{H_x^1}^2 + \varepsilon^2 \|I_{N_{k+1}} \partial_t v_\varepsilon\|_{L_x^2}^2 \right] + \frac{1}{4} \int_{\mathbb{T}^2} (\mathbb{E}[(I_{N_{k+1}} v_\varepsilon)^2])^2 dx \\
&\lesssim \frac{1}{2} N_{k+1}^{2\delta} \mathbb{E} \left[\|v_\varepsilon\|_{H_x^{1-\delta}}^2 \right] + \frac{1}{2} \varepsilon^2 N_{k+1}^{2\delta} \mathbb{E} \left[\|\partial_t v_\varepsilon\|_{H_x^{-\delta}}^2 \right] + \frac{1}{4} \mathbb{E} \left[\|v_\varepsilon\|_{H_x^{1-\delta}}^4 \right] \\
&\lesssim \frac{1}{2} N_{k+1}^{2\delta} \mathbb{E} \left[\|I_{N_k} v_\varepsilon\|_{H_x^1}^2 + \varepsilon^2 \|I_{N_k} \partial_t v_\varepsilon\|_{L_x^2}^2 \right] + \frac{1}{4} \mathbb{E} \left[\|I_{N_k} v_\varepsilon\|_{H_x^1}^4 \right] \\
&\lesssim N_{k+1}^{2\delta} E(I_{N_k} \mathbf{v}_\varepsilon)(t) + E(I_{N_k} \mathbf{v}_\varepsilon)(t)^2 \\
&\leq c(N_{k+1}^{2\delta} N_k^\alpha + N_k^{2\alpha})
\end{aligned} \tag{5.34}$$

where $c > 0$ is chosen to be sufficiently large such that the inequalities hold.

From $\mathbf{v}_\varepsilon(0) = (\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$, (5.6), (5.8), Minkowski's integral inequality, Sobolev's embedding Theorem, and the fact that $\beta > 2\delta$, it follows that we can choose $N_0 \in \mathbb{N}$ such that

$$E(I_{N_0} \mathbf{v}_\varepsilon)(0) \lesssim N_0^{2\delta} \|(\phi_0, \phi_1)\|_{L_\omega^2, \mathcal{H}_x^{1-\delta}}^2 \leq N_0^\beta.$$

Choosing $N_0 \gg 1$ sufficiently large, we can apply Proposition 5.13 which yields

$$E(I_{N_0} \mathbf{v}_\varepsilon)(t) \leq N_0^\alpha$$

for all $0 \leq t \leq t + \tau$. Note that τ depends only on α, β and δ .

From (5.34) and (5.33) it follows

$$E(I_{N_1} \mathbf{v}_\varepsilon)(\tau) \leq N_1^\beta.$$

Applying Proposition 5.13 again, we have

$$E(I_{N_1} \mathbf{v}_\varepsilon)(t) \leq N_1^\alpha$$

for all $0 \leq t \leq 2\tau$.

Iterating this argument $k = \lfloor \frac{T}{\tau} \rfloor + 1$ times, we obtain the bound

$$E(I_{N_k} \mathbf{v}_\varepsilon)(t) \leq N_k^\alpha \tag{5.35}$$

for all $0 \leq t \leq T$.

From Proposition 4.1, we know there exists a time $T_0 = T_0(\|(\phi_0, \phi_1)\|_{L_\omega^2, \mathcal{H}_x^{1-\delta}}) > 0$ and a solution $v \in L^2(\Omega; C([0, 1] \times [0, T_0]; H^{1-\delta}(\mathbb{T}^2)))$ and $\partial_t v_\varepsilon \in L^2(\Omega; C([0, T_0]; H^{1-\delta}(\mathbb{T}^2)))$ for all $\varepsilon \in (0, 1]$, to the Residual SDMFV in the sense of (2.20) with respect to the initial data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$. (5.35) yields bounds for $\|(v_\varepsilon(T_0), \partial_t v_\varepsilon(T_0))\|_{L_\omega^2, \mathcal{H}_x^{1-\delta}}$ only depending on $\|(\phi_0, \phi_1)\|_{L_\omega^2, \mathcal{H}_x^{1-\delta}}$, however, $\|\partial_t v_\varepsilon(T_0)\|_{L_\omega^2, \mathcal{H}_x^{-\delta}}$ is not uniformly bounded in $\varepsilon \in$

$(0, 1]$. Consequently, to obtain a bound uniform in ε , we must adjust the local well-posedness proof from Proposition 4.1 to account for this fact.

Define the space

$$X := L^\infty((0, 1]; L^2(\Omega; C([T_0, T_1]; H^{1-\delta}(\mathbb{T}^2))))),$$

for a time $T_1 > 0$, we specify later. Note that the function $w_\varepsilon \in X$ is uniformly bounded in $L^2(\Omega; C([T_0, T_1]; H^{1-\delta}(\mathbb{T}^2)))$ in the approximation parameter ε . Consider the closed ball $\overline{B_R} \subset X$ with radius $R > 0$. We define the map $\Gamma : \overline{B_R} \rightarrow \overline{B_R}$ as in (4.1), setting the initial data to $\phi_0 = v(T_0)$ and $\phi_1 = \partial_t v(T_0)$.

Lemma 3.5 (3.5), (3.6), and (5.35) yield

$$\begin{aligned} \|P_\varepsilon(t)(\phi_0, \phi_1)\|_X^2 &= \|(\varepsilon^{-2} + \partial_t)\mathcal{D}_\varepsilon(t)\phi_0 + \mathcal{D}_\varepsilon(t)\phi_1\|_X^2 \\ &\leq 2 \sup_{\varepsilon \in (0,1]} \mathbb{E} \left[\sup_{t \in [T_0, T_1]} (\|(\varepsilon^{-2} + \partial_t)\mathcal{D}_\varepsilon(t)\phi_0\|_{H_x^{1-\delta}}^2 + \|\mathcal{D}_\varepsilon(t)\phi_1\|_{H_x^{1-\delta}}^2) \right] \\ &\lesssim 2 \sup_{\varepsilon \in (0,1]} \mathbb{E} \left[\sup_{t \in [T_0, T_1]} (2\|\phi_0\|_{H_x^{1-\delta}}^2 + \varepsilon^2\|\phi_1\|_{H_x^{-\delta}}^2) \right] \\ &\lesssim 4 \sup_{\varepsilon \in (0,1]} \mathbb{E} \left[\|I\phi_0\|_{H_x^1}^2 + \varepsilon^2\|I\phi_1\|_{L_x^2}^2 \right] \\ &\lesssim 8E(I_{N_k}\mathbf{v})(T_0) \lesssim 8N_k^\alpha. \end{aligned}$$

Analogously to Proposition 4.1, it follows that

$$\|\Gamma v\|_X \leq CN_k^{\frac{\alpha}{2}} + C\sqrt{T_1 - T_0}(1 + \|v\|_X)^3,$$

and also, for all $v, w \in \overline{B_R}$

$$\|\Gamma v - \Gamma w\|_X \leq C\sqrt{T_1 - T_0}\|v - w\|_X(1 + R^2)$$

for some constant $C > 0$ sufficiently large and independent of T_0, T_1 , and ε .

Fixing $R := 2CN_k^{\frac{\alpha}{2}}$, we can choose T_1 such that the time increment $\tau := T_1 - T_0 > 0$ satisfies

$$\tau < \min \left(N_k^{\frac{\alpha}{2}}(1 + 2CN_k^{\frac{\alpha}{2}})^{-3}, C^{-1}(1 + 4C^2N_k^\alpha)^{-1} \right)$$

to show that $\Gamma : \overline{B_R} \rightarrow \overline{B_R}$ is a well-defined contraction map. Hence, from Banach's fixedpoint theorem it follows that there exists $v \in \overline{B_R}$ such that $\Gamma v = v$. By the same arguments as in Proposition 4.1, $(v_\varepsilon, \partial_\varepsilon v)$ is a unique solution to the Residual SDFW (2.20) in $L^2(\Omega; C(T_0, T_1); \mathcal{H}^{1-\delta}(\mathbb{T}^2))$ for all $\varepsilon \in (0, 1]$ and by the definition of X it follows that

$$\sup_{\varepsilon \in (0,1]} \|v_\varepsilon\|_{L_\omega^2 C_{[T_0, T_1]} H_x^{1-\delta}} < \infty.$$

Since $\tau > 0$ has a fixed size (as N_k^α is the global a priori bound), we can iterate this local well-posedness argument to obtain a global solution. $\square$

Remark 5.15. Note that we require the initial data to be in $L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$ with $0 < \delta < \frac{1}{5}$ for the GWP of the Residual SDMFw, in contrast to the LWP result, which only requires initial data in the same space with $0 < \delta \leq \frac{1}{2}$.

Finally, we show that there exists a globally well-defined family of functions $\{u_\varepsilon\}_{\varepsilon \in (0,1]}$ that satisfies the mean field SKA in the sense of Definition 2.4.

Theorem 5.16. *Let $0 < \delta < \frac{1}{5}$ and $T > 0$. Given initial data $(\phi_0, \phi_1) \in L^2(\Omega; \mathcal{H}^{1-\delta}(\mathbb{T}^2))$ there exists for each $\varepsilon \in (0, 1]$ a solution $u_\varepsilon \in L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ in the sense of Definition 2.2 for any $\sigma > 0$ and a solution u_0 in the sense of Definition 2.3 in the same space. Moreover, $\{u_\varepsilon\}_{\varepsilon \in (0,1]}$ converges to u_0 in $L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ as $\varepsilon \rightarrow 0$.*

Proof. Let $\varepsilon \in (0, 1]$ and $\sigma > 0$. From Proposition 3.7 and Proposition 5.14, it follows that there exists $\Psi_\varepsilon \in L^2(\Omega; C([0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$ and $v_\varepsilon \in L^2(\Omega; C([0, T]; H^{1-\delta}(\mathbb{T}^2)))$. Thus, we obtain $u_\varepsilon := v_\varepsilon + \Psi_\varepsilon \in L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ satisfying Definition 2.2.

Let $u_0 := v_0 + \Psi_0 \in L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ satisfy Definition 2.3 (whose existence is justified in [4] Theorem 3.6). From Proposition 5.14, we know that $\|v_\varepsilon\|_{L^\infty_{\omega} C_T H_x^{1-\delta}}$ is bounded uniformly in ε . Therefore, recalling Remark 4.4, we can apply Proposition 4.3 to obtain the convergence $v_\varepsilon \rightarrow v_0$ in $L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$. Combining this with Proposition 3.7, which establishes that $\Psi_\varepsilon \rightarrow \Psi_0$ in $L^2(\Omega; C([0, T]; W^{-\sigma, \infty}(\mathbb{T}^2)))$, we conclude that $u_\varepsilon \rightarrow u_0$ in $L^2(\Omega; C([0, T]; H^{-\sigma}(\mathbb{T}^2)))$ as $\varepsilon \rightarrow 0$. $\square$

6 Invariant Dynamics of the Mean Field SKA

The first goal of this chapter is to prove that there is a unique invariant measure of the SDMFw under the condition that this measure satisfies $M_\mu := \mathbb{E}_\mu[:u^2:] \in H^{-\sigma}(\mathbb{T}^2)$ for $\sigma \in (0, \frac{1}{2})$, where we normalize with respect to the Wick renormalization defined in (2.25). In Section 2.3, we already established that proving this uniqueness boils down to understanding the linear SDE:

$$dX(t) := d \begin{pmatrix} u_\varepsilon(t) \\ \partial_t u_\varepsilon(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\varepsilon^{-2}L & -\varepsilon^{-2} \end{pmatrix} X(t)dt + \sqrt{2} \begin{pmatrix} 0 \\ \varepsilon^{-2} \end{pmatrix} dW_t,$$

where we define the operator $L := \Lambda + M_\mu$, with $\Lambda := 1 - \Delta$.

To prove existence of a unique invariant measure, we proceed as follows: In Section 6.1, we fix M_μ as an element in $H^{-\sigma}(\mathbb{T}^2)$ and show that there exists a mild solution to the linear SDE. Since we are analyzing a linear stochastic differential equation, we employ the framework described in [20]. To adapt this formalism, we first define an appropriate Hilbert space $\mathcal{H}_L$ and then show that there exists a mild solution to the linear SDE on this Hilbert space. In order to show that (2.26) admits a unique mild solution, we must verify two conditions ([20, Chapter 11.2]):

i) The operator

$$A := \begin{pmatrix} 0 & 1 \\ -\varepsilon^{-2}L & -\varepsilon^{-2} \end{pmatrix}$$

generates a strongly continuous semigroup e^{tA} on $\mathcal{H}_L$.

- ii) The stochastic convolution integral is well-defined, meaning $\int_0^t e^{(t-s)A} B dW_s \in L^2(\Omega; \mathcal{H}_L)$ for all $t \geq 0$, where $B := \sqrt{2} \begin{pmatrix} 0 \\ \varepsilon^{-2} \end{pmatrix}$. By the Itô isometry, this is equivalent to showing that the trace of the covariance operator Q_t is finite (see [20, Chapter 4.4]), namely,

$$\text{Tr}(Q_t) := \text{Tr} \left(\int_0^t e^{sA} B B^* (e^{sA})^* ds \right) < \infty \quad \text{for all } t \geq 0.$$

In Section 6.2, we first prove that for a fixed $M_\mu \in H^{-\sigma}(\mathbb{T}^2)$, the linear SDE admits a unique invariant measure that we can explicitly characterize. In a second step, we show that this invariant measure is equal to $\rho_\varepsilon := \mathcal{N} \left(0, \begin{pmatrix} \Lambda^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix} \right)$ for any fixed $\varepsilon \in (0, 1]$.

As the second aim of this chapter, we show in Section 6.3 that $\rho_\varepsilon \rightarrow \rho_0$ as $\varepsilon \rightarrow 0$ in the sense of Definition 2.6.

6.1 Existence of a Mild Solution

To show that A generates a strongly continuous semigroup, we must first understand the behavior of the operator L and define the Hilbert space $\mathcal{H}_L$. Note that L is just the operator Λ perturbed by a real-valued multiplication operator M_μ . Since Λ is a well-understood operator on $L^2(\mathbb{T}^2)$ with many nice properties, it would be ideal if L inherited these properties from Λ . To demonstrate that this is indeed the case, we employ the KLMN Theorem ([24, Theorem 10.21]). If we can satisfy the following assumptions:

- $\langle M_\mu u, v \rangle_{L_x^2}$ is a densely defined symmetric form on $L^2(\mathbb{T}^2)$,
- $\langle M_\mu u, v \rangle_{L_x^2}$ is relatively bounded, with relative bound $0 < a < 1$ with respect to $\langle \Lambda u, v \rangle_{L_x^2}$, specifically,

$$\langle M_\mu u, u \rangle_{L_x^2} \leq a \langle \Lambda u, u \rangle_{L_x^2} + b \|u\|_{L_x^2}^2$$

for all u in the form domain $\mathcal{Q}(\Lambda)$ of Λ for some $b > 0$,

then, $L = \Lambda + M_\mu$ is a self-adjoint operator bounded from below on $L^2(\mathbb{T}^2)$, and the domain of its quadratic form, which we denote by $\mathcal{Q}(L)$, is equal to the form domain of Λ . Since M_μ is a real-valued multiplication operator, it is symmetric. Consequently, we only need to show its relative boundedness (the proof of which will also imply that $\langle M_\mu u, v \rangle_{L_x^2}$ is densely defined), which follows from the following lemma:

Lemma 6.1. *Suppose $M_\mu \in H^{-\sigma}(\mathbb{T}^2)$ for some $\sigma \in (0, 1)$. Then the quadratic form associated with M_μ is relatively bounded with respect to Λ . Specifically, for any arbitrarily small $\delta > 0$, there exists a constant $C_\delta > 0$ such that for all $u \in H^1(\mathbb{T}^2)$:*

$$|\langle M_\mu u, u \rangle_{L_x^2}| \leq \delta \langle \Lambda u, u \rangle_{L_x^2} + C_\delta \|u\|_{L_x^2}^2.$$

Proof. Let $\varepsilon > 0$ be such that $2\varepsilon + \sigma \leq 1$. By the duality pairing between $H^{-1+\varepsilon}(\mathbb{T}^2)$ and $H^{1-\varepsilon}(\mathbb{T}^2)$, Lemma 3.1 2 with $p \geq \frac{2}{1-\varepsilon}$, and Sobolev's inequality, we have

$$|\langle M_\mu u, u \rangle_{L_x^2}| \leq \|M_\mu u\|_{H_x^{-1+\varepsilon}} \|u\|_{H_x^{1-\varepsilon}} \lesssim \|M_\mu\|_{W_x^{-1+\varepsilon, p}} \|u\|_{H_x^{1-\varepsilon}}^2 \lesssim \|M_\mu\|_{H_x^{-\sigma}} \|u\|_{H_x^{1-\varepsilon}}^2. \quad (6.1)$$

Interpolating between fractional Sobolev spaces, for any $\tilde{\delta} > 0$, there exists a constant $C(\tilde{\delta}) > 0$ such that

$$\|u\|_{H_x^{1-\epsilon}}^2 \leq \tilde{\delta}\|u\|_{H_x^1}^2 + C(\tilde{\delta})\|u\|_{L_x^2}^2.$$

Substituting this bound back into 6.1, we obtain

$$|\langle M_\mu u, u \rangle_{L_x^2}| \leq C_0 \|M_\mu\|_{H_x^{-\sigma}} \left(\tilde{\delta}\|u\|_{H_x^1}^2 + C(\tilde{\delta})\|u\|_{L_x^2}^2 \right),$$

for some constant $C_0 > 0$. Recognizing that $\|u\|_{H^1}^2 = \langle \Lambda u, u \rangle_{L^2}$, we can choose $\tilde{\delta}$ sufficiently small such that the statement follows. $\square$

Next, we define the Hilbert space $\mathcal{H}_L$ and show that the eigenvalues of L and Λ have the same asymptotic growth behavior, which guarantees the existence of a mild solution to (2.26) as we will see later. Since we have established that L is a self-adjoint operator on $L^2(\mathbb{T}^2)$ bounded from below, there exists $c > 0$ such that $L_c := L + c$ is strictly positive, and the Spectral Theorem ([24, Proposition 5.12]) implies that $(L + c)^{-s}$ is a well-defined self-adjoint bounded operator on $L^2(\mathbb{T}^2)$ for any $s \geq 0$. We define $\mathcal{H}_L$ as the closure of $L^2(\mathbb{T}^2) \times L^2(\mathbb{T}^2)$ with respect to the norm induced by the inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle_{L_c} := \langle L_c^{-\sigma} u_1, v_1 \rangle_{L_x^2} + \langle L_c^{-1-\sigma} u_2, v_2 \rangle_{L_x^2},$$

with respect to the normalized inner product on $L^2(\mathbb{T}^2)$:

$$\langle u, v \rangle_{L_x^2} := (2\pi)^{-2} \int_{\mathbb{T}^2} u(x) \overline{v(x)} dx.$$

The next Lemma establishes that the forms $\langle L_c u, v \rangle_{L_x^2}$ and $\langle \Lambda u, v \rangle_{L_x^2}$ are equivalent, implying through the min-max principle that the eigenvalues of these operators are also equivalent.

Lemma 6.2. *Suppose $M_\mu \in H^{-\sigma}(\mathbb{T}^2)$ for $\sigma \in (0, 1)$. There exists $c > 0$ such that $\Lambda := 1 - \Delta$ and $L_c := M_\mu + \Lambda + c$ are form equivalent, in the sense that $\langle \Lambda u, u \rangle_{L_x^2} \sim \langle L_c u, u \rangle_{L_x^2}$ for all $u \in \mathcal{Q}(\Lambda)$. Furthermore, the eigenvalues $\lambda_n(L_c)$ of L_c and $\lambda_n(\Lambda)$ are equivalent, in the sense that there exist $C_1, C_2 > 0$ such that*

$$C_1 \lambda_n(\Lambda) \leq \lambda_n(L_c) \leq C_2 \lambda_n(\Lambda) \quad \text{for all } n \in \mathbb{N}.$$

In particular, the eigenvalues belong to the same asymptotic growth class.

Proof. Lemma 6.1 and the KLMN Theorem yield that L is a self-adjoint operator on $L^2(\mathbb{T}^2)$ that is bounded from below. Thus, there exists $c > 0$ such that $L_c = L + c$ is strictly positive. Note, that Lemma 6.1 applies to L_c and that L_c remains self-adjoint. From Lemma 6.1, it follows that the quadratic form associated with $M_\mu + c$ is relatively bounded by the form associated to Λ with relative bound $a < 1$. Inserting this relative bound yields the following inequalities:

$$(1 - a) \langle \Lambda u, u \rangle_{L_x^2} - b \|u\|_{L_x^2}^2 \leq \langle L_c u, u \rangle_{L_x^2} \leq (1 + a) \langle \Lambda u, u \rangle_{L_x^2} + b \|u\|_{L_x^2}^2$$

for some $b > 0$.

Absorbing the $L^2(\mathbb{T}^2)$ terms into the strictly positive forms, namely, using that $\|u\|_{L_x^2}^2 \leq \langle \Lambda u, u \rangle_{L_x^2}$ for the upper bound and that $\|u\|_{L_x^2}^2 \leq \langle L_c u, u \rangle_{L_x^2}$ for the lower bound, guarantees the existence of constants $C_1, C_2 > 0$ such that

$$C_1 \langle \Lambda u, u \rangle_{L_x^2} \leq \langle L_c u, u \rangle_{L_x^2} \leq C_2 \langle \Lambda u, u \rangle_{L_x^2} \quad (6.2)$$

for all $u \in \mathcal{Q}(\Lambda)$.

By the Min-Max Principle, it follows directly that the asymptotic behavior of the eigenvalues is the same. For the operator Λ , the n -th eigenvalue $\lambda_n(\Lambda)$ (ordered such that $\lambda_1 \leq \lambda_2 \leq \dots$) is given by

$$\lambda_n(\Lambda) = \min_{\substack{U \subset \mathcal{Q}(\Lambda) \\ \dim(U)=n}} \max_{\substack{u \in U \\ \|u\|=1}} \langle \Lambda u, u \rangle_{L_x^2}.$$

Since we have established the form equivalence, we can apply this directly to obtain for any n -dimensional subspace $U \subset \mathcal{Q}(\Lambda)$:

$$\max_{\substack{u \in U \\ \|u\|=1}} \langle L_c u, u \rangle_{L_x^2} \leq \max_{\substack{u \in U \\ \|u\|=1}} C_2 \langle \Lambda u, u \rangle_{L_x^2} = C_2 \max_{\substack{u \in U \\ \|u\|=1}} \langle \Lambda u, u \rangle_{L_x^2}.$$

Taking the minimum over all such n -dimensional subspaces U on both sides yields

$$\lambda_n(L_c) \leq C_2 \lambda_n(\Lambda).$$

Applying the same logic to the lower bound $C_1 \langle \Lambda u, u \rangle_{L_x^2} \leq \langle L_c u, u \rangle_{L_x^2}$, we obtain $C_1 \lambda_n(\Lambda) \leq \lambda_n(L_c)$.

Combining these inequalities gives

$$C_1 \lambda_n(\Lambda) \leq \lambda_n(L_c) \leq C_2 \lambda_n(\Lambda) \quad \text{for all } n \in \mathbb{N}.$$

□

Using this lemma, we can show that A generates a strongly continuous semigroup on $\mathcal{H}_L$ and that $\text{Tr } Q_t < \infty$ for all $t > 0$. In particular, we can show that a mild solution to (2.26) exists.

Proposition 6.3. *Suppose $M_\mu \in H^{-\sigma}(\mathbb{T}^2)$ for $\sigma \in (0, 1)$. Then a unique mild solution to (2.26) exists and is given by*

$$X(t) = e^{tA} X(0) + \int_0^t e^{(t-s)A} B dW_s.$$

Proof. First, we show that A generates a strongly continuous semigroup on $\mathcal{H}_L$. Lemma 6.1 and the KLMN Theorem establish that L is a self-adjoint operator on $L^2(\mathbb{T}^2)$ bounded from below, hence, there exists a constant $c \geq 0$ such that the shifted operator $L_c := L + c$ is a self-adjoint, strictly positive operator. We can decompose $A := A_c + B_c$, where

$$A_c := \begin{pmatrix} 0 & 1 \\ -\varepsilon^{-2} L_c & -\varepsilon^{-2} \end{pmatrix} \quad \text{and} \quad B_c := \begin{pmatrix} 0 & 0 \\ \varepsilon^{-2} c & 0 \end{pmatrix}.$$

Define the Hilbert space $\mathcal{H}_\varepsilon$ as the closure of $L^2(\mathbb{T}^2) \times L^2(\mathbb{T}^2)$ with respect to the inner product:

$$\langle \mathbf{u}, \mathbf{v} \rangle_\varepsilon := \varepsilon^{-2} \langle L_c^{-\sigma} u_1, v_1 \rangle_{L_x^2} + \langle L_c^{-1-\sigma} u_2, v_2 \rangle_{L_x^2}.$$

Note that this Hilbert space is topologically equivalent to $\mathcal{H}_L$ as the inner products are equivalent. To show that A_c is the infinitesimal generator of a strongly continuous semigroup we apply the Lumer-Phillips Theorem ([25, Chapter 1 Theorem 4.3]). We show that A_c is dissipative with respect to the $\langle \mathbf{u}, \mathbf{v} \rangle_{L_\varepsilon}$ -inner product. For any $\mathbf{u} = (u_1, u_2)^T \in \mathcal{D}(A_c)$, we compute

$$\begin{aligned} \langle A_c \mathbf{u}, \mathbf{u} \rangle_\varepsilon &= \left\langle \begin{pmatrix} u_2 \\ -\varepsilon^{-2} L_c u_1 - \varepsilon^{-2} u_2 \end{pmatrix}, \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \right\rangle_\varepsilon \\ &= \varepsilon^{-2} \langle L_c^{-\sigma} u_2, u_1 \rangle_{L_x^2} - \varepsilon^{-2} \langle L_c^{-\sigma} u_1, u_2 \rangle_{L_x^2} - \varepsilon^{-2} \langle L_c^{-1-\sigma} u_2, u_2 \rangle_{L_x^2}. \end{aligned}$$

Taking the real part, the self-adjointness of L_c ensures that the cross-terms cancel, yielding

$$\operatorname{Re} \langle A_c \mathbf{u}, \mathbf{u} \rangle_\varepsilon = -\varepsilon^{-2} \langle L_c^{-1-\sigma} u_2, u_2 \rangle_{L_x^2} \leq 0,$$

since L_c is strictly positive. Therefore, A_c is dissipative in the space $\mathcal{H}_\varepsilon$. Since L_c is self-adjoint and strictly positive definite, it is also straightforward to verify that $\operatorname{Ran}(\lambda I - A_c) = \mathcal{H}_\varepsilon$ for $\lambda > 0$. By the Lumer-Phillips Theorem, A_c generates a strongly continuous contraction semigroup on $\mathcal{H}_\varepsilon$. Because the norms are equivalent, A_c also generates a strongly continuous semigroup on the original space $\mathcal{H}_L$ (however, it might not be a contraction anymore).

Finally, we need to show that $A = A_c + B_c$ also generates a strongly continuous semigroup. First, note that the operator B_c maps $\mathcal{H}_L$ into itself. Since L^{-1} is a bounded operator, we have $\langle L_c^{-1-\sigma} u, u \rangle_{L_x^2} \lesssim \langle L_c^{-\sigma} u, u \rangle_{L_x^2}$. Consequently, for $\mathbf{u} \in \mathcal{H}_L$, we obtain

$$\|B_c \mathbf{u}\|_{\mathcal{H}_L} = \varepsilon^{-2} c \langle L_c^{-1-\sigma} u_1, u_1 \rangle_{L_x^2} \lesssim \varepsilon^{-2} c \|\mathbf{u}\|_{\mathcal{H}_L}.$$

This proves that B_c is a bounded linear operator on $\mathcal{H}_L$. By Theorem 1.1 in [25, Chapter 3], the operator $A = A_c + B_c$ generates a strongly continuous semigroup on $\mathcal{H}_L$.

In the second part of this proof, we show that the stochastic convolution is well-defined by proving $\operatorname{Tr}(Q_t) < \infty$ for all $t \geq 0$. First, observe that W_t is defined as a cylindrical Wiener process on $L^2(\mathbb{T}^2)$. Consequently, the operator $B = \sqrt{2} \begin{pmatrix} 0 \\ \varepsilon^{-2} \end{pmatrix}$ maps $L^2(\mathbb{T}^2) \rightarrow \mathcal{H}_L$. By the standard trace property $\operatorname{Tr}(e^{sA} B (e^{sA} B)^*) = \operatorname{Tr}((e^{sA} B)^* e^{sA} B)$, we have for any orthonormal basis $(f_n)_{n \in \mathbb{N}}$ of $L^2(\mathbb{T}^2)$:

$$\operatorname{Tr}(Q_t) = \operatorname{Tr} \left(\int_0^t e^{sA} B B^* (e^{sA})^* ds \right) = \int_0^t \operatorname{Tr}(B^* (e^{sA})^* e^{sA} B) ds = \sum_{m \in \mathbb{N}} \int_0^t \|e^{sA} B f_m\|_{\mathcal{H}_L}^2 ds. \quad (6.3)$$

From Lemma 6.2 and the fact that the eigenvalues for Λ converge to ∞ , it follows that the eigenvalues of L_c also converge to infinity. In particular, the resolvent of L_c is compact. By the Spectral Theorem, there exists an orthonormal basis $(e_m)_{m \in \mathbb{N}}$ of $L^2(\mathbb{T}^2)$ consisting of eigenfunctions of L_c , such that $L_c e_m = \mu_m e_m$. We evaluate the trace using $e_m = f_m$ in (6.3).

Next, we compute $\|e^{sA}Be_m\|_{\mathcal{H}_L}^2$ explicitly:

By construction, $e^{sA}Be_m$ corresponds to the solution of the ODE

$$\begin{cases} \partial_t x_m = y_m \\ \partial_t y_m + \varepsilon^{-2} y_m + \varepsilon^{-2} L x_m = 0 \\ (x_m(0), y_m(0)) = (0, \sqrt{2}\varepsilon^{-2} e_m). \end{cases}$$

The explicit solution is given by

$$\begin{aligned} x_m(t) &= c_m(t) e_m := \frac{\sqrt{2}}{\varepsilon^2 \omega_m} e^{-\frac{t}{2\varepsilon^2}} \sinh(\omega_m t) e_m \\ \partial_t x_m(t) &= \partial_t c_m(t) e_m := \sqrt{2}\varepsilon^{-2} e^{-\frac{t}{2\varepsilon^2}} \left(\cosh(\omega_m t) - \frac{1}{2\varepsilon^2 \omega_m} \sinh(\omega_m t) \right) e_m \end{aligned} \quad (6.4)$$

where $\omega_m := \frac{\sqrt{1-4\varepsilon^2(\mu_m-c)}}{2\varepsilon^2}$. In particular, $e^{sA}B$ leaves the subspace spanned by $\begin{pmatrix} 0 \\ e_m \end{pmatrix}$ invariant.

Using this explicit solution, we obtain

$$\begin{aligned} \|e^{sA}Be_m\|_{\mathcal{H}_L}^2 &= |c_m(s)|^2 \langle L_c^{-\sigma} e_m, e_m \rangle_{L_x^2} + |\partial_t c_m(s)|^2 \langle L_c^{-1-\sigma} e_m, e_m \rangle_{L_x^2} \\ &= |c_m(s)|^2 \mu_m^{-\sigma} + |\partial_t c_m(s)|^2 \mu_m^{-1-\sigma}. \end{aligned} \quad (6.5)$$

By Lemma 6.2, the eigenvalues μ_m of L_c grow asymptotically at the same rate as the eigenvalues of Λ , meaning $\mu_m \rightarrow \infty$. Thus, for sufficiently large m , the discriminant becomes negative, and ω_m becomes purely imaginary. Consequently, the solution decays exponentially. Combining this with the observation that $|\omega_m|^{-2} \lesssim \mu_m^{-1}$, there exists a constant $c_\varepsilon > 0$ and a number $M > 0$ such that for all $m > M$:

$$\|e^{sA}Be_m\|_{\mathcal{H}_L}^2 \lesssim e^{-c_\varepsilon s} \mu_m^{-1-\sigma}. \quad (6.6)$$

Since there are only finitely many terms $m \leq M$ the bound

$$\begin{aligned} \|e^{sA}Be_m\|_{\mathcal{H}_L}^2 &= |c_m(s)|^2 \mu_m^{-\sigma} + |\partial_t c_m(s)|^2 \mu_m^{-1-\sigma} =: C(s, m, \varepsilon) \\ &\leq \sup_{\substack{s \in [0, T] \\ 0 \leq n \leq M}} C(n, s, \varepsilon) < \infty \end{aligned} \quad (6.7)$$

suffices to show that the sum over all finite $m \in \mathbb{N}$ is bounded.

Let $M > 0$ be sufficiently large such that for all $m > M$ the bound in (6.6) holds, and for all $m \leq M$ the bound derived in (6.7) holds. Inserting these bounds into (6.3), we obtain

$$\begin{aligned} \text{Tr}(Q_t) &\lesssim tM \sup_{\substack{s \in [0, T] \\ 0 \leq n \leq M}} C(n, s, \varepsilon) + \int_0^t e^{-c_\varepsilon s} \sum_{\substack{m \in \mathbb{N} \\ m \geq M}} \mu_m^{-1-\sigma} ds \\ &\leq tM \sup_{\substack{s \in [0, T] \\ 0 \leq n \leq M}} C(n, s, \varepsilon) + c_\varepsilon^{-1} \sum_{m \in \mathbb{N}} \mu_m^{-1-\sigma} < \infty. \end{aligned}$$

The final sum converges because the eigenvalues of Λ and the eigenvalues of L_c have the same asymptotic behavior and the sum converges with respect to the eigenvalues of Λ . $\square$

6.2 Conditional Uniqueness of the Invariant Measure

In this section, under the condition $M_\mu := \mathbb{E}_\mu[:u^2:] \in H^{-\sigma}(\mathbb{T}^2)$ for $\sigma \in (0, \frac{1}{2})$, we prove that the unique invariant measure to (2.26) is $\rho_\varepsilon := \mathcal{N}\left(0, \begin{pmatrix} \Lambda^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}\right) \in \mathcal{P}(\mathcal{H}^{-\sigma}(\mathbb{T}^2))$. First, we prove that any invariant measure to (2.26) must be a Gaussian measure on $\mathcal{H}_L$. This is a standard consequence of the linearity of the equation, as established in [20, Theorem 11.7].

Proposition 6.4. *Let $\sigma \in (0, \frac{1}{2})$ and $\nu \in \mathcal{P}(\mathcal{H}_L)$. Assume $M_\mu \in H^{-\sigma}(\mathbb{T}^2)$. Suppose $X(0) \sim \nu$ and that ν is an invariant measure for the flow*

$$X(t) = e^{tA}X(0) + \int_0^t e^{(t-s)A}B dW_s,$$

which is well-defined by Proposition 6.3. Specifically, we assume $\text{Law}(X(t)) = \nu$ for all $t \geq 0$. Then the following hold:

1. *L is strictly positive.*
2. *$\|e^{tA}\|_{\mathcal{H}_L} \leq Me^{-\omega t}$ for some $\omega > 0, M \geq 1$, and ν is the unique invariant measure for the flow $X(t)$.*
3. *$\nu = \mathcal{N}\left(0, \begin{pmatrix} L^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}\right)$.*

Proof. From Theorem 11.7 (iii) in [20], the existence of an invariant measure implies that $\sup_{t \geq 0} \text{Tr}(Q_t) < +\infty$. We use this to show that L must be strictly positive. Suppose toward contradiction that L is not strictly positive. Arguing as in Proposition 6.3, we see that L has compact resolvent and is bounded from below, hence, there exists a lowest eigenvalue $\mu_0 \leq 0$ with a corresponding eigenfunction e_0 . Recall the explicit solution for $e^{sA}Be_0$ discussed in Proposition 6.3. Note, however, that here e_0 is an eigenfunction of L and not of $L + c$ for some $c > 0$ in contrast to Proposition 6.3. If $\mu_0 \leq 0$ then $\omega_0 = \frac{\sqrt{1-4\varepsilon^2\mu_0}}{2\varepsilon} \geq \frac{1}{2\varepsilon^2}$ is positive. Expanding the hyperbolic sine, we see that the solution $c_0(t)$ does not decay in t . It grows either linearly (if $\mu_0 = 0$) or exponentially (if $\mu_0 < 0$). In either case, the time integral diverges:

$$\int_0^t \|e^{sA}Be_0\|_{\mathcal{H}_L}^2 ds \rightarrow \infty \quad \text{as } t \rightarrow \infty.$$

Because $\int_0^t \|e^{sA}Be_0\|_{\mathcal{H}_L}^2 ds \leq \text{Tr}(Q_t)$, this contradicts the uniform bound $\sup_{t \geq 0} \text{Tr}(Q_t) < \infty$.

It follows that L is strictly positive. Consequently, as established in Proposition 6.3, the real parts of the characteristic roots for all modes $m \in \mathbb{N}$ are strictly negative. From (6.5)

and (6.4), it follows that there exist constants $M \geq 1$ and $\omega > 0$ such that the semigroup decays exponentially:

$$\|e^{tA}u\|_{\mathcal{H}_L} \leq Me^{-\omega t}\|u\|_{\mathcal{H}_L} \quad \text{for all } t \geq 0 \text{ and } u \in \mathcal{H}_L. \quad (6.8)$$

From Proposition 11.10 in [20], this strict exponential decay guarantees that ν is the unique invariant measure for the flow $X(t)$.

Furthermore, Theorem 11.7 in [20] establishes that this unique invariant measure takes the form $\mathcal{N}(0, \overline{Q})$, where the covariance operator $\overline{Q}$ is given by

$$\overline{Q}u := \lim_{t \rightarrow \infty} \int_0^t e^{sA}BB^*(e^{sA})^*u \, ds \quad \text{for } u \in \mathcal{H}_L.$$

We finish the proof by computing $\overline{Q}$ explicitly. Let $\{e_n\}_{n \in \mathbb{N}}$ be an orthonormal basis of $L^2(\mathbb{T}^2)$ of eigenfunctions of L such that $Le_n = \mu_n e_n$. We define the L -invariant 2-dimensional subspaces $V_n := \text{span}\{(e_n, 0)^T, (0, e_n)^T\}$ of $\mathcal{H}_L$. Let $\overline{Q}_n := \overline{Q}|_{V_n}$. On this subspace, A restricts to the 2×2 matrix

$$A_n := A|_{V_n} = \begin{pmatrix} 0 & 1 \\ -\varepsilon^{-2}\mu_n & -\varepsilon^{-2} \end{pmatrix}.$$

By the construction of e^{tA} and the L -invariance of V_n it follows that $e^{tA}|_{V_n} = e^{tA_n}$. Furthermore, the basis elements of V_n remain orthogonal in $\mathcal{H}_L$. Consequently, the union over $n \in \mathbb{N}$ of V_n spans $\mathcal{H}_L$.

By the properties of the strongly continuous semigroup e^{tA} , we can differentiate the matrix exponential to obtain

$$\frac{d}{dt} \left(e^{tA_n}BB^*e^{tA_n^*} \right) = A_n e^{tA_n}BB^*e^{tA_n^*} + e^{tA_n}BB^*e^{tA_n^*}A_n^*.$$

Integrating both sides with respect to t over $[0, \infty)$ yields

$$\int_0^\infty \frac{d}{dt} \left(e^{tA_n}BB^*e^{tA_n^*} \right) dt = \int_0^\infty \left(A_n e^{tA_n}BB^*e^{tA_n^*} + e^{tA_n}BB^*e^{tA_n^*}A_n^* \right) dt.$$

By the Fundamental Theorem of Calculus and since due to the exponential decay derived in (6.8), the boundary term at infinity vanishes, the left side evaluates to

$$\left[e^{tA_n}BB^*e^{tA_n^*} \right]_0^\infty = 0 - BB^* = -BB^*.$$

Because the matrices A_n and A_n^* are independent of time and recalling the definition of $\overline{Q}_n$, we obtain

$$\begin{aligned} -BB^* &= A_n \left(\int_0^\infty e^{tA_n}BB^*e^{tA_n^*} dt \right) + \left(\int_0^\infty e^{tA_n}BB^*e^{tA_n^*} dt \right) A_n^* \\ &= A_n \overline{Q}_n + \overline{Q}_n A_n^*. \end{aligned}$$

Recall that B corresponds to the vector $(0, \sqrt{2}\varepsilon^{-2})^T$, meaning $BB^* = \begin{pmatrix} 0 & 0 \\ 0 & 2\varepsilon^{-4} \end{pmatrix}$. A direct

computation shows that the unique solution to this 2×2 -matrix equation is

$$\overline{Q}_n = \begin{pmatrix} \mu_n^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix} = \begin{pmatrix} L^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix} \Big|_{V_n}.$$

Since this holds for every subspace V_n which together span $\mathcal{H}_L$, we conclude that

$$\overline{Q} = \begin{pmatrix} L^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}.$$

□

The previous proposition characterizes the explicit form of any invariant measure of (2.26): if we assume there exists an invariant measure μ of (2.26), then it is the unique invariant measure of the related linear SDE for a fixed M_μ , as explicitly characterized above. In the following, we show that any such invariant measure μ must equal ρ_ε . In particular, this implies that ρ_ε is the unique invariant measure of the SDMFV. Before proceeding with the proof, we record two helpful Lemmas.

The first lemma is a standard result for Gaussian random variables with operator-valued covariance matrices, namely that the expansion of the Gaussian variable is independent of the chosen basis. This is needed to evaluate $M_\mu = \mathbb{E}_\mu[:u^2:]$ explicitly.

Lemma 6.5. *Let $\mu = \mathcal{N}\left(0, \begin{pmatrix} L^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}\right)$ be a Gaussian measure on $\mathcal{H}_L$, and let $b_n(x) := e^{in \cdot x}$ for $n \in \mathbb{Z}^2$ be the standard orthonormal Fourier basis of $L^2(\mathbb{T}^2)$. For any finite $N \in \mathbb{N}$, the point-wise variance of the truncated field is given by*

$$\mathbb{E}_\mu[(\mathbf{P}_N u)^2] = \sum_{|n| \leq N} \sum_{|m| \leq N} \langle L^{-1} \overline{b_n}, b_m \rangle_{L_x^2} b_n b_m$$

which converges formally to the (possibly divergent) series

$$\lim_{N \rightarrow \infty} \mathbb{E}_\mu[(\mathbf{P}_N u)^2] = \sum_{n \in \mathbb{Z}^2} L^{-1}(\overline{b_n}) b_n.$$

Proof. Notice that for any $\mathbf{u} \sim \mu$ the distribution of the first component is $u \sim \mathcal{N}(0, L^{-1})$. Since L is self-adjoint and has purely discrete spectrum we can choose a real-valued orthonormal basis $\{e_j\}_{j \in \mathbb{N}}$ on $L^2(\mathbb{T}^2)$ consisting of eigenfunctions of L , such that $Le_j = \mu_j e_j$. Let $\{\beta_j\}_{j \in \mathbb{N}}$ be a sequence of independent, standard real-valued Gaussian random variables. Then

$$u = \sum_{j \in \mathbb{N}} \mu_j^{-\frac{1}{2}} e_j \beta_j \sim \mathcal{N}(0, L^{-1}).$$

Recall the definition of the truncated Fourier projection $\mathbf{P}_N$

$$\mathbf{P}_N u = \mathcal{F}^{-1}(\mathbf{1}_{\langle n \rangle \leq N} \widehat{f}(n)) = \sum_{|n| \leq N} \langle u, b_n \rangle_{L_x^2} b_n.$$

Squaring this projection and taking the expectation yields

$$\mathbb{E}_\mu[(\mathbf{P}_N u)^2] = \sum_{|n|, |m| \leq N} \mathbb{E}_\mu[\langle u, b_n \rangle_{L_x^2} \langle u, b_m \rangle_{L_x^2}] b_n b_m.$$

Because the Gaussian variables β_j are independent and centered, the cross-terms vanish, and utilizing the spectral decomposition of L^{-1} , we obtain

$$\begin{aligned} \mathbb{E}_\mu[\langle u, b_n \rangle_{L_x^2} \langle u, b_m \rangle_{L_x^2}] &= \sum_{j \in \mathbb{N}} \mu_j^{-1} \langle e_j, b_n \rangle_{L_x^2} \langle e_j, b_m \rangle_{L_x^2} \\ &= \sum_{j \in \mathbb{N}} \mu_j^{-1} \langle \overline{b_n}, e_j \rangle_{L_x^2} \langle e_j, b_m \rangle_{L_x^2} \\ &= \left\langle \sum_{j \in \mathbb{N}} \mu_j^{-1} \langle \overline{b_n}, e_j \rangle_{L_x^2} e_j, b_m \right\rangle_{L_x^2} \\ &= \langle L^{-1} \overline{b_n}, b_m \rangle_{L_x^2}. \end{aligned}$$

Inserting this back into the finite sum and taking the limit $N \rightarrow \infty$, we obtain

$$\begin{aligned} \lim_{N \rightarrow \infty} \mathbb{E}_\mu[(\mathbf{P}_N u)^2] &= \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \sum_{|m| \leq N} \langle L^{-1} \overline{b_n}, b_m \rangle_{L_x^2} b_n b_m \\ &= \sum_{n \in \mathbb{Z}^2} \left(\sum_{m \in \mathbb{Z}^2} \langle L^{-1} \overline{b_n}, b_m \rangle_{L_x^2} b_m \right) b_n \\ &= \sum_{n \in \mathbb{Z}^2} L^{-1}(\overline{b_n}) b_n \end{aligned}$$

where the last two equations hold only in a formal sense as the limit is not necessarily well-defined (see the discussion in Section 2.2). $\square$

This next Lemma provides standard algebraic inequalities for the Japanese bracket $\langle x \rangle := (1 + |x|^2)^{1/2}$.

Lemma 6.6. *For the Japanese bracket $\langle \cdot \rangle$, the following inequalities hold for all $x, y \in \mathbb{R}^d$.*

1. *Let $s \geq 0$. Then $\langle x + y \rangle^s \leq 2^{\frac{s}{2}} \langle x \rangle^s \langle y \rangle^s$.*
2. *Let $s < 0$. Then $\langle x + y \rangle^s \leq 2^{-\frac{s}{2}} \langle x \rangle^s \langle y \rangle^{-s}$.*

Proof. By standard algebraic calculation and the triangle inequality, we have

$$\langle x + y \rangle \leq \sqrt{2} \langle x \rangle \langle y \rangle.$$

For $s \geq 0$, the first statement follows directly by raising both sides to the power of s .

For the second statement, let $s < 0$, which implies $r := -s > 0$. Using the first statement, we have

$$\langle x \rangle^r = \langle (x + y) - y \rangle^r \leq 2^{\frac{r}{2}} \langle x + y \rangle^r \langle -y \rangle^r = 2^{\frac{r}{2}} \langle x + y \rangle^r \langle y \rangle^r.$$

Dividing both sides by $\langle x + y \rangle^r \langle x \rangle^r$ yields

$$\langle x + y \rangle^{-r} \leq 2^{\frac{r}{2}} \langle x \rangle^{-r} \langle y \rangle^r.$$

□

Finally, we are prepared to prove the conditional uniqueness of the invariant measure of the SDMFV.

Theorem 6.7. *Let $\sigma \in (0, \frac{1}{2})$. Then $\rho_\varepsilon := \mathcal{N}\left(0, \begin{pmatrix} \Lambda^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}\right)$ is the unique invariant measure of (2.26) in the class $W := \{\mu \in \mathcal{P}(\mathcal{H}^{-\sigma}(\mathbb{T}^2)) \mid \mathbb{E}_\mu[:u^2:] \in H^{-\sigma}(\mathbb{T}^2)\}$.*

Proof. Suppose there exists an invariant measure $\mu \in W$ for a solution to (2.26). From Proposition 6.4, we know $\mu = \mathcal{N}\left(0, \begin{pmatrix} L^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}\right)$. The idea of this proof is to show that $M_\mu := \mathbb{E}_\mu[:u^2:] \in L^2(\mathbb{T}^2)$, which ensures that the integral $\int_{\mathbb{T}^2} M_\mu(x)^2 dx$ is well-defined. Then, we prove that this integral is equal to 0. This implies $L = \Lambda$, and by the characterization in Proposition 6.4, we directly obtain $\mu = \rho_\varepsilon$ and $\mathcal{H}_L = \mathcal{H}^{-\sigma}(\mathbb{T}^2)$.

To show $M_\mu \in L^2(\mathbb{T}^2)$, we compute M_μ explicitly. Let $b_n(x) := e^{in \cdot x}$ for $n \in \mathbb{Z}^2$ denote the orthonormal basis of $L^2(\mathbb{T}^2)$.

By Lemma 6.5 we can formally expand $u \sim \mathcal{N}(0, L^{-1})$ with respect to the basis b_n to obtain

$$\lim_{N \rightarrow \infty} \mathbb{E}_\mu[(P_N u)^2] = \sum_{n \in \mathbb{Z}^2} L^{-1}(b_n) \overline{b_n}.$$

Inserting this into the definition of M_μ , we obtain

$$M_\mu = \sum_{n \in \mathbb{Z}^2} (L^{-1} - \Lambda^{-1})(b_n) b_{-n}. \quad (6.9)$$

To show that $M_\mu \in L^2(\mathbb{T}^2)$, we compute the Fourier coefficients $\widehat{M}_\mu(m)$ for all $m \in \mathbb{Z}^2$. From the resolvent identities

$$\begin{aligned} L^{-1} - \Lambda^{-1} &= -L^{-1}(L - \Lambda)\Lambda^{-1} = -L^{-1}M_\mu\Lambda^{-1}, \\ L^{-1} &= \Lambda^{-1} - \Lambda^{-1}M_\mu L^{-1}, \end{aligned}$$

we can write $L^{-1} - \Lambda^{-1}$ in the symmetric form

$$L^{-1} - \Lambda^{-1} = -\Lambda^{-1}M_\mu\Lambda^{-1} + \Lambda^{-1}M_\mu L^{-1}M_\mu\Lambda^{-1}.$$

Using this algebraic equality and the fact that $b_n b_m = b_{n+m}$, we derive the following for $m \in \mathbb{Z}^2$:

$$\begin{aligned} \widehat{M}_\mu(m) &= \sum_{n \in \mathbb{Z}^2} \langle (L^{-1} - \Lambda^{-1})b_n \cdot b_{-n}, b_m \rangle_{L_x^2} \\ &= - \sum_{n \in \mathbb{Z}^2} \langle \Lambda^{-1}M_\mu\Lambda^{-1}b_n, b_{n+m} \rangle_{L_x^2} + \sum_{n \in \mathbb{Z}^2} \langle \Lambda^{-1}M_\mu L^{-1}M_\mu\Lambda^{-1}b_n, b_{n+m} \rangle_{L_x^2} \\ &=: I(m) + R(m). \end{aligned}$$

Since $\Lambda^{-1}b_n = \langle n \rangle^{-2}b_n$ and Λ^{-1} is self-adjoint, we can rewrite $I(m)$ as follows:

$$\begin{aligned} I(m) &= - \sum_{n \in \mathbb{Z}^2} \langle \Lambda^{-1}M_\mu \Lambda^{-1}b_n, b_{n+m} \rangle_{L_x^2} \\ &= - \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2} \langle n+m \rangle^{-2} \langle M_\mu b_n, b_{n+m} \rangle_{L_x^2} \\ &= - \widehat{M}_\mu(m) \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2} \langle n+m \rangle^{-2}. \end{aligned}$$

Define $a(m) := \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2} \langle n+m \rangle^{-2}$ which is finite for any $m \in \mathbb{Z}^2$ by Lemma 3.2. Thus,

$$\widehat{M}_\mu(m) = -a(m)\widehat{M}_\mu(m) + R(m).$$

Dividing by $1 + a(m) > 1$, we obtain

$$\widehat{M}_\mu(m) = \frac{R(m)}{1 + a(m)}.$$

In particular, since $(1 + a(m))^{-1} < 1$, it suffices to bound $|R(m)|$, as $|\widehat{M}_\mu(m)| \leq |R(m)|$.

Using $\Lambda^{-1}b_n = \langle n \rangle^{-2}b_n$, the self-adjointness of Λ^{-1} , and the fact that M_μ is a real-valued multiplication operator, yields

$$\begin{aligned} R(m) &= \sum_{n \in \mathbb{Z}^2} \langle \Lambda^{-1}M_\mu L^{-1}M_\mu \Lambda^{-1}b_n, b_{n+m} \rangle_{L_x^2} \\ &= \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2} \langle n+m \rangle^{-2} \langle L^{-1}(M_\mu b_n), M_\mu b_{n+m} \rangle_{L_x^2}. \end{aligned} \tag{6.10}$$

To bound the inner product in the last expression, we show that we gain up to two derivatives through L^{-1} : Intuitively, we gain two derivatives through Λ^{-1} . Since L and Λ are form equivalent, we expect this property to transfer to L^{-1} . Specifically, we show that $\|L^{-1}u\|_{H_x^s} \lesssim \|u\|_{H_x^{-1}}$ for $s \in [0, 1]$: By the Heinz inequality for self-adjoint operators ([24, Proposition 10.14]) the form equivalence of L and Λ translates to the form equivalence of L^α and Λ^α for $\alpha \in [0, 1]$. Since L and Λ are self-adjoint operators the Spectral Theorem implies that the operators L^α and Λ^α remain self-adjoint operators, therefore we can apply Corollary 10.12 in [24] to see that $L^{-\alpha}$ and $\Lambda^{-\alpha}$ are form equivalent. Furthermore, since $L^{-\alpha}$ is a bounded, self-adjoint operator on $L^2(\mathbb{T}^2)$, we have that $\langle L^{-\alpha}u, u \rangle_{L_x^2} \leq \|u\|_{L_x^2}^2$. These considerations yield for $s \in [0, 1]$:

$$\begin{aligned} \|L^{-1}u\|_{H_x^s}^2 &= \langle \Lambda^s L^{-1}u, L^{-1}u \rangle_{L_x^2} \lesssim \langle L^s L^{-1}u, L^{-1}u \rangle_{L_x^2} \\ &= \langle L^{-1+s} L^{-\frac{1}{2}}u, L^{-\frac{1}{2}}u \rangle_{L_x^2} \leq \langle L^{-1}u, u \rangle_{L_x^2} \\ &\lesssim \langle \Lambda^{-1}u, u \rangle_{L_x^2} = \|u\|_{H_x^{-1}}^2. \end{aligned} \tag{6.11}$$

Next, we apply this derivative gain to evaluate the inner product in (6.10). We utilize the duality pairing between $H^\sigma(\mathbb{T}^2)$ and $H^{-\sigma}(\mathbb{T}^2)$, the fact that b_n acts as a frequency shift combined with Lemma 6.6. Since (6.11) guarantees a sufficient derivative gain (using

$\sigma < 1$), and $M_\mu \in H^{-\sigma}(\mathbb{T}^2)$, we obtain

$$\begin{aligned}
\langle L^{-1}(M_\mu b_n), M_\mu b_{n+m} \rangle_{L_x^2} &\leq \|L^{-1}(M_\mu b_n)\|_{H^\sigma} \|M_\mu b_{n+m}\|_{H^{-\sigma}} \\
&= \|L^{-1}(M_\mu b_n)\|_{H^\sigma} \left(\sum_{j \in \mathbb{Z}^2} \langle j + n + m \rangle^{-2\sigma} |\widehat{M_\mu}(j)|^2 \right)^{\frac{1}{2}} \\
&\leq 2^{\frac{\sigma}{2}} \langle n + m \rangle^\sigma \|L^{-1}(M_\mu b_n)\|_{H^\sigma} \|M_\mu\|_{H^{-\sigma}} \\
&\lesssim 2^{\frac{\sigma}{2}} \langle n + m \rangle^\sigma \|M_\mu b_n\|_{H^{-\sigma}} \\
&\lesssim 2^\sigma \langle n + m \rangle^\sigma \langle n \rangle^\sigma.
\end{aligned}$$

Plugging this estimate into (6.10) and applying Lemma 3.2 (utilizing $2 - \sigma \in (1, 2)$), we obtain

$$|R(m)| \lesssim \sum_{n \in \mathbb{Z}^2} \langle n \rangle^{-2} \langle n + m \rangle^{-2} \langle n + m \rangle^\sigma \langle n \rangle^\sigma \lesssim \langle m \rangle^{-2+2\sigma}.$$

Combining this with $|\widehat{M_\mu}(m)|^2 \leq |R(m)|^2$ and noting that $\sigma \in (0, \frac{1}{2})$ implies $-4 + 4\sigma < -2$, it follows that

$$\|M_\mu\|_{L_x^2}^2 = \sum_{m \in \mathbb{Z}^2} |\widehat{M_\mu}(m)|^2 \leq \sum_{m \in \mathbb{Z}^2} |R(m)|^2 \lesssim \sum_{m \in \mathbb{Z}^2} \langle m \rangle^{-4+4\sigma} < \infty.$$

Finally, we show that $M_\mu = 0$. Inserting (6.9) for M_μ and using the resolvent identity $L^{-1} - \Lambda^{-1} = -L^{-1}M_\mu\Lambda^{-1}$, we have

$$\begin{aligned}
0 &\leq \int_{\mathbb{T}^2} M_\mu(x)^2 dx = \int_{\mathbb{T}^2} M_\mu(x) M_\mu(x) dx \\
&= \int_{\mathbb{T}^2} M_\mu(x) \sum_{n \in \mathbb{Z}^2} (L^{-1} - \Lambda^{-1})(b_n)(x) b_{-n}(x) dx \\
&= - \sum_{n \in \mathbb{Z}^2} \langle M_\mu L^{-1} M_\mu \Lambda^{-1} b_n, b_n \rangle_{L_x^2} \\
&= - \sum_{n \in \mathbb{Z}^2} \langle L^{-1} M_\mu \Lambda^{-\frac{1}{2}} b_n, M_\mu \Lambda^{-\frac{1}{2}} b_n \rangle_{L_x^2} \\
&\leq 0,
\end{aligned}$$

where the last inequality holds because the quadratic form associated with L^{-1} is positive definite. Consequently, this series of inequalities yields $M_\mu = 0$. $\square$

6.3 Convergence of SDMFw's Invariant Measure to the SMFW's Invariant Measure

As discussed in Section 2.3, we show that the SKA also occurs for the respective probability measures. Let $\sigma > 0$ and $\pi : \mathcal{H}^{-\sigma}(\mathbb{T}^2) \rightarrow H^{-1-\sigma}(\mathbb{T}^2)$ be the canonical projection onto the second component. To prove convergence, we define a random variable $\psi_\varepsilon \sim \mathcal{N}(0, \varepsilon^{-2}I) = \pi_* \rho_\varepsilon$ on $H^{-1-\sigma}(\mathbb{T}^2)$. Let $\{h_n\}_{n \in \mathbb{Z}^2}$ be a family of independent random variables such that for

$n \neq 0$, h_n is a complex-valued centered Gaussian random variable with variance 2 satisfying $h_{-n} = \overline{h_n}$ for $n \in \mathcal{J}$, and for $n = 0$, h_0 is a real-valued centered Gaussian random variable with variance 1. We assume these are independent of the cylindrical Wiener process W_t defined in (2.13). Define

$$\psi_\varepsilon(x) := \varepsilon^{-1}h_0 + \frac{1}{\sqrt{2}} \sum_{n \in \mathbb{Z}^2 \setminus \{0\}} \varepsilon^{-1}h_n e^{in \cdot x} \sim \mathcal{N}(0, \varepsilon^{-2}I).$$

First, we show that $(\mathbf{P}_N \psi_\varepsilon)_* \mathbb{P}$ converges weakly to $(\psi_\varepsilon)_* \mathbb{P} = \pi_* \rho_\varepsilon$ as $N \rightarrow \infty$, where $\mathbf{P}_N$ is the truncated Fourier projection defined in (2.11). This weak convergence is implied by the following standard statement.

Lemma 6.8. *For fixed $\varepsilon \in (0, 1]$, we have*

$$\mathbf{P}_N \psi_\varepsilon \rightarrow \psi_\varepsilon \quad \text{in } L^2(\Omega; H^{-1-\sigma}(\mathbb{T}^2)),$$

as $N \rightarrow \infty$.

Proof. For $s \geq 0$ and $N \in \mathbb{N}$, define the truncated random variable

$$\psi_{N,s}(x) := \sum_{\substack{n \in \mathbb{Z}^2 \\ \langle n \rangle \leq N}} \frac{h_n}{\langle n \rangle^s} e^{in \cdot x}, \quad (6.12)$$

where h_n is defined as above. Let $\sigma \in \mathbb{R}$. We will show that $\psi_{N,s}$ is a Cauchy sequence in $L^2(\Omega; H^\sigma(\mathbb{T}^2))$. Let $M, N \in \mathbb{N}$ with $M \leq N$. Using the independence of h_n and the fact that $\mathbb{E}[|h_n|^2] \leq 2$, we compute the expectation of the difference

$$\begin{aligned} \mathbb{E} \left[\|\psi_{N,s} - \psi_{M,s}\|_{H_x^\sigma}^2 \right] &= \sum_{\substack{n \in \mathbb{Z}^2 \\ M < \langle n \rangle \leq N}} \langle n \rangle^{2\sigma} \mathbb{E} \left[\left| \frac{h_n}{\langle n \rangle^s} \right|^2 \right] \\ &\leq 2 \sum_{\substack{n \in \mathbb{Z}^2 \\ M < \langle n \rangle \leq N}} \langle n \rangle^{2\sigma-2s}. \end{aligned}$$

This sum converges if and only if $\sigma < s - 1$. Hence, for any $\sigma < s - 1$, the limit $\lim_{N \rightarrow \infty} \psi_{N,s}$ is well-defined in $L^2(\Omega; H^\sigma(\mathbb{T}^2))$. Since our target space requires $-1 - \sigma < -1$, the statement follows directly. $\square$

We are now ready to prove the convergence of the probability measures.

Theorem 6.9. *Let $\sigma > 0$, $\varepsilon > 0$, and $\pi : \mathcal{H}^{-\sigma}(\mathbb{T}^2) \rightarrow H^{-1-\sigma}(\mathbb{T}^2)$ be the canonical projection onto the second component. Let $\rho_\varepsilon := \mathcal{N}\left(0, \begin{pmatrix} \Lambda^{-1} & 0 \\ 0 & \varepsilon^{-2} \end{pmatrix}\right)$ and $\rho_0 := \mathcal{N}(0, \Lambda^{-1})$. Then $\rho_\varepsilon \rightarrow \rho_0$ as $\varepsilon \rightarrow 0$ in the sense of Definition 2.6.*

Proof. Let $f \in C_0(H^{-1-\sigma}(\mathbb{T}^2))$ be arbitrary. By the definition, f is bounded, thus there exists $C > 0$ such that $|f| \leq C$. Furthermore, for any $\delta > 0$, there exists a $K > 0$ such that $|f(u)| < \delta$ for all $\|u\|_{H_x^{-1-\sigma}} > K$.

Using the fact that $(\pi_*\rho_\varepsilon)$ is a probability measure, therefore it is bounded by 1, a Chebyshev-like inequality, and the fact that $(\mathbf{P}_N\psi_\varepsilon)_*\mathbb{P}$ converges weakly to $(\psi_\varepsilon)_*\mathbb{P}$ (by Lemma 6.8), we obtain

$$\begin{aligned}
\left| \int_{H^{-1-\sigma}(\mathbb{T}^2)} f(u) d(\pi_*\rho_\varepsilon)(u) \right| &\leq \delta(\pi_*\rho_\varepsilon)(\|u\|_{H_x^{-1-\sigma}} > K) + C(\pi_*\rho_\varepsilon)(\|u\|_{H_x^{-1-\sigma}} \leq K) \\
&= \delta + C(\pi_*\rho_\varepsilon)(\|u\|_{H_x^{-1-\sigma}} \leq K) \\
&\leq \delta + Ce^{K^2} \int_{H^{-1-\sigma}(\mathbb{T}^2)} e^{-\|u\|_{H_x^{-1-\sigma}}^2} d(\psi_\varepsilon)_*\mathbb{P}(u) \\
&= \delta + Ce^{K^2} \lim_{N \rightarrow \infty} \int_{\Omega} e^{-\|\mathbf{P}_N\psi_\varepsilon(\omega)\|_{H_x^{-1-\sigma}}^2} d\mathbb{P}(\omega).
\end{aligned}$$

To compute the last term, we denote $\mathcal{J}_N := \{n \in \mathcal{J} \mid \langle n \rangle \leq N\}$. This is a finite subset of $\mathbb{Z}^2$. Furthermore, by the cylindrical construction of the random variables h_n , namely that $h_{-n} = \overline{h_n}$ for $n \in \mathcal{J}$, we have $|h(n)|^2 = |h(-n)|^2$. Consequently, we obtain

$$\begin{aligned}
&\lim_{N \rightarrow \infty} \int_{\Omega} e^{-\|\mathbf{P}_N\psi_\varepsilon(\omega)\|_{H_x^{-1-\sigma}}^2} d\mathbb{P}(\omega) \\
&= \lim_{N \rightarrow \infty} \int_{\Omega} \exp \left(-\langle 0 \rangle^{-2-2\sigma} \varepsilon^{-2} |h_0(\omega)|^2 - \sum_{n \in \mathcal{J}_N} \langle n \rangle^{-2-2\sigma} \varepsilon^{-2} |h_n(\omega)|^2 \right) d\mathbb{P}(\omega) \\
&= \lim_{N \rightarrow \infty} (1 + 2\varepsilon^{-2})^{-\frac{1}{2}} \prod_{n \in \mathcal{J}_N} (1 + 2\varepsilon^{-2} \langle n \rangle^{-2-2\sigma})^{-1} \\
&= (1 + 2\varepsilon^{-2})^{-\frac{1}{2}} \lim_{N \rightarrow \infty} \prod_{n \in \mathcal{J}_N} \left(1 - \frac{2}{\varepsilon^2 \langle n \rangle^{2+2\sigma} + 2} \right).
\end{aligned}$$

This infinite product converges to a strictly positive number if and only if the series

$$\sum_{n \in \mathbb{Z}^2} \frac{2}{\varepsilon^2 \langle n \rangle^{2+2\sigma} + 2} < \infty$$

converges. This holds true because $2 + 2\sigma > 2$. Note, that the product converges for all $\varepsilon \in [0, 1]$, since for $\varepsilon = 0$ the product equals 0. Finally, as $(1 + 2\varepsilon^{-2})^{-\frac{1}{2}} \rightarrow 0$ for $\varepsilon \rightarrow 0$ and since the infinite product is uniformly bounded in ε the final expression converges to 0. Consequently, the integral converges to 0, implying that $\rho_\varepsilon \rightarrow \rho_0$ in the sense of Definition 2.6. $\square$

7 Appendix

Here, we collect some technical results necessary for this thesis that do not help the understanding or the readability of the main subject of this thesis.

7.1 Garsia-Rodemich-Rumsey Lemma and Application

The first Lemma is an extension of the Garsia-Rodemich-Rumsey result to multiple parameters. The Lemma is based on a more general version by [26, Lemma 1], however we edited it to suit our application.

Lemma 7.1. *Let $(E, \|\cdot\|)$ be a normed space, $\gamma > 0$, and let $f \in C([0, 1]^d; E)$. Consider a continuous, convex, and strictly increasing function $\Phi : \mathbb{R} \rightarrow [0, \infty)$. Furthermore, assume $\Phi(0) = 0$ and $\Phi(x) \rightarrow \infty$ as $x \rightarrow \infty$.*

Then,

$$\int_{[0,1]^d} \int_{[0,1]^d} \Phi \left(\frac{\|f(\mathbf{x}) - f(\mathbf{x}')\|}{|\mathbf{x} - \mathbf{x}'|^\gamma} \right) d\mathbf{x} d\mathbf{x}' \leq F, \quad (7.1)$$

implies for all $\mathbf{x}, \mathbf{x}' \in [0, 1]^d$

$$\|f(\mathbf{x}) - f(\mathbf{x}')\| \leq 8\gamma \int_0^{|\mathbf{x} - \mathbf{x}'|} \Phi^{-1} \left(\frac{F}{u^{2d}} \right) u^{\gamma-1} du. \quad (7.2)$$

Proof. Follows directly analogous to Lemma 1 in [26]. $\square$

In the next Lemma, we apply this Garsia-Rodemich-Rumsey result to the exponential function. Combining this with the Chebychev inequality, we can obtain a uniform exponential bound for the stochastic convolution Ψ .

Lemma 7.2. *Let $(E, \|\cdot\|)$ be a normed space and let $f \in C([0, 1]^2; E)$. Let $c_0, \alpha \geq 1$, and $\gamma \in (0, 1)$. Suppose that*

$$\int_{[0,1]^2} \int_{[0,1]^2} \exp \left[c_0 \left(\frac{\|f(\mathbf{x}) - f(\mathbf{x}')\|}{|\mathbf{x} - \mathbf{x}'|^\gamma} \right)^\alpha \right] d\mathbf{x} d\mathbf{x}' =: F < \infty. \quad (7.3)$$

Then,

$$\exp \left[\frac{c_0}{(8\gamma)^\alpha} \left(\sup_{\substack{\mathbf{x}, \mathbf{x}' \in [0,1]^2 \\ \mathbf{x} \neq \mathbf{x}'}} \frac{\|f(\mathbf{x}) - f(\mathbf{x}')\|}{\zeta(|\mathbf{x} - \mathbf{x}'|)} \right)^\alpha \right] \leq \max(F, e) \quad (7.4)$$

where $\zeta(x) := \int_0^x \left(2 \log \left(1 + \frac{4}{u^4} \right) \right)^{\frac{1}{\alpha}} u^{\gamma-1} du$ is bounded by $\zeta(1) < \infty$.

Proof. We follow closely the arguments of Lemma 2.2 in [15]. We apply Lemma 7.1 with $\Phi(t) = e^{c_0 t^\alpha} - 1$, which satisfies the assumptions of Lemma 7.1 (specifically, convexity is satisfied as $\alpha \geq 1$). Furthermore, by assumption, (7.1) is satisfied using Φ , thus, Lemma

7.1 yields for all $\mathbf{x}, \mathbf{x}' \in [0, 1]^2$

$$\|f(\mathbf{x}) - f(\mathbf{x}')\| \leq 8\gamma c_0^{-\frac{1}{\alpha}} \int_0^{|\mathbf{x}-\mathbf{x}'|} \left(\log \left(1 + \frac{4F}{u^4} \right) \right)^{\frac{1}{\alpha}} u^{\gamma-1} du.$$

Let $A \geq e - 1$ and $B \geq e$. Then the inequality

$$\log(1 + AB) \leq \log(1 + A) + \log(B) \leq 2\log(1 + A) \cdot \log B$$

implies that

$$\|f(\mathbf{x}) - f(\mathbf{x}')\| \leq 8\gamma c_0^{-\frac{1}{\alpha}} \zeta(|\mathbf{x} - \mathbf{x}'|) (\log(\max(F, e)))^{\frac{1}{\alpha}}, \quad (7.5)$$

as $\frac{4}{u^4} \geq e - 1$, for $0 < u \leq 1$.

Applying Hölder's inequality with $p, p' > 0$, $\frac{1}{p} + \frac{1}{p'} = 1$ such that $p(\gamma - 1) > -1$ (which utilizes that $\gamma \in (0, 1)$), and the fact that $\log x$ converges slower to infinity than any positive root of x as $x \rightarrow \infty$, we obtain

$$\begin{aligned} \zeta(|\mathbf{x} - \mathbf{x}'|) &= \int_0^{|\mathbf{x}-\mathbf{x}'|} \left(2\log \left(1 + \frac{4}{x^4} \right) \right)^{\frac{1}{\alpha}} x^{\gamma-1} dx \\ &\leq \left(\int_0^1 \left(2\log \left(1 + \frac{4}{x^4} \right) \right)^{\frac{p'}{\alpha}} dx \right)^{\frac{1}{p'}} \left(\int_0^1 x^{p\gamma-p} dx \right)^{\frac{1}{p}} < \infty, \end{aligned}$$

uniformly in $\mathbf{x}, \mathbf{x}' \in [0, 1]^2$. Finally, the desired estimate (7.4) follows directly from (7.5). $\square$

7.2 Operator Theory

Here, we recall fundamental concepts of operator theory on Hilbert spaces. First the definition of a (self-)adjoint operator:

Definition 7.3 (Adjoint and Self-Adjoint Operator ([24] Chapter 1.2)). *Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces with inner products $\langle \cdot, \cdot \rangle_{\mathcal{H}_1}, \langle \cdot, \cdot \rangle_{\mathcal{H}_2}$, respectively, and let $A: \mathcal{D}(A) \subset \mathcal{H}_1 \rightarrow \mathcal{H}_2$ be a densely defined linear operator.*

The adjoint operator A^ is defined on the domain $\mathcal{D}(A^*)$, which consists of all $y \in \mathcal{H}_2$ for which there exists a $z \in \mathcal{H}_1$ such that*

$$\langle Ax, y \rangle_{\mathcal{H}_2} = \langle x, z \rangle_{\mathcal{H}_1} \quad \text{for all } x \in \mathcal{D}(A).$$

For such $y \in \mathcal{D}(A^)$, we define $A^*y = z$.*

The operator A is called self-adjoint if its domain coincides with the domain of its adjoint, and the operators agree on this common domain. That is,

$$\mathcal{D}(A) = \mathcal{D}(A^*) \quad \text{and} \quad Ax = A^*x \quad \text{for all } x \in \mathcal{D}(A).$$

Secondly, the spectrum of operators on Hilbert spaces.

Definition 7.4 (Resolvent, Resolvent Set, and Spectrum ([24] Chapter 2.2)). *Let $\mathcal{H}$ be a complex Hilbert space and let $A: \mathcal{D}(A) \subset \mathcal{H} \rightarrow \mathcal{H}$ be a closed linear operator. Let I denote*

the identity operator on $\mathcal{H}$.

The resolvent set of A , denoted by $\rho(A)$, is defined as the set of all complex numbers $\lambda \in \mathbb{C}$ for which the operator $(\lambda I - A)$ is a bijection from $\mathcal{D}(A)$ onto $\mathcal{H}$, and its inverse is a bounded linear operator defined on all of $\mathcal{H}$.

For any $\lambda \in \rho(A)$, the bounded inverse operator

$$R(\lambda, A) = (\lambda I - A)^{-1}$$

is called the resolvent of A at the point λ .

The spectrum of A , denoted by $\sigma(A)$, is defined as the complement of the resolvent set in the complex plane. That is,

$$\sigma(A) = \mathbb{C} \setminus \rho(A).$$

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